

Hematopoiesis and Myeloproliferative Neoplasms: hydrodynamic limit and statistical analysis

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CHAPTER 1

Introduction

1.1 Biological context

1.1.1 Hematopoiesis

Hematopoiesis is the biological process through which all blood cells are produced. Mature blood cells perform specific and essential functions in the body and are categorized into three main types: erythrocytes (red blood cells), which transport oxygen and carbon dioxide; thrombocytes (platelets) that facilitate clotting; and leukocytes (white blood cells), which drive the immune system. The total number of non-nucleated cells (erythrocytes and platelets) in the male human body is of around 2.9×10^{13} , which accounts to around 80% of the total number of cells in the body ([30]).

The process of hematopoiesis begins in the bone marrow with hematopoietic stem cells (HSCs), a multipotent type of immature cell that has the capacity to develop into any blood cell type. The transition from a HSC to a mature cell relies on two fundamental mechanisms: proliferation and differentiation. Division or proliferation is characterized by a mother cell dividing into two daughter cells, and thus increasing the total cell population. On the other hand, differentiation is the process through which cells become more and more specialized, until finally reaching full maturity. As cells differentiate, they progressively lose their capacity to become any type of mature cell and commit to a specific lineage. Mature myeloid blood cells do not undergo division, their populations are replenished through differentiation of progenitor cells. Furthermore, cellular death occurs in both mature and immature cells to maintain

balance.

Hematopoiesis is tightly regulated through intrinsic and extrinsic factors that either promote or inhibit division and differentiation. This regulation is essential as it allows to maintain a sufficient quantity of HSC throughout the life of an individual while also ensuring the production of the needed number of each type of mature cell. Regulation arises from signals from the microenvironment that influence division, differentiation and lineage commitment. These signals include cytokines, often secreted by mature cells as part of a feedback loop, which have a specific and possibly different effect on different maturation stages as explained in [71]. These growth factors do not act alone, they interact with each other to further promote or inhibit their effect. Intrinsically, cells produce transcription factors that form regulatory networks competing and interacting with each other to ultimately determine the fate of the cell.

Differentiation in hematopoiesis

The exact underlying mechanisms in hematopoiesis are still unknown with many long-held hypotheses being constantly challenged. An example of this is found in differentiation, as it has been commonly assumed as a step-wise process where cells transition through a fixed number of cell types before reaching full maturity. This reasoning arises from the observation of cell surface proteins and transplantation using bulk cell analysis. These procedures often assume that cells sharing the same phenotype (surface markers) have identical functions ([18]).

As described in [18] and illustrated in 1.1, the classical hematopoiesis step-wise approach considers that among HSCs we find two types of cells: long term (LT)-HSC and short term (ST)-HSC. LT-HSCs are quiescent cells with a long term lifelong self-renewal ability, while ST-HSC have a short-term self-renewal ability as they will eventually differentiate. ST-HSC differentiate into multipotent progenitors (MPPs). The first major lineage commitment occurs after MPPs, where cells either differentiate into a Common Myeloid Progenitor (CMP) with myeloid, erythroid and megakaryocytic potential or into a Common Lymphoid Progenitor (CLP) with lymphoid potential. A further commitment occurs when the CMP differentiate into megakaryocyte-erythrocyte progenitor MEP, eventually producing red blood cells or platelets, or into granulocyte-macrophage progenitor (GMP). According to the models under this approach, only a small and fixed number of cell types transitions are needed after these progenitors to reach a mature cell.

As new technologies of observation appear at single-cell level (single-cell RNA-Seq analysis), the observation of cells becomes more detailed which allows to revisit the classical step-wise approach with a different perspective. Previously defined types of maturation stages in hematopoiesis development are able to divide into more subtypes. This is the case of HSC where in [11] and [83] they prove the existence of an intermediary stage of cell maturation IT-HSC between LT-HSCs and ST-HSCs. Additionally,

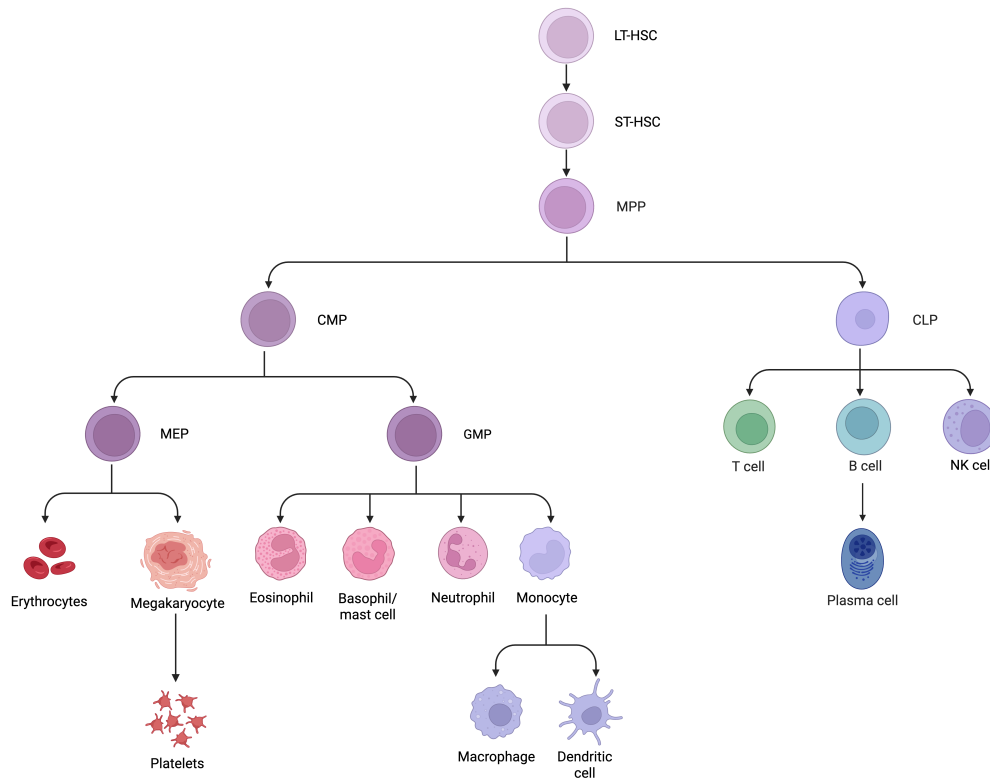


Figure 1.1: The classical hematopoiesis steps. *Image created with BioRender.com.*

as shown in [60] and [82] MPP can further be divided into 4 categories based on their immunophenotype, cell cycle status, differentiation ability, resistance to treatment and bone marrow abundance.

Furthermore, in [62] it is explained that the phenotype of cells widely fluctuates over time implying important heterogeneity among cells rather than clearly defined phenotypes that allow to separate into different cell types. Based on these findings, the authors in [62] and [85] suggest that hematopoiesis is a continuous process as an alternative of the classical step-wise hypothesis. Then, differentiation would be a progressive process from HSC to mature cells, and instead of maturation types we would be talking about a maturation level in each lineage.

1.1.2 Myeloproliferative Neoplasms

Myeloproliferative neoplasms (MPNs) are blood cancers issued from a mutation at stem cell level which induces deregulation in proliferation resulting in an overproduction of myeloid mature blood cells. The lineage emerging from these mutated stem cells will

carry the same mutation. The excess of mature cells can potentially have important implications in the health of an individual. MPNs are divided into three main diseases: Polycythemia Vera (PV), Essential Thrombocythemia (ET) and Primary Myelofibrosis (PMF). PV presents an enlargement of the red blood cells mass, ET is characterized by an excess of platelets and PMF by bone marrow fibrosis ([74]).

These diseases have a clear age structure with a growing number of cases for older ages. This can be observed in the two age of detection data sets we work with in Chapter 3 as seen in Figure 1.4. Indeed, very few cases are present for younger individuals, the majority of them corresponding to older individuals. Other data sets confirm this structure, with a median age of detection of around 65 years (see [67]).

A driver mutation is a mutation that is sufficient to develop the disease. In MPN, the most common driver mutation is JAK2V617F followed by JAK2 exon12, CALR and MPL ([78]). In each pathology, this mutation is presented differently: in ET most cells are heterozygous, meaning that the mutation is present in only one allele; in PV a mitotic recombination induces that most cells carry the mutation in both alleles ([70]); while in PMF the mutation is coupled with the acquisition of other mutations such as TET2 or ASXL1. Additional mutations can arise before or after the driver mutation, commonly in already mutated cells (see [45]). The accumulation of these additional mutations can induce different dynamics in the mutant cells leading to an increase of the probability of the evolution towards more serious diseases, a decrease of the survival probability ([75], [81]) or a higher resistance of cells to treatment ([37]).

Three main cytoreductive treatments are administered to patients diagnosed with MPN: Interferon- α (IFN), Hydroxyurea (HU) and Ruxolitinib (Ruxo). Each treatment has a different mode of action on the cancerous population. IFN works by promoting differentiation of mutated cells, eventually exhausting the cell. HU inhibits proliferation of cells, which affects the growth of the mutant population. Ruxo works as a JAK1/JAK2 inhibitor which suppresses the mutant cells.

1.2 Mathematical context

1.2.1 Mathematics and biology

Mathematical modeling and statistical analysis are powerful tools that allow to represent, make predictions and better understand biological phenomena. A large variety of tools has been developed in this aim. From individual-based models to macroscopic descriptions of system behavior, each modeling approach has a particular interest in mind.

Biological systems are inherently complex, as they are composed of multiple interconnected networks which may evolve over different scales in time and/or space. Advances in observational techniques lead to an increased amount of details about the

system across different scales to become available, further contributing to this complexity. Mathematical modeling provides the framework to gather this information in different scales and explain how microscopic behavior can give rise to the macroscopic observations ([8]).

Obtaining biological data is often costly and time-consuming. Mathematics here play a crucial role, as they allow to deduce biological insight both from direct measurements and indirectly from non-invasive observations ([58]). As explained in [8], *in vivo* experiments may be difficult to conduct, and while an *in vitro* approach provides a much simpler alternative, it only offers a partial and isolated representation of the complete biological system. Mathematical modeling helps reduce the gap between *in vivo* and *in vitro* observations by integrating *in vitro* experimental results into models that better reflect the *in vivo* environment. By comparing models with data, it is possible to study different assumptions on the structure and mechanism of a system, and conclude based on which ones better explain the data. Furthermore, the mathematical representation of a system provides a framework to perform simulations under a variety of conditions, without having to conduct the associated experiments.

As presented in [19], important reciprocal scientific developments arise from the intersection of mathematics and biology: mathematics has provided valuable insights into biology, while biological questions have inspired mathematical developments.

1.2.2 Mathematical models for hematopoiesis

A large variety of approaches to model hematopoiesis have been developed, with different mathematical approaches and each aiming to capture a particular biological phenomena. Commonly, these models consider a compartmental structure, where each compartment represents a distinct type of cell corresponding to a maturation stage. Some examples of deterministic compartmental models are [15], [40] and [73]. Stochastic compartmental modeling naturally arises as a birth and death process, where divisions can be considered as births and cells can migrate or “die” towards a higher level of differentiation. This type of modeling originates with [76], where they study the dynamics of stem cells through a birth and death process.

Differentiation is often linked to division as done in [15], [22], [40], [51]. In this scenario, each of the two daughter cells resulting from a division event remains the same type as the mother with a certain probability, or differentiates and commits to a specific lineage. Regulation in proliferation and/or division is often included through feedback loops depending on the mature cells population size as in [23], [40] and [51] or through cytokines having its own dynamics as in [3]. In stress hematopoiesis, important changes in sizes of the compartments arise and regulatory dynamics are much stronger and do not only arise from mature cells, this can be seen in [3] and [15]. Hematological diseases can induce oscillatory dynamics in the sizes of the cells population, a topic studied in [1] and [40]. In [15], [14] and [22], they focus on the

significant amplification in size between the stem and mature cells populations.

Works like [15], [14], [22], [40], [51] and [73] consider a discrete differentiation, where stem cells differentiate a fixed number of times to become mature cells. On the other hand, models like the one presented in [3] take a continuous approach, where cells spend a determined amount of time at stem and immature levels. The authors in [23] connect the discrete and continuous approaches to differentiation. In this aim, they derive the continuous differentiation limit of a deterministic compartmental model for cellular differentiation with discrete differentiation regulated through a feedback loop. We develop in Chapter 2 a stochastic multi-scale compartmental model and its hydrodynamic continuous limit. We introduce regulation in both the differentiation and regulation rates and extend the space of solutions for the limiting system to the set of finite non-negative measures on $[0, 1]$ that do not necessarily have a density. The relation between our work and [23] is done in the particular case where densities exist at each time for the limiting measures gathering the immature compartments. Our result proves the uniqueness of the limiting continuous system obtained in [23].

This thesis combines theoretical modeling and data-driven approaches to deepen the understanding of hematopoiesis and MPN dynamics. Chapter 2 addresses the modeling of hematopoiesis as a continuum process through the continuous limit of a stochastic compartmental model. Chapter 3 and 4 are devoted to the statistical modeling and analysis of clinical data for MPN. Chapter 3 introduces an age-structured model for the emergence and development of MPN, highlighting an age-dependent active mutation rate. Chapter 4 develops a non-linear mixed-effects framework to study the influence of additional mutations on disease progression and treatment response through longitudinal sequencing data.

1.3 Main results of the thesis

1.3.1 Results from Chapter 2

This chapter consists of the preprint *Hematopoiesis as a continuum: from stochastic compartmental model to hydrodynamic limit*, a joint work with Vincent Bansaye, Stéphane Giraudier and Sylvie Méléard [7].

Hematopoiesis starts with hematopoietic stem cells (denoted HSCs) that go through differentiation and become more and more specialized until reaching full maturity. Immature cells can also proliferate (cell division) or die, while the most mature myeloid cell do not divide and are only created through differentiation. Proliferation and differentiation of immature cells are regulated mostly through cytokines secreted by mature cells ([71]). Through biological observations, the process of hematopoiesis has been

assumed to have a discrete step-wise structure, where cells differentiate a fixed number of times to reach full maturity. New techniques of observation at single cell level suggest that hematopoiesis differentiation is rather a progressive continuous process (see [18], [85])

Inspired by the step-wise to continuous differentiation approaches to differentiation in hematopoiesis, this chapter presents a stochastic slow-fast compartmental model for hematopoiesis with regulation, and derives its hydrodynamic limit to model continuous hematopoiesis.

We characterize each cell by a quantitative trait $x \in [0, 1]$ representing its maturation level: $x = 0$ corresponds to HSCs, $x = 1$ to mature cells and for $x \in (0, 1)$ we are talking about immature cells. We consider initially a compartmental model with N types of cells between HSC and mature cells, and study the limiting behavior when $N \rightarrow \infty$. A cell of type $i \in \{1, \dots, N\}$ has a maturation level of $x = i/N$. In this framework, HSCs have a maturation level of $x = 1/N$, approaching $x = 0$ as N grows. Then, as $N \rightarrow \infty$, we obtain a continuum of types corresponding to the interval $[0, 1]$.

We define a continuous in time pure jump Markov process with N compartments integrating self-renewal and cell differentiation. The process $(X(t), t \geq 0) = ((X_1(t), \dots, X_N(t)))_{t \geq 0}$ defines $X_i(t)$ as the number of cells of type i at time t . As regulation of proliferation and differentiation often arises from molecules secreted by mature cells, we include a feedback loop depending on the size of the mature cells population. The continuous non-negative functions $(x, z) \rightarrow r(x, z)$, $(x, z) \rightarrow m(x, z)$ defined on $[0, 1] \times [0, \infty)$ represent the proliferation and division rates for a cell of maturation level x when the mature population is of size z . These positive functions are assumed to be bounded from above and m is also bounded from below, amongst other hypotheses.

The model for the N subpopulations of cells considers the following transitions:

- Each cell of type i , ($1 \leq i \leq N - 1$) divides at rate $r(i/N, X_N/N)$ creating two cells of same type,
- Each cell of type 1 becomes a cell of type 2 at rate $m(1/N, X_N/N)$,
- Each cell of type i , ($2 \leq i \leq N - 1$) becomes a cell of type $i + 1$ at rate $Nm(i/N, X_N/N)$,
- Each cell of type N dies at rate d ,

where d is a positive constant.

The parameter N not only reflects the number of compartments, but also the speed of differentiation in immature compartments. As the number of compartments grows,

the more maturation stages the cell has to pass by to become a mature cell. To keep the time scale of mature cells creation of order one, immature cells have to go faster through these maturation stages, which explains the acceleration in the differentiation rates. This implies that each immature compartment is both filled and emptied very fast. On the other hand, the populations of stem and mature cells are of the same order of magnitude. We take N as a scaling parameter that represents the large order of magnitude of the number of stem cells. Then, we proceed to re-scale the system as follows

$$X_1^N(t) = \frac{X_1(t)}{N}, \quad X_i^N(t) = X_i(t), \quad X_N^N(t) = \frac{X_N(t)}{N}.$$

Additionally, the immature compartments $2 \leq i \leq N-1$ are gathered in the stochastic measure μ_t^N defined for any $t \geq 0$ as

$$\mu_t^N = \frac{1}{N} \sum_{i=2}^{N-1} X_i^N(t) \delta_{i/N}.$$

We then work with the triplet of processes $(X_1^N(t), \mu_t^N, X_N^N(t))_{t \geq 0}$ composed of the empirical measure containing all the immature compartments, and the boundaries that correspond to the cell and mature cells population sizes. We consider the trajectorial representation of these processes using Poisson point measures, and then work with their semimartingale decomposition. After computations to write the equations in such a way that they depend on all X_i^N for $2 \leq i \leq N-1$ only through the measure μ^N , we obtain the following system for any function $f \in \mathcal{C}([0, 1])$ and for any $t \geq 0$,

$$\begin{aligned} X_1^N(t) &= X_1^N(0) + \int_0^t \left[r\left(\frac{1}{N}, X_N^N(s)\right) - m\left(\frac{1}{N}, X_N^N(s)\right) \right] X_1^N(s) ds + M_1^N(t) \\ \langle \mu_t^N, f \rangle &= \langle \mu_0^N, f \rangle + f\left(\frac{2}{N}\right) \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds \\ &\quad + \int_0^t \langle \mu_s^N, f r(\cdot, X_N^N(s)) \rangle ds + \int_0^t \langle \mu_s^N, (\Delta_{1/N} f) m(\cdot, X_N^N(s)) \rangle ds \\ &\quad - f(1) \left[\langle \mu_0^N, 1 \rangle - \langle \mu_t^N, 1 \rangle + \int_0^t \langle \mu_s^N, r(\cdot, X_N^N(s)) \rangle ds \right. \\ &\quad \left. + \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds \right] + M^{N,f}(t) \\ X_N^N(t) &= X_N^N(0) + \langle \mu_0^N, 1 \rangle - \langle \mu_t^N, 1 \rangle + \int_0^t \langle \mu_s^N, r(\cdot, X_N^N(s)) \rangle ds \\ &\quad + \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds - d \int_0^t X_N^N(s) ds + M^N(t), \end{aligned} \tag{1.3.1}$$

where $\Delta_{1/N} f$ corresponds to the discrete derivative of f defined by

$$\Delta_h f(x) = \frac{f(x+h) - f(x)}{h}.$$

We prove that the martingales M_1^N , $M^{N,f}$, M^N are square-integrable with quadratic variations of order $1/N$ and will hence go to 0 when N goes towards infinity.

We have hence set the framework to study the hydrodynamic limit as the parameter N reaches infinity. To ensure convergence, we assume boundedness of the initial condition $(X_1^N(0), \mu_0^N, X_N^N(0))$ such that

$$\sup_N \mathbb{E} \left[\left(X_1^N(0) + \langle \mu_0^N, 1 \rangle + X_N^N(0) \right)^2 \right] < \infty.$$

Let (M_F, w) denote the set of finite non-negative measures on $[0, 1]$ endowed with the weak topology. We further assume that the initial condition $(X_1^N(0), \mu_0^N, X_N^N(0))$ converges in law (and in probability) in $\mathbb{R}^+ \times (M_F, w) \times \mathbb{R}^+$ to the deterministic triplet (a_0, μ_0, z_0) , where $a_0 \in (0, \infty)$, $\mu_0 \in M_F$ and $z_0 \in [0, \infty)$.

Under these assumptions, the main result of our work states the convergence to a unique limit as stated below.

Theorem 1.3.1. *For each $T > 0$, the sequence of processes $(X_1^N, \mu^N, X_N^N)_N$ converges in probability in $\mathbb{D}([0, T], \mathbb{R}^+ \times (M_F, w) \times \mathbb{R}^+)$ to the unique deterministic triplet $(a, \mu, z) \in \mathcal{C}([0, T], \mathbb{R}^+ \times (M_F, w) \times \mathbb{R}^+)$ satisfying for any function $f \in \mathcal{C}^1([0, 1])$ and $t \in [0, T]$,*

$$\left\{ \begin{array}{l} a(t) = a_0 + \int_0^t [r(0, z(s)) - m(0, z(s))] a(s) ds; \\ \langle \mu_t, f \rangle = f(0) \int_0^t m(0, z(s)) a(s) ds + \langle \mu_0, f \rangle + \int_0^t \langle \mu_s, f r(\cdot, z(s)) \rangle ds \\ \quad + \int_0^t \langle \mu_s, f' m(\cdot, z(s)) \rangle ds; \\ \quad - f(1) \left[\langle \mu_0, 1 \rangle - \langle \mu_t, 1 \rangle + \int_0^t \langle \mu_s, r(\cdot, z(s)) \rangle ds + \int_0^t m(0, z(s)) a(s) ds \right]; \\ z(t) = z_0 + \langle \mu_0, 1 \rangle - \langle \mu_t, 1 \rangle + \int_0^t \langle \mu_s, r(\cdot, z(s)) \rangle ds + \int_0^t m(0, z(s)) a(s) ds \\ \quad - d \int_0^t z(s) ds; \\ \mu_t(\{0\}) = \mu_t(\{1\}) = 0 \quad \text{for almost every } t \in [0, T]. \end{array} \right. \quad (1.3.2)$$

The proof of this convergence result is constructed through a tightness-identification-uniqueness procedure, where the main difficulty lies in the uniqueness proof, given the non-linearity of the system through the differentiation and proliferation rates.

A first result bounds the first and second moment such that

$$\sup_N \mathbb{E} \left[\sup_{t \leq T} \left(X_1^N(t) + \langle \mu_t^N, 1 \rangle + X_N^N(t) \right)^2 \right] \leq C_T,$$

with $C_T > 0$ a positive constant depending only on T . This result ensures the non-explosion in finite time of the process.

Then, we prove that for each $T > 0$ the sequence of laws of the triplet $(X_1^N(t), \mu_t^N, X_N^N(t), t \in [0, T])_N$ is tight in $\mathcal{P}(\mathbb{D}([0, T], \mathbb{R}^+ \times (M_F, w) \times \mathbb{R}^+))$. This implies the existence of a converging subsequence. We proceed to identify this limiting process as System (1.3.2).

To overcome the non-linearity of the system, we consider for any $(t, x) \in \mathbb{R}^2$ the flow generated by the solution $M^z(\cdot, t, x)$ of the (non-autonomous) Cauchy problem defined for $s \in \mathbb{R}$ by

$$\frac{\partial M^z}{\partial s}(s, t, x) = m(M^z(s, t, x), z(s)), \quad M^z(t, t, x) = x.$$

This flow $M^z(\cdot, t, x)$ captures the dynamics of the maturation of a cell having a maturation level of x at time t .

Then, we consider the measure that combines the three elements of the triplet (a, μ, z) defined for each $t \geq 0$ by

$$\nu_t = a(t)\delta_0 + \mu_t + z(t)\delta_1.$$

We use the flow to compute the associated mild equation and find that for $\tilde{f} \in \mathcal{C}_b^1(\mathbb{R})$, $(\nu_t)_t$ is solution of

$$\begin{aligned} \langle \nu_t, \tilde{f} \rangle &= \langle \nu_0, \tilde{f}(M^z(t, 0, \cdot)) \rangle + \int_0^t \tilde{f}(M^z(t, s, 0)) r(0, z(s)) a(s) ds \\ &\quad + \int_0^t \langle \mu_s, \tilde{f}(M^z(t, s, \cdot)) r(\cdot, z(s)) \rangle ds - d \int_0^t \tilde{f}(M^z(t, s, 1)) z(s) ds. \end{aligned}$$

This mild equation allows us to prove uniqueness of System (1.3.2) and Theorem 1.3.1 is proven.

We highlight that the analysis is performed in the space of measure-valued solutions, which is more challenging than working in a functional space, as this space is larger and hence uniqueness is harder to establish.

By using the mild equation, we further prove that if the limiting initial measure μ_0 admits a density, then, this property propagates through time and the measure μ_t admits a density $u(t, \cdot)$ with respect to the Lebesgue measure. This result allows to find an associated system of partial differential equations (PDE) such that the function

(a, u, z) is weak solution of

$$\begin{aligned} \frac{d}{dt}a(t) &= [r(0, z(t)) - m(0, z(t))] a(t) \\ \partial_t u(x, t) + \partial_x [m(x, z(t)) u(x, t)] &= r(x, z(t)) u(x, t) \\ \frac{d}{dt}z(t) &= m(1, z(t)) u(1, t) - dz(t), \end{aligned} \quad (1.3.3)$$

for $t \geq 0$ and $x \in [0, 1]$, with boundary conditions

$$\begin{aligned} u(0, t) &= a(t) \\ u(1, t) &= \frac{1}{m(1, z(t))} \left[\int_0^1 \frac{d}{dt}u(x, t) dx - \int_0^1 r(x, z(t)) u(x, t) dx - m(0, z(t)) a(t) \right], \end{aligned}$$

and initial conditions $(a_0, z_0) \in (\mathbb{R}_+)^2$.

This PDE system is equivalent to the one found in [23]. As we prove in Theorem 1.3.1 uniqueness on the set of finite non-negative measures on $[0, 1]$, it follows that uniqueness holds when this measures has a density, proving uniqueness of System (1.3.3) and hence of the system obtained in [23].

We then perform numerical simulations on the stochastic initial System (1.3.1) for $N = 50, 200$ compartments and the limiting PDE System (1.3.3). The stochastic system is simulated by directly drawing event counts from Poisson processes for discrete time steps. For the PDE system, we construct a numerical scheme in which derivatives are replaced by finite backward differences, integrals by Riemann sums and the PDE is approximated using an upwind method. Our aim in these simulations is to illustrate the dynamics of the amplification through time as the maturity level increases. To do so, we started with an initial condition with only very few HSCs. With these simulations, we want to exemplify the biologically observed amplification in size that occurs as the maturity level increases. The initial condition taken consists of only a small number of stem cells. The code for these simulations is presented in Appendix A.

The plots in Figure 1.2 and Figure 1.3 illustrate the distribution through time of immature cells depending on the maturation level for both the discrete and continuous system. In particular, Figure 1.2 shows the average of 50 simulations of the discrete system (1.3.1) with $N = 50$ and $N = 200$ and Figure 1.3 also shows the dynamics of the mature cells population. In both systems, we can observe a propagation front that progressively fills all maturation stages. Furthermore, there is an amplification in the number of cells as the maturation level increases. In Figure 1.3 we confirm this amplification given that the number of mature cells is much bigger than the one of

stem cells.

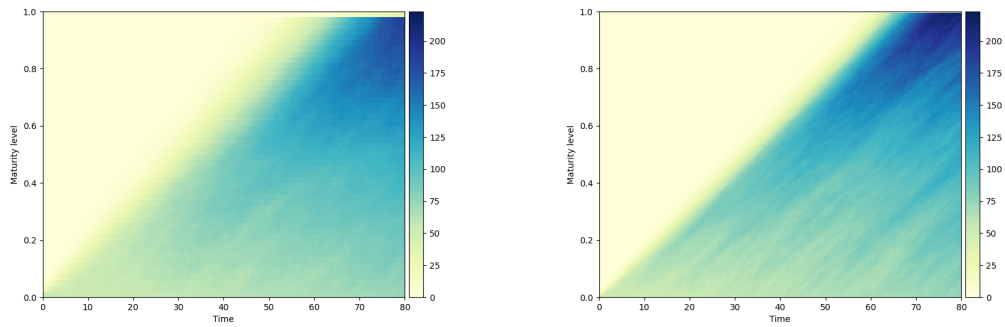


Figure 1.2: Dynamics of immature cells populations by their maturity level for System (1.3.1) for 50 (left) and 200 (right) compartments.

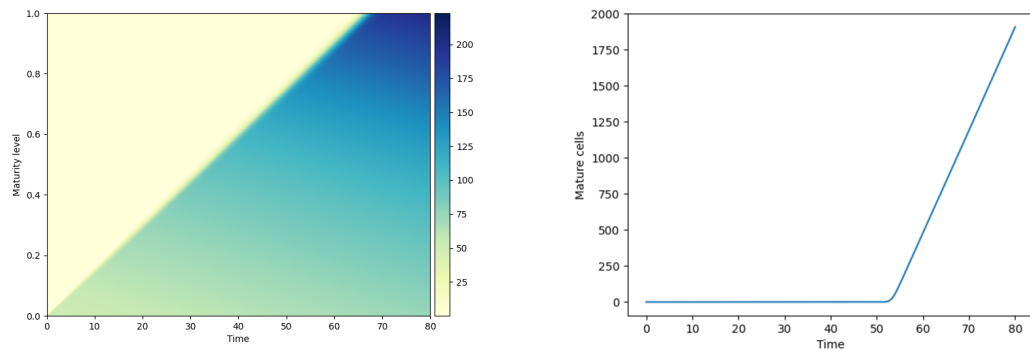


Figure 1.3: Left: dynamics of immature cells populations by their maturity level for System (1.3.3). Right: dynamics of the mature cells population size for System (1.3.3).

Perspectives

This model can be extended to include different biological considerations. Indeed, we work under the framework of unperturbed hematopoiesis (commonly known as steady-state) as opposed to stress hematopoiesis, where we can assume that cells differentiate to the immediate next maturation stage and regulation only arises from mature cells. Regardless, differentiation and regulation can behave very differently in stress hematopoiesis as regulation from immature cells also arises; plus, it has been

reported that jumps in differentiation stages occur given that ST-HSCs can directly create stress erythroid progenitors (see [29]). These jumps have also been observed in megakaryocytes, a type of mature cells, that emerge from LT-HSCs in a small number of intermediary steps. Then, we can envision to extend this model to factor these jumps where stem cells skip some maturation levels, differentiating directly to a more advanced maturation level. A different type of regulation depending on immature cells can also be considered.

We prove the convergence of a stochastic slow-fast system to a deterministic system with boundary conditions. We aim in the future to characterize the speed of convergence by bounding the pathwise deviation between the stochastic process and its deterministic limit; as well as the associated fluctuations using a tightness-identification procedure as, for example, [14].

1.3.2 Results from Chapter 3

This chapter corresponds to the preprint *Classical JAK2V617F+ Myeloproliferative Neoplasms emergence and development based on real life incidence and mathematical modeling*, a joint work with Vincent Bansaye, Evelyne Lauret, Morgane Mounier, Valérie Ugo, Sylvie Méléard and Stéphane Giraudier [25]. It is currently under revision at *Mathematical Biosciences*.

Inspired by the age structure in MPNs, this work uses age of detection data to develop an age-dependent model for the time to emergence of the first JAK2V617F mutation and the following development time for the disease until diagnosis. Constructed through an initial cohort, this model is selected amongst age-independent models as well as an alternative age-dependent models as it is the only one able to explain the data. We validate this model on a second cohort. This study allows us to better understand the dynamics of MPN emergence and development and to obtain quantitative results on these times.

We work with two cohorts of age of diagnosis for PV and ET in the presence of the JAKV617F mutation. We use an initial cohort consisting of the cancer registry of the Côte d'Or region in France from 1990 to 2020 to compare the different models. Then, to validate the selected model, we consider a second cohort from FIMBANK. We begin by adjusting both data sets to factor in that some individuals could have died before being diagnosed and are thus not present in the data sets. This adjustment can be observed in Figure 1.4.

We consider age of diagnosis as the sum of two independent time periods:

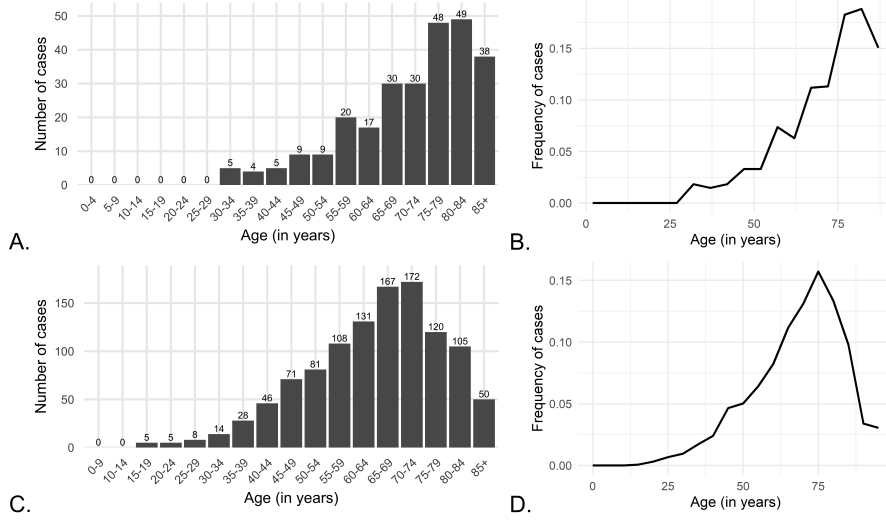


Figure 1.4: **A**: Incidence by age of JAK2V617F Classical MPN (ET and PV, excluding myelofibrosis) in the Cote d'Or Regional Registry Database. **B**: Incidence by age of JAK2V617F classical MPN (ET and PV, excluding myelofibrosis) in the Cote d'Or Regional Registry Database after adjustment excluding other causes of death before the MPN diagnosis. **C**: Incidence by age of JAK2V617F Classical MPN (ET and PV, excluding myelofibrosis) in the National BCB-FIMBANK Cohort. **D**: Incidence by age of JAK2V617F classical MPN (ET and PV, excluding myelofibrosis) in the National BCB-FIMBANK Cohort after adjustment excluding other causes of death before the MPN diagnosis.

- Emergence time, denoted T_1 : the time from embryonic development for the first JAK2V617F mutant cell to appear and enter in proliferation;
- Expansion time, denoted T_2 : the time for clonal expansion once the first cancerous cell begins to proliferate until diagnosis.

We proceed to construct an age-dependent model, where T_1 follows a Gompertz distribution. We select a Gompertz model as it allows to capture the dynamics of the data, whereas age-independent models and an alternative Weibull age-dependent model fail to give an adequate fit to the data. This is discussed in more detail later.

To define the law of the emergence time T_1 , we consider τ as the rate of mutation or activation of an existing mutation at stem cell level and assume that each activated mutant cell has a probability p to proliferate and then be detected. Under these hypotheses, T_1 is an exponential random variable with parameter $\delta = \tau p$. The parameter δ will be referred to as active mutation rate and combines mutation appearance, activation of a quiescent stem cell and success probability of expansion. We consider the survival analysis approach where the rate δ corresponds to the hazard function

defined as the instantaneous rate of active mutation (successful mutation appearance and activation) at age t , given that it did not occur before (see, for example, page 85 of [42]).

To account for the age structure of MPNs, given by a rising number of diagnoses with age that can be observed in Figure 1.4 as well as in other data sets (see [67]), we take an age-dependent active mutation rate $\delta = \delta(t)$. We consider the Gompertz model (see [42], pages 90-92) for the active mutation rate δ , which is given by

$$\delta_G(t) = A \exp(kt),$$

where $A > 0$, $k > 0$, and t is the age of an individual.

We use the survival function associated to $\delta(t)$ to obtain the density of T_1 (see page 85 of [42]) which is given for an age t by

$$f_1(t) = \delta(t) \exp\left(-\int_0^t \delta(s) ds\right) = A \exp\left(\frac{A}{k}\right) \exp(kt) \exp\left(-\frac{A}{k} \exp(kt)\right).$$

As a high number of divisions are needed to reach detection size (10^6 divisions), the variability in division time can be averaged. Hence, the expansion time T_2 is considered as similar between all individuals. We hence analyze both a lognormally distributed T_2 which gives a unimodal positive distribution and a fixed T_2 . By independence of the two variables T_1 and T_2 , the resulting distribution for the age of detection T_M is computed as the convolution of the densities f_1 and f_2 of T_1 and T_2 , respectively.

The two models with a random and fixed expansion time are fitted to the data using a generalized Expectation Maximization (EM)-algorithm, where the maximization part uses one Newton-Raphson step to obtain the new set of parameters. The code for this algorithm is presented in Appendix A. Both models seem to give a good fit to the data (see Figure 1.5). We then proceed to perform a Chi-square goodness-of-fit test on both estimations. These tests result in not rejecting both models. Given the good fit of both models, we use the Bayesian information criterion (BIC) to compare them. This criterion indicates that the model with fixed expansion time should be selected.

We further validate the model with a Gompertz active mutation rate and fixed expansion time in two steps: using the second data set; and testing the fit of age-independent models and an alternative age-dependent model. We estimate the parameters of the model on the second data set through the same EM-algorithm, and perform a Chi-square goodness-of-fit. This test results in not rejecting the model. Additionally, we study three age-independent models where the active mutation rate δ is fixed in time, these models consider both a random and fixed expansion time, and account for individual variability on the active mutation rate. After estimation of the parameters, all three models give a bad fit for the data as seen in Figure 1.6.

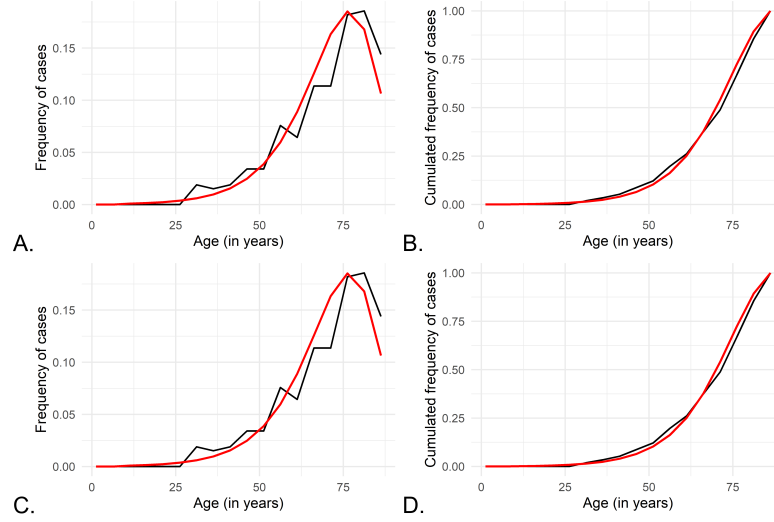


Figure 1.5: **A**: Frequency of cases for the data (in black) and probability density function of the estimation (in red) for the model with fixed expansion time. **B**: Cumulated frequencies of cases for the data (black) and cumulative distribution of the estimation (in red) for the model with fixed expansion time. **C**: Frequency of cases for the data (in black) and probability density function of the estimation (in red) for the model with lognormal MPN growing time. **D**: Cumulated frequencies of cases for the data (black) and cumulative distribution of the estimation (in red) for the model with lognormal MPN growing time.

leading to their rejection. We consider an alternative age structure in the emergence time through a Weibull active mutation rate given by $\delta_W(t) = \frac{\gamma}{\lambda} \left(\frac{t}{\lambda}\right)^{\gamma-1}$ where $\gamma > 0$, $\lambda > 0$, and the broader model with lognormal T_2 . This model is also rejected.

The rejection of all three age-independent mutation rate models imply that the hypotheses taken are not enough to properly explain the data. The good fit of the Gompertz model with an exponential increase in both data sets and the rejection of the Weibull model with a polynomial increase indicate an important age structure in the emergence of MPN.

This age-dependent model with a fixed expansion time further allows us to obtain quantitative results on the diseases by applying the model to the data. We obtain a mean emergence time of 63 years with a standard deviation of 13 years and an expansion time of 8.8, giving a mean detection age of 71.8 years. Additionally, we compare PV and ET separately and find a difference of 1.4 years in the emergence time between the two diseases. This result can be explained by the time taken for the needed mitotic recombination to happen in PV, as most cells present the mutation on

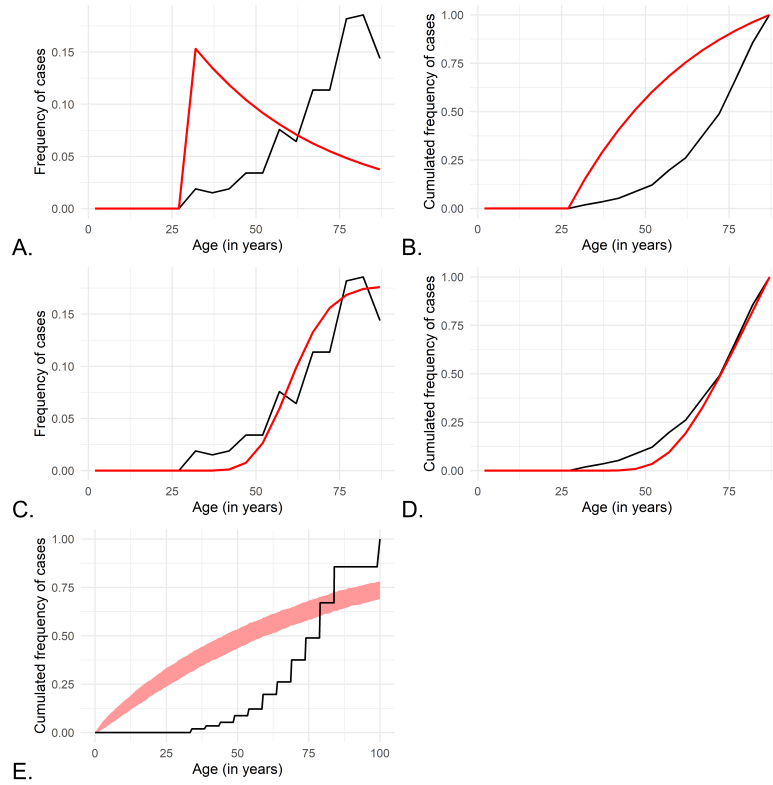


Figure 1.6: **A:** Frequency of cases for the data (in black) and probability density function of the estimation (in red) for the age-independent model with fixed expansion time. **B:** Cumulated frequencies of cases for the data from the (black) and cumulative distribution of the estimation (in red) for the age-independent model with fixed expansion time. **C:** Frequency of cases for the data (in black) and probability density function of the estimation (in red) for the age-independent model with lognormal MPN growing time. **D:** Cumulated frequencies of cases for the data (in black) and cumulative density of the estimation (in red) for the age-independent model with lognormal MPN growing time. **E:** Cumulated frequencies of cases for the data (in black) and 90% confidence interval of the estimation (in red) for the age-independent model with individual variability.

both alleles, which differs from ET where most cells present the mutation in only one allele.

1.3.3 Results from Chapter 4

This work in progress emerges from a tight collaboration with the medical doctors from Hôpital Saint-Louis Stéphane Giraudier and Nabih Maslah. Using longitudinal sequencing data collected from their patients, this analysis aims to gain valuable information for treatment selection. This work is done in collaboration with Charlotte Dion-Blanc.

MPNs originate from a driver mutation that induces an accelerated proliferation leading to an excess of a type of mature blood cells. This work focuses on the JAK2V617F driver mutation, the most common in MPNs. Cells carrying JAK2V617F can also carry additional mutations that appear before or after the driver mutation, and cause changes in the proliferation rate or the sensitivity of the cell to certain treatments. This study focuses on identifying the effect that different additional mutations have on the growth and treatment response in MPN using clinical sequencing data from diagnosed patients. This information aims to support the treatment selection depending on the mutations carried by a patient.

We study exclusively the three main cytoreductive treatments: Interferon- α (IFN), Hydroxyurea (HU) and Ruxolitinib (Ruxo). Given their different mechanisms, we assume that if more than one treatment is administered at the same time, the resulting effect is simply the sum of the effects of all undergoing treatments.

The data used in this study consists of bulk sequencing analysis of blood samples of $N = 244$ diagnosed JAK2V617F MPN patients with at least four samples taken at different times. At each sample, we have the following informations:

- Age and sex of the patient;
- Time elapsed since the first sample;
- Diagnosis and undergoing treatment(s) at the time of the sample;
- The percentage of alleles carrying the JAK2V617F mutation;
- For 53 additional mutations: the percentage of alleles carrying each mutation.

A preliminary work consisted on cleaning, filtering and formatting the data. Data is available here: <https://github.com/ana-fernandezbaranda/additional-mutations>.

We define a mixed-effect model for the dynamics of the percentage of mutated JAK2V617F alleles integrating natural growth, treatment effect and the influence of additional mutations on both growth and response to treatment. The software Monolix is used to perform the parameter estimation and the statistical analysis of this model. The estimation is based on the SAEM algorithm.

We construct an initial model for the dynamics of the percentage of alleles with the JAK2V617F mutation. This model combines natural growth of the population through a generalized logistic function, and the growth reduction of each treatment depending on their mode of action. Let $P(t)$ be the percentage of JAK2V617F mutated alleles at time t which is measured in days. Then, the model for the dynamics of $P(t)$ is given by

$$\begin{aligned} \frac{dP}{dt}(t) = & C P(t) \left(1 - \left(\frac{P(t)}{100} \right)^\gamma \right) [1 - HU(t) K_{HU} - IFN(t - 182) K_{IFN}] \\ & - Ruxo(t) K_{Ruxo} P(t), \end{aligned}$$

where $C, \gamma > 0$, $K_l > 0$ for $l \in \{IFN, HU, Ruxo\}$. We take the generalized logistic function for the growth of the population as it groups a variety of models such as the logistic if $\gamma = 1$ or the Gompertz when γ approaches 0 and further allows to bound $P(t)$ which cannot exceed 100.

To take into account inter-individual variability arising from individual characteristics that can affect the speed of growth of proliferation and from the treatment dose and frequency, we consider the parameters C , γ , K_{IFN} , K_{HU} , K_{Ruxo} as random effects. As they are all strictly positive, we take them as samples from lognormal distributions, each with its own parameters.

We analyze the addition of the two available covariates, age and sex, to the growth parameters C and γ using the *Stochastic Approximation for Model Building Algorithm* (SAMBA) tool implemented in Monolix. Presented in [61], this powerful tool allows to select the best covariate model without having to fit the complete model to the data. The model selection is based on minimization of a penalized likelihood. We find that the best model to fit the data includes an age-dependency on the growth parameters. This indicates no significant dependence of these parameters on sex. We know that $\log(C^n)$ and $\log(\gamma^n)$ for $1 \leq n \leq N$ are normally distributed as C and γ are lognormal

random variables. The age-dependency is introduced as follows

$$\begin{aligned}\log(C^n) &= \log(C_{pop}) + \beta_C \log\left(\frac{age_n}{\widehat{Age}}\right) + \eta_C^n \\ \log(\gamma^n) &= \log(\gamma_{pop}) + \beta_\gamma \log\left(\frac{age_n}{\widehat{Age}}\right) + \eta_\gamma^n,\end{aligned}$$

where $\eta_C^n \sim \mathcal{N}(0, \omega_C^2)$, $\eta_\gamma^n \sim \mathcal{N}(0, \omega_\gamma^2)$; C_{pop} and γ_{pop} are the medians of the distribution of C^n and γ^n ; β_C and β_γ are parameters to be estimated. In this construction, the population parameters C and γ are defined for a typical individual of the population of age $\widehat{Age}$, the weighted age average of the population.

As a summary, this base model integrates:

- a Generalized logistic growth function with age-dependent parameters C and γ ,
- the effect of the three administered treatments: Interferon- α , Hydroxyurea and Ruxolitinib.

The parameters of the model without additional mutations are estimated using the data with the software Monolix. We estimate the parameters using a randomly selected sample consisting of 80% of the individuals, obtaining a good fit with the data and a small number of outliers (7.56%) (see Figure 1.7). Then, we use this estimation to predict the individual parameters of the remaining 20% of individuals, which also succeeds to give a good estimation with only a 7.41% of outliers. This cross-validation verifies the choice of the natural growth model.

Having validated the base model, we proceed to adapt it to factor in the effects of additional mutations. In this aim, we define $\mathcal{M}$ as the set of all additional mutations and $P^i(t)$ for $i \in \mathcal{M}$ as the subpopulation of all JAK2V617F cells also carrying mutation i (and possibly other additional mutations). Four parameters are associated to each additional mutation $i \in \mathcal{M}$ which gives us information on its impact:

- $C_i \in \mathbb{R}$: growth advantage or disadvantage of mutation i
- $K_l^i \in \mathbb{R}$ for any treatment $l \in \{HU, IFN, Ruxo\}$: the effect of mutation i on treatment l .

If $C_i > 0$, it means that the speed of growth of a JAK2V617F population carrying exclusively mutation i (and JAK2V617F) is faster than the one of a JAK2V617F population without any mutations. If $C_i < 0$ it means that the speed of growth of a JAK2V617F population carrying exclusively mutation i is slower than the one of a JAK2V617F population without any mutations. If $K_l^i > 0$ for some $l \in \{HU, IFN, Ruxo\}$, then

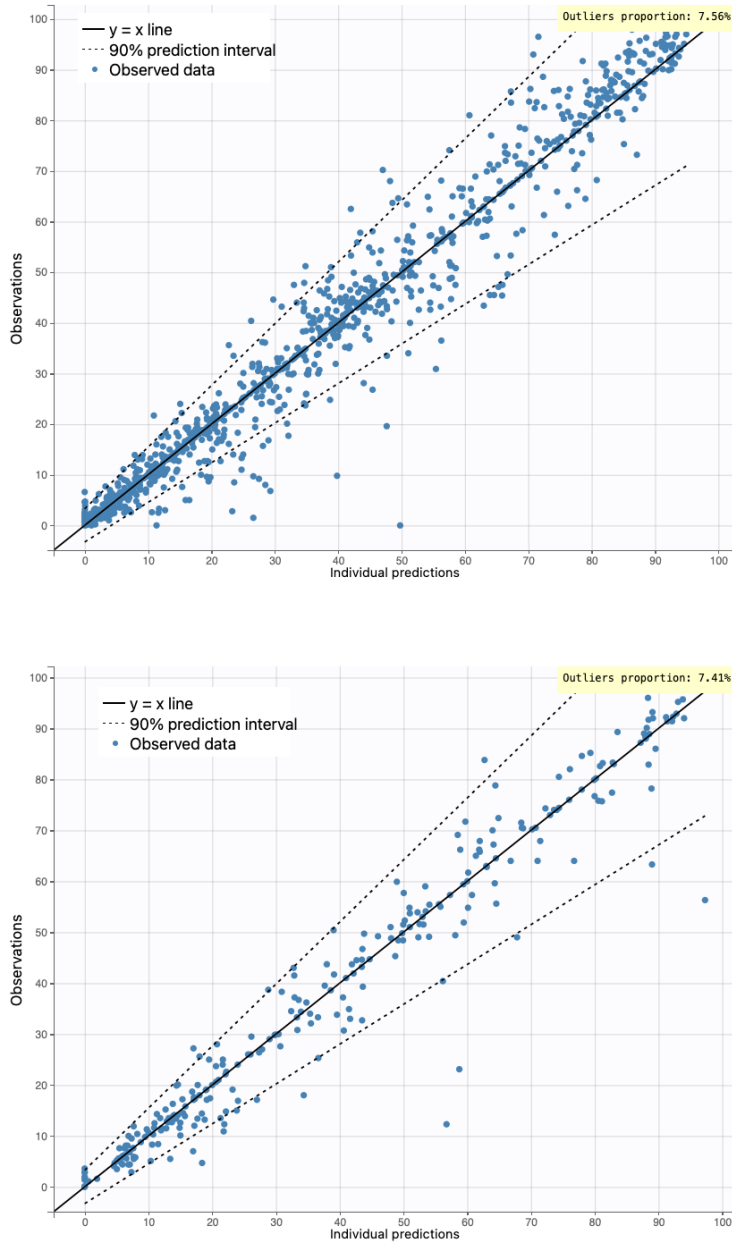


Figure 1.7: Observations vs. predictions in the model without additional mutations with 90% confidence interval. The estimation was done with 80% of patients (top) and applied to remaining 20% of patients (bottom).

a JAK2V617F cell carrying exclusively mutation i (and JAK2V617F) has a better response to treatment l than a JAK2V617F cell without any additional mutation, while $K_l^i < 0$ it implies that mutation i makes cells more resistant to treatment l . Then, the parameters K_l^i , $l \in \{HU, IFN, Ruxo\}$ give important insight that can be useful on the choice of treatment.

We assume that the effects of the additional mutations are independent from each other. Then, if a cell has the two mutations $i, j \in \mathcal{M}$, the effect on growth is $C_i + C_j$ and the effect on treatment $l \in \{HU, IFN, Ruxo\}$ is $K_l^i + K_l^j$. We also assume that new mutations arrive in the most mutated cells which implies that the number of cells carrying (non-exclusively) mutations $i, j \in \mathcal{M}$ is equal to $\min(P^i(t), P^j(t))$. After computations, the model becomes

$$\begin{aligned} \frac{dP}{dt}(t) = C g(P(t)) & \left[\left(P(t) + \sum_{i \in \mathcal{M}} P^i(t) C_i \right) [1 - HU(t) K_{HU} - IFN(t) K_{IFN}] \right. \\ & - \sum_{i \in \mathcal{M}} P^i(t) (HU(t) K_{HU}^i + IFN(t) K_{IFN}^i) \\ & \left. - \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{M}} C_i (HU(t) K_{HU}^j + IFN(t) K_{IFN}^j) \min(P^i(t), P^j(t)) \right] \\ & - Ruxo(t) \left[P(t) K_{Ruxo} + \sum_{i \in \mathcal{M}} P^i(t) K_{Ruxo}^i \right]. \end{aligned} \quad (1.3.4)$$

We consider the four parameters associated to each additional mutation $i \in \mathcal{M}$, C_i , K_{IFN}^i , K_{HU}^i , K_{Ruxo}^i as fixed. A summary of all parameters in the model is presented in Table 1.1.

Parameter	Domain	Distribution	Description
C	$(0, \infty)$	lognormal	Related to the natural growth
β_C	$(-\infty, \infty)$		Age effect on parameter C
γ	$(0, \infty)$	lognormal	Related to the natural growth
β_γ	$(-\infty, \infty)$		Age effect on parameter γ
K_l	$(0, \infty)$	lognormal	Response to treatment $l \in \{HU, IFN, Ruxo\}$
C_i	$(-\infty, \infty)$	Fixed	Effect on growth of mutation $i \in \mathcal{M}$
K_l^i	$(-\infty, \infty)$	Fixed	Effect on response to treatment $l \in \{HU, IFN, Ruxo\}$ of mutation $i \in \mathcal{M}$

Table 1.1: Parameters of the model.

We study the effect on growth and treatments IFN and HU of the three most prevalent mutations in the data: *TET2* (48.8% of patients), *DNMT3A* (39.3%) and

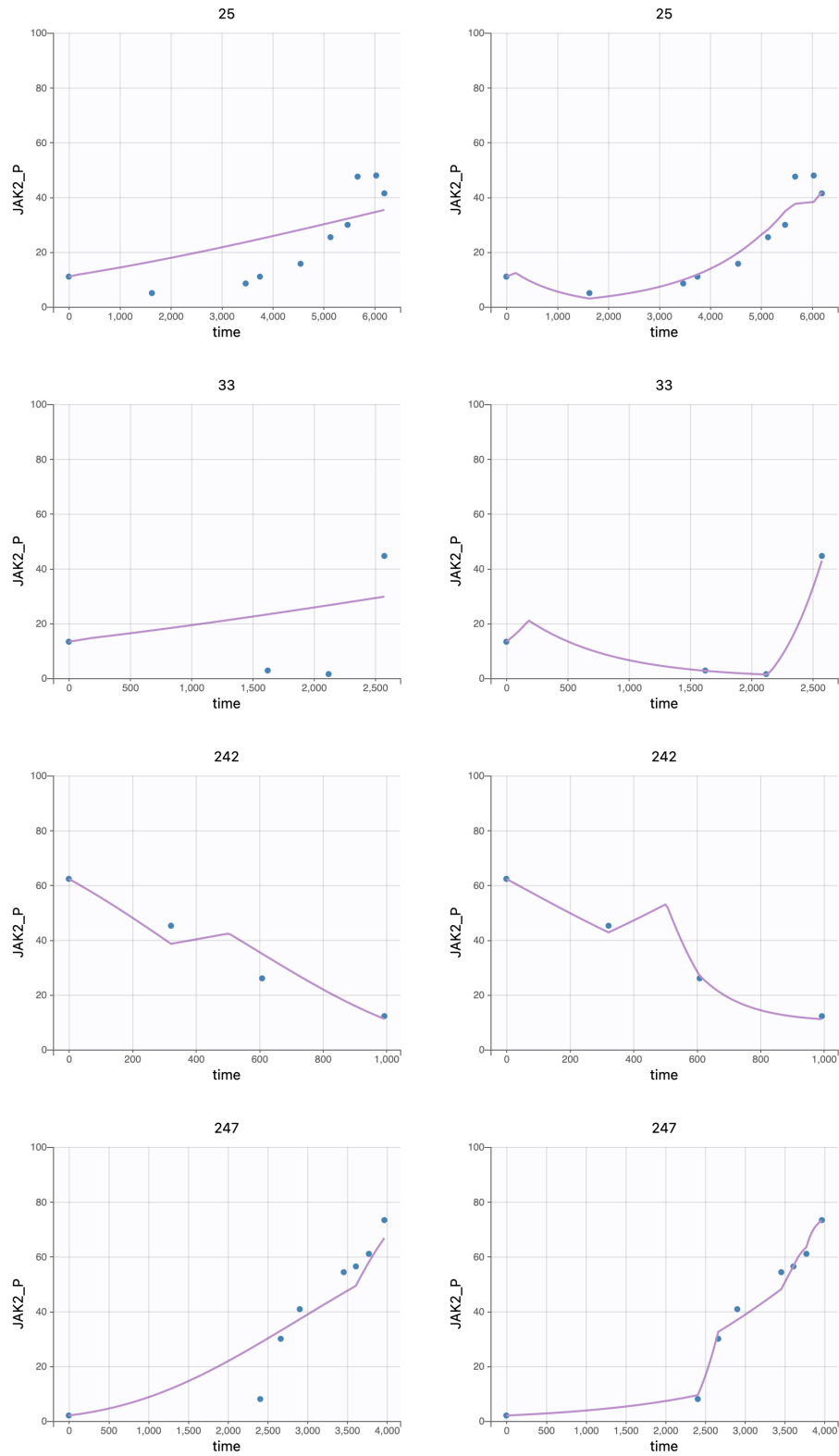
ASXL1 exon 12 (29.9%). This allows to compare the two treatments when at least one of these mutations is present. The effect of the mutations in treatment Ruxo is not included as it was not possible to obtain good estimates with finite confidence intervals for the related parameters.

We work with the set of additional mutations $\mathcal{M}_1 = \{TET2, DNMT3A, ASXL1\}$. We include the effect of these three mutations on growth and treatment response in IFN and HU using (1.3.4). Then, the model is given by

$$\begin{aligned} \frac{dP}{dt}(t) = C g(P(t)) & \left[\left(P(t) + \sum_{i \in \mathcal{M}_1} P^i(t) C_i \right) [1 - HU(t) K_{HU} - IFN(t) K_{IFN}] \right. \\ & - \sum_{i \in \mathcal{M}_1} P^i(t) (HU(t) K_{HU}^i + IFN(t) K_{IFN}^i) \\ & \left. - \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{M}} C_i (HU(t) K_{HU}^j + IFN(t) K_{IFN}^j) P^{i,j}(t) \right] \\ & - Ruxo(t) P(t) K_{Ruxo}. \end{aligned}$$

We perform the estimation using the same data set with 80% of the individuals. In Figure 1.8, we can see an important gain in the prediction for certain individuals carrying at least one of these mutations when adding the additional mutations compared to the model without additional mutations. Indeed, we observe that this new model captures fluctuations in the observations that were previously badly predicted. Using this estimation in the remaining 20%, we also find an improvement in the predictions for some patients carrying at least one of these mutations, as seen in Figure 1.9.

The values of the parameters obtained in this estimation, and their respective 95% confidence interval is presented in Table 1.2.



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Figure 1.8: Observation points and estimation curve of four patients for the model without additional mutations (left) and for the model with three additional mutations (right).

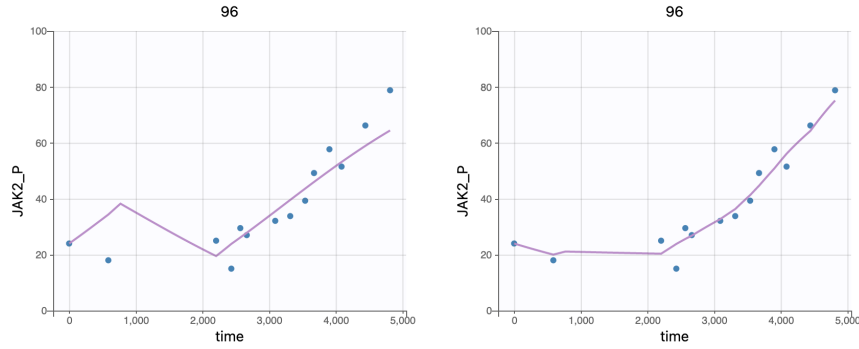


Figure 1.9: Observation points and estimation curve of a patient without additional mutations (left) and with three additional mutations (right).

Parameter	Value	Confidence interval
C_{TET2}	12.76	[8.8 , 16.73]
K_{IFN}^{TET2}	-2.08	[-2.28, -1.87]
K_{HU}^{TET2}	-4.45	[-4.73, -4.17]
C_{DNMT3A}	-0.0041	[-0.016, 0.0077]
K_{IFN}^{DNMT3A}	-33.72	[-42.29, -25.16]
K_{HU}^{DNMT3A}	-6.41	[-8.89, -3.92]
C_{ASXL1}	5.93	[5.18, 6.68]
K_{IFN}^{ASXL1}	-37.62	[-43.8, -31.44]
K_{HU}^{ASXL1}	-2.32	[-2.9, -1.73]

Table 1.2: Estimation of the parameters and 95% confidence interval.

These results can be interpreted as:

- $C_{TET2}, C_{ASXL1} > 0$: mutations $TET2$ and $ASXL1$ promote faster growth of the JAK2V617F population, with the effect of $TET2$ being much bigger than the one of $ASXL1$.
- $C_{DNMT3A} \approx 0$: mutation $DNMT3A$ does not have a significant effect on the growth of the JAK2V617F mutation.

- $K_l^i < 0$ for $l \in \{IFN, HU\}$ and $i \in \{TET2, ASXL1, DNMT3A\}$: all three mutations induce a worse treatment response for IFN and HU.
- $K_{HU}^{TET2} < K_{IFN}^{TET2} < 0$: mutation $TET2$ has a more significant impact on the response to HU than the one to IFN.
- $K_{IFN}^i < K_{HU}^i < 0$ for $i \in \{ASXL1, DNMT3A\}$ mutations $DNMT3A$ and $ASXL1$ have a much more significant impact on the response to IFN than the one to HU.

These results suggest that under the presence of $TET2$ or $ASXL1$ one should expect a faster disease evolution, particularly for $TET2$. When the $TET2$ mutation is present alone, treatment IFN should be more effective than HU. On the contrary, with the existence of $DNMT3A$ or $ASXL1$, one can expect a better response to HU than IFN. In the case where both $TET2$ and $DNMT3A$ or $TET2$ and $ASXL1$ are present, the best treatment will depend on the percentage of alleles with each mutation. Though the effect of $TET2$ on HU is much smaller than the ones of $DNMT3A$ and $ASXL1$ on IFN, indicating that HU would probably give the best results, regardless of the proportion of alleles carrying each mutation.

Perspectives

This work sets the framework to identify the impact of additional mutations, giving important insight on treatment selection. We characterize the effect of the three most prevalent additional mutations on treatments IFN and HU. More analysis remains to be done. The effect of additional mutations on treatment Ruxo could not be included, as it was not possible to find good estimates. We aim to solve this problem by discussing whether to modify the model to include the mechanisms of this treatment differently, or we could think of using a simplification of the model to facilitate the estimation. We aim to continue including more additional mutations in our analysis. Furthermore, in this work we characterize the effects of each individual mutation, we intend to extend this information to mutational profiles, where an individual may have an accumulation of additional mutations, each in a different proportion.

1.4 Résultats principaux de la thèse

1.4.1 Résultats du Chapitre 2

Ce chapitre correspond au preprint *Hematopoiesis as a continuum: from stochastic compartmental model to hydrodynamic limit*, en collaboration avec Vincent Bansaye, Stéphane Giraudier et Sylvie Méléard [\[7\]](#).

L'hématopoïèse commence par des cellules souches hématopoïétiques (CSH) qui se différencient en devenant de plus en plus spécialisées jusqu'à atteindre leur pleine maturité. Les cellules immatures peuvent aussi proliférer (division cellulaire) ou mourir, tandis que les cellules myéloïdes matures ne se divisent pas et sont créées uniquement par différenciation. La prolifération et la différenciation des cellules immatures sont régulées surtout par des cytokines sécrétées par les cellules matures ([71]). À partir des observations biologiques, on a supposé que le processus de l'hématopoïèse avait une structure discrète, où les cellules se différencient un nombre fixé de fois pour arriver à la maturité. De nouvelles techniques d'observation à l'échelle *single cell* suggèrent que la différenciation dans l'hématopoïèse est plutôt un processus continu progressif.

Inspiré par l'hypothèse biologique d'une différenciation continue dans l'hématopoïèse, ce chapitre présente un modèle stochastique à compartiments lent-rapide pour l'hématopoïèse avec régulation, et obtient sa limite hydrodynamique pour modéliser ce phénomène comme un continuum.

Nous caractérisons chaque cellule par un trait quantitatif $x \in [0, 1]$ qui représente son niveau de maturation : $x = 0$ correspond aux CSHs, $x = 1$ aux cellules matures, et tout $x \in (0, 1)$ aux cellules immatures. Nous considérons initialement un modèle à compartiments avec N types de cellules entre les CSH et les cellules matures, et étudions le comportement limite quand $N \rightarrow \infty$. Une cellule de type $i \in \{1, \dots, N\}$ a un niveau de maturation de $x = i/N$. Dans ce contexte, les CSHs ont un niveau de maturation de $x = 1/N$, se rapprochant à $x = 0$ à mesure que N augmente. Quand $N \rightarrow \infty$, nous obtenons alors un continuum de types correspondant à l'intervalle $[0, 1]$.

Nous commençons par construire un processus de Markov de sauts purs, continu en temps, avec N compartiments intégrant l'auto-renouvellement et la différenciation cellulaire. Le processus $(X(t), t \geq 0) = ((X_1(t), \dots, X_N(t)))_{t \geq 0}$ définit $X_i(t)$ comme le nombre de cellules de type i au temps t . La régulation de la prolifération et de la différenciation est intégrée à travers une boucle de rétroaction dépendant de la taille de la population de cellules matures. Les fonctions continues non négatives $(x, z) \rightarrow r(x, z)$ et $(x, z) \rightarrow m(x, z)$, définies sur $[0, 1] \times [0, \infty)$, représentent les taux de prolifération et de différenciation pour une cellule avec un niveau de maturation x lorsque la population de cellules matures est de taille z . Ces fonctions positives sont supposées bornées supérieurement, et m est également bornée inférieurement, parmi d'autres hypothèses.

Le modèle pour les N sous-populations de cellules considère les transitions suivantes :

- Chaque cellule de type i , ($1 \leq i \leq N - 1$) se divise à taux $r(i/N, X_N/N)$, en créant deux cellules de même type,

- Chaque cellule de type 1 devient une cellule de type 2 à taux $m(1/N, X_N/N)$,
- Chaque cellule de type i , ($2 \leq i \leq N - 1$) devient une cellule de type $i + 1$ à taux $Nm(i/N, X_N/N)$,
- Chaque cellule de type N meurt à taux d ,

où d est une constante positive.

Le paramètre N reflète non seulement le nombre de compartiments, mais aussi la vitesse de différenciation dans les compartiments immatures. Plus le nombre de compartiments augmente, plus nombreux sont les stades de maturation que la cellule doit traverser pour devenir mature. Pour maintenir l'échelle de temps de création des cellules matures d'ordre un, les cellules immatures doivent traverser ces stades de maturation plus rapidement, ce qui explique l'accélération des taux de différenciation. Cela implique que chaque compartiment immature est à la fois rempli et vidé très vite. D'autre part, les populations de cellules souches et de cellules matures sont du même ordre de grandeur. Nous prenons N comme un paramètre d'échelle représentant l'ordre de grandeur élevé du nombre de cellules souches. Nous procédons à la mise à l'échelle du système de la façon suivante :

$$X_1^N(t) = \frac{X_1(t)}{N}, \quad X_i^N(t) = X_i(t), \quad X_N^N(t) = \frac{X_N(t)}{N}.$$

De plus, les compartiments immatures $2 \leq i \leq N - 1$ sont regroupés dans la mesure stochastique μ_t^N définie pour tout $t \geq 0$ comme

$$\mu_t^N = \frac{1}{N} \sum_{i=2}^{N-1} X_i^N(t) \delta_{i/N}.$$

Nous travaillons alors avec le triplet de processus $(X_1^N(t), \mu_t^N, X_N^N(t))_{t \geq 0}$ composé de la mesure empirique qui contient tous les compartiments immatures, et les bords qui correspondent aux tailles des populations de cellules souches et matures. Nous considérons la représentation trajectorielle de ces processus à travers des mesures ponctuelles de Poisson, et travaillons avec leur décomposition en semi-martingales. Après des calculs permettant d'écrire les équations de telle sorte qu'elles ne dépendent de tous les X_N^i pour $2 \leq i \leq N - 1$ qu'à travers la mesure μ^N , nous obtenons le système suivant

pour toute fonction $f \in \mathcal{C}([0, 1])$ et pour tout $t \geq 0$,

$$\begin{aligned}
 X_1^N(t) &= X_1^N(0) + \int_0^t \left[r\left(\frac{1}{N}, X_N^N(s)\right) - m\left(\frac{1}{N}, X_N^N(s)\right) \right] X_1^N(s) ds + M_1^N(t) \\
 \langle \mu_t^N, f \rangle &= \langle \mu_0^N, f \rangle + f\left(\frac{2}{N}\right) \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds \\
 &\quad + \int_0^t \langle \mu_s^N, f r(\cdot, X_N^N(s)) \rangle ds + \int_0^t \langle \mu_s^N, (\Delta_{1/N} f) m(\cdot, X_N^N(s)) \rangle ds \\
 &\quad - f(1) \left[\langle \mu_0^N, 1 \rangle - \langle \mu_t^N, 1 \rangle + \int_0^t \langle \mu_s^N, r(\cdot, X_N^N(s)) \rangle ds \right. \\
 &\quad \left. + \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds \right] + M^{N,f}(t) \tag{1.4.1} \\
 X_N^N(t) &= X_N^N(0) + \langle \mu_0^N, 1 \rangle - \langle \mu_t^N, 1 \rangle + \int_0^t \langle \mu_s^N, r(\cdot, X_N^N(s)) \rangle ds \\
 &\quad + \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds - d \int_0^t X_N^N(s) ds + M^N(t),
 \end{aligned}$$

où $\Delta_{1/N} f$ correspond à la dérivée discrète de f définie par

$$\Delta_h f(x) = \frac{f(x+h) - f(x)}{h}.$$

Nous prouvons que les martingales M_1^N , $M^{N,f}$, M^N sont de carré intégrable, que leurs variations quadratiques sont d'ordre $1/N$ et vont donc converger vers 0 quand N tend vers l'infini.

Nous avons alors établi le cadre permettant d'étudier la limite hydrodynamique lorsque le paramètre N tend vers l'infini. Afin de garantir la convergence, nous supposons que la condition initiale $(X_1^N(0), \mu_0^N, X_N^N(0))$ est bornée, de telle sorte que

$$\sup_N \mathbb{E} \left[\left(X_1^N(0) + \langle \mu_0^N, 1 \rangle + X_N^N(0) \right)^2 \right] < \infty.$$

Nous supposons en plus que la condition initiale $(X_1^N(0), \mu_0^N, X_N^N(0))$ converge en loi (et en probabilité) dans $\mathbb{R}^+ \times (M_F, w) \times \mathbb{R}^+$ vers le triplet déterministe (a_0, μ_0, z_0) , où $a_0 \in (0, \infty)$, $\mu_0 \in M_F$ et $z_0 \in [0, \infty)$.

Sous ces hypothèses, le résultat principal de notre travail établit la convergence vers une unique limite, comme énoncé ci-dessous.

Theorem 1.4.1. *Pour chaque $T > 0$, la séquence des processus $(X_1^N, \mu^N, X_N^N)_N$ converge en probabilité dans $\mathbb{D}([0, T], \mathbb{R}^+ \times (M_F, w) \times \mathbb{R}^+)$ vers l'unique triplet déterministe*

$(a, \mu, z) \in \mathcal{C}([0, T], \mathbb{R}^+ \times (M_F, w) \times \mathbb{R}^+)$ vérifiant pour toute fonction $f \in \mathcal{C}^1([0, 1])$ and $t \in [0, T]$,

$$\left\{ \begin{array}{l} a(t) = a_0 + \int_0^t [r(0, z(s)) - m(0, z(s))] a(s) ds ; \\ \langle \mu_t, f \rangle = f(0) \int_0^t m(0, z(s)) a(s) ds + \langle \mu_0, f \rangle + \int_0^t \langle \mu_s, f r(\cdot, z(s)) \rangle ds \\ \quad + \int_0^t \langle \mu_s, f' m(\cdot, z(s)) \rangle ds ; \\ \quad - f(1) [\langle \mu_0, 1 \rangle - \langle \mu_t, 1 \rangle + \int_0^t \langle \mu_s, r(\cdot, z(s)) \rangle ds + \int_0^t m(0, z(s)) a(s) ds] ; \\ z(t) = z_0 + \langle \mu_0, 1 \rangle - \langle \mu_t, 1 \rangle + \int_0^t \langle \mu_s, r(\cdot, z(s)) \rangle ds + \int_0^t m(0, z(s)) a(s) ds \\ \quad - d \int_0^t z(s) ds ; \\ \mu_t(\{0\}) = \mu_t(\{1\}) = 0 \quad \text{for almost every } t \in [0, T]. \end{array} \right. \quad (1.4.2)$$

La preuve de ce résultat de convergence est construite par une procédure de tension-identification-unicité, où la difficulté principale réside dans la preuve d'unicité en raison de la non-linéarité du système induite par les taux de différenciation et de prolifération.

Un premier résultat borne le deuxième moment de façon que

$$\sup_N \mathbb{E} \left[\sup_{t \leq T} \left(X_1^N(t) + \langle \mu_t^N, 1 \rangle + X_N^N(t) \right)^2 \right] \leq C_T,$$

avec $C_T > 0$ une constante positive qui ne dépend que de T . Ce résultat garantit la non-explosion du processus en temps fini, et permet de démontrer la tension.

Nous prouvons ensuite que pour chaque $T > 0$ la suite des lois du triplet $(X_1^N(t), \mu_t^N, X_N^N(t), t \in [0, T])_N$ est tendue dans $\mathcal{P}(\mathbb{D}([0, T], \mathbb{R}^+) \times \mathbb{D}([0, T], (M_F, w)) \times \mathbb{D}([0, T], \mathbb{R}^+))$. Ce résultat implique l'existence de valeurs d'adhérence. Nous identifions ces valeurs comme le Système (1.4.2).

Afin de surmonter la non-linéarité du système, nous introduisons pour tout $(t, x) \in \mathbb{R}^2$ le flot issu de la solution $M^z(\cdot, t, x)$ du problème de Cauchy, défini pour $s \in \mathbb{R}$ par

$$\frac{\partial M^z}{\partial s}(s, t, x) = m(M^z(s, t, x), z(s)), \quad M^z(t, t, x) = x.$$

Ce flot $M^z(\cdot, t, x)$ capture la dynamique de la maturation d'une cellule ayant un niveau de maturation x au temps t .

Nous considérons la mesure qui combine les trois éléments du triplet (a, μ, z) définie pour chaque $t \geq 0$ par

$$\nu_t = a(t)\delta_0 + \mu_t + z(t)\delta_1.$$

Nous utilisons le flot pour calculer l'équation *mild*, nous trouvons pour $\tilde{f} \in \mathcal{C}_b^1(\mathbb{R})$, que $(\nu_t)_t$ est solution de

$$\begin{aligned} \langle \nu_t, \tilde{f} \rangle &= \langle \nu_0, \tilde{f}(M^z(t, 0, \cdot)) \rangle + \int_0^t \tilde{f}(M^z(t, s, 0)) r(0, z(s)) a(s) ds \\ &\quad + \int_0^t \langle \mu_s, \tilde{f}(M^z(t, s, \cdot)) r(\cdot, z(s)) \rangle ds - d \int_0^t \tilde{f}(M^z(t, s, 1)) z(s) ds. \end{aligned}$$

Cette équation *mild* nous permet de prouver l'unicité du Système (1.4.2) et le Théorème 1.4.1 est prouvé.

À travers l'équation *mild*, nous démontrons de plus que si la mesure limite initiale μ_0 admet une densité, alors cette propriété se propage dans le temps et la mesure μ_t admet une densité $u(t, \cdot)$ par rapport à la mesure de Lebesgue. Ce résultat permet d'établir un système d'équations aux dérivées partielles (EDP) associé tel que le triplet (a, u, z) est solution faible de

$$\begin{aligned} \frac{d}{dt}a(t) &= [r(0, z(t)) - m(0, z(t))] a(t) \\ \partial_t u(x, t) + \partial_x [m(x, z(t)) u(x, t)] &= r(x, z(t)) u(x, t) \\ \frac{d}{dt}z(t) &= m(1, z(t)) u(1, t) - dz(t), \end{aligned} \quad (1.4.3)$$

pour $t \geq 0$ et $x \in [0, 1]$, avec les conditions de bord

$$\begin{aligned} u(0, t) &= a(t) \\ u(1, t) &= \frac{1}{m(1, z(t))} \left[\int_0^1 \frac{d}{dt}u(x, t) dx - \int_0^1 r(x, z(t)) u(x, t) dx - m(0, z(t)) a(t) \right], \end{aligned}$$

et les conditions initiales $(a_0, z_0) \in (\mathbb{R}_+)^2$.

Nous effectuons ensuite des simulations sur le modèle stochastique initial (1.4.1) avec $N = 50, 200$ compartiments et sur le Système (1.4.3) EDP limite. Le système stochastique est simulé par une procédure d'acceptation-rejet où le temps est discrétisé. Pour le système d'EDP, nous construisons un schéma numérique où les dérivées sont remplacées par des différences finies rétrogrades, les intégrales par des sommes de Riemann, et l'EDP est approchée par une méthode *upwind*. Le code utilisé pour ces simulations est disponible à l'Appendix A. Les simulations visent à illustrer l'amplification observée de la taille des populations à mesure que le niveau de maturité croît. La condition initiale choisie consiste uniquement en un petit nombre de cellules souches.

Les graphiques présentés dans la Figure 1.2 et la Figure 1.3 illustrent la distribution au cours du temps des cellules immatures en fonction du niveau de maturation pour le système discret et continu. En particulier, la Figure 1.3 montre la moyenne de 50 simulations du système discret (1.4.1) avec $N = 50$ et $N = 200$, tandis que Figure 1.3 montre aussi la dynamique de la population des cellules matures. Dans les deux systèmes, nous observons un front de propagation qui remplit progressivement tous les niveaux de maturation. De plus, il y a une amplification du nombre de cellules à

mesure que le niveau de maturation augmente. Dans la Figure [1.3](#), nous confirmons cette amplification étant donné que le nombre de cellules matures est beaucoup plus grand que celui des cellules souches.

Perspectives

Ce modèle peut être étendu pour inclure différentes considérations biologiques. En effet, nous travaillons sous l'hypothèse d'une hématopoïèse non perturbée (communément appelée *steady-state*) par opposition à l'hématopoïèse de stress, où nous pouvons supposer que les cellules se différencient vers le stade de maturation immédiatement suivant et que la régulation provient uniquement des cellules matures. Cependant, la différenciation et la régulation peuvent se comporter de manière très différente lors d'une hématopoïèse de stress, car une régulation par les cellules immatures apparaît également ; de plus, il a été observé que des sauts dans les stades de différenciation surviennent étant donné que les CSH à court terme peuvent directement créer des progéniteurs érythroïdes de stress ([\[29\]](#)). Ces sauts ont aussi été observés pour les mégacaryocytes, un type de cellule mature, qui émergent à partir de CSH à long terme en très peu d'étapes intermédiaires. Nous pouvons alors envisager d'étendre notre modèle pour prendre en compte ces sauts, où les cellules souches sautent des niveaux de maturation, pour se différencier directement vers un stade plus avancé. Un type de régulation différent dépendant de cellules immatures peut aussi être considéré.

Nous avons obtenu un résultat de convergence à partir d'un système lent-rapide vers un système déterministe avec des conditions de bord. Nous visons dans le futur de caractériser la vitesse de convergence en bornant la déviation trajectorielle entre le processus stochastique et sa limite déterministe ; ainsi que les fluctuations associées en utilisant une procédure de tension-identification comme, par exemple, [\[14\]](#).

1.4.2 Résultats du Chapitre 3

Ce chapitre correspond au preprint *Classical JAK2V617F+ Myeloproliferative Neoplasms emergence and development based on real life incidence and mathematical modeling*, en collaboration avec Vincent Bansaye, Evelyne Lauret, Morgane Mounier, Valérie Ugo, Sylvie Méléard and Stéphane Giraudier [\[25\]](#). Il est actuellement sous révision à *Mathematical Biosciences*.

Inspiré par la structure en âge des syndromes myéloprolifératifs (SMP), ce travail utilise des données d'âge de diagnostic pour développer un modèle âge-dépendant pour le temps d'émergence de la première cellule JAK2V617F et le développement

subséquent de la maladie jusqu'au diagnostic. Construit à partir d'une cohorte initiale, ce modèle est choisi parmi des modèles âge-indépendants ainsi qu'un modèle alternatif âge-dépendant étant donné qu'il est le seul pouvant expliquer les données. Ce modèle nous permet de mieux comprendre les dynamiques d'émergence et développement des SMP et d'obtenir des résultats quantitatifs sur ces temps.

Nous travaillons avec deux cohortes d'âge de diagnostic pour la polyglobulie vaquez (PV) et la thrombocythémie essentielle (TE) en présence de la mutation JAK2V617F. Nous utilisons une cohorte initiale qui consiste en le registre du cancer de la région Côte d'Or en France de 1990 à 2020 pour comparer les différents modèles. Pour ensuite valider le modèle choisi, nous utilisons une deuxième cohorte de FIMBANK. Nous commençons par ajuster les données pour tenir compte du fait que certains individus ont pu mourir avant d'être diagnostiqués et ne sont donc pas présents dans les données. Cet ajustement peut être observé dans la Figure 1.4.

Nous considérons l'âge de diagnostic comme la somme de deux composantes indépendantes :

- Le temps d'émergence, noté T_1 : le temps depuis le développement embryonnaire nécessaire pour la première cellule JAK2V617F à apparaître et entrer en prolifération ;
- Le temps d'expansion, noté T_2 : la temps pour l'expansion clonale une fois la première cellule mutante commence à proliférer jusqu'au diagnostic.

Nous procédons à la construction d'un modèle âge-dépendant, où T_1 suit une loi de Gompertz. Nous choisissons un modèle de Gompertz car il a permis de capturer la dynamique des données, alors que les modèles âge-indépendants et un modèle alternatif âge-dépendant de Weibull n'ont pas réussi à donner un ajustement adéquat aux données. Ce point est abordé plus en détail ultérieurement.

Pour définir les lois du temps d'émergence T_1 , nous considérons τ comme le taux de mutation ou d'activation d'une mutation existante au niveau des cellules souches, et supposons que chaque cellule mutante activée a une probabilité p de proliférer et éventuellement être détectée. Nous prouvons que T_1 suit une loi exponentielle de paramètre $\delta = \tau p$. Nous nous référons au paramètre δ comme le taux de mutation active, qui combine l'apparition de la mutation, l'activation d'une cellule souche quiescente et la probabilité de succès d'expansion. Nous considérons l'approche de l'analyse de survie où le taux δ correspond à la fonction de hasard définie comme le taux instantané de mutation active (apparition et activation de la mutation) à un âge t , étant donné que ça ne s'est pas produit avant (voir, par exemple, page 85 de [42]).

Pour tenir compte de la structure en âge des données, donnée par un nombre croissant de diagnostics avec l'âge qui peuvent être observés dans la Figure 1.4 comme dans d'autres jeux de données (voir [67]), nous prenons un taux de mutation active qui

dépend de l'âge $\delta = \delta(t)$. Nous considérons le modèle de Gompertz (voir [42], pages 90-92) pour le taux de mutation active δ , donné par

$$\delta_G(t) = A \exp(kt),$$

où $A > 0$, $k > 0$, et t c'est l'âge de l'individu.

Nous utilisons la fonction de survie associé à $\delta(t)$ pour obtenir la densité de T_1 (voir page 85 de [42]) qui est donnée pour un âge t par

$$f_1(t) = \delta(t) \exp\left(-\int_0^t \delta(s) ds\right) = A \exp\left(\frac{A}{k}\right) \exp(kt) \exp\left(-\frac{A}{k} \exp(kt)\right).$$

D'autre part, le temps d'expansion T_2 est considérée comme similaire entre tous les individus : étant donné le nombre élevé de divisions nécessaires pour atteindre la taille de détection (10^6 divisions), la variabilité des durées de divisions peut être moyennée. Nous analysons donc à la fois une variable T_2 suivant une loi lognormale, ce qui donne une distribution unimodale, et une valeur T_2 fixe. Par l'indépendance des deux variables, la distribution résultante pour l'âge de détection T_M est calculée comme la convolution des densités f_1 et f_2 de T_1 et T_2 , respectivement. Ainsi, la densité de T_M pour $t \geq 0$ est donnée par :

$$f_M(t) = \int_0^t f_1(s) f_2(t-s) ds.$$

Les deux modèles avec un temps d'expansion fixe et aléatoire sont ajustés aux données en utilisant un algorithme d'Espérance-Maximisation (EM) généralisée, où l'étape de maximisation utilise un pas de Newton-Raphson pour obtenir le nouveau jeu de paramètres. Le code pour cet algorithme est présenté dans l'Appendix A. Les deux modèles semblent offrir un bon ajustement aux données (voir Figure 1.5). Nous procédons alors à un test d'ajustement de Chi-deux sur les deux estimations. Ces tests mènent à ne pas rejeter aucun des deux modèles. Compte tenu du bon ajustement des deux modèles, nous utilisons le critère d'information bayésien (BIC) pour les comparer. Ce critère indique que le modèle avec un temps d'expansion fixe doit être choisi.

Nous validons en outre le modèle avec un taux de mutation active Gompertz et un temps d'expansion fixe en deux étapes : en utilisant le deuxième jeu de données ; et en testant des différents modèles sans dépendance en âge et un modèle alternatif avec dépendance en âge. Nous estimons les paramètres du modèle sur le deuxième jeu de données à travers le même algorithme EM et appliquons un test d'adéquation Chi-deux. Ce test ne rejette pas le modèle. De plus, nous étudions trois modèles sans dépendance en âge où le taux de mutation active δ est fixé au cours du temps, ces modèles considèrent un temps d'expansion fixe et aléatoire, et prennent en compte la variabilité individuelle dans le taux de mutation active. Après l'estimation des paramètres, les trois modèles donnent un mauvais ajustement aux données comme vu dans la Figure 1.6, menant à

leur rejet. Nous considérons une structure en âge alternative du temps d'émergence à travers un taux de mutation active Weibull donnée par $\delta_W(t) = \frac{\gamma}{\lambda} \left(\frac{t}{\lambda}\right)^{\gamma-1}$ où $\gamma > 0$, $\lambda > 0$, et le modèle plus vaste avec un T_2 lognormal. Ce modèle est aussi rejeté.

Le bon ajustement du modèle choisi dans les deux jeux de données, et le rejet des autres trois modèles sans dépendance en âge et du modèle avec une dépendance en âge Weibull démontrent et caractérisent une structure en âge qui augmente exponentiellement dans l'émergence des SMP en suivant un modèle de Gompertz.

Ce modèle avec dépendance en âge et temps d'expansion fixe nous permet en outre d'obtenir des résultats quantitatifs sur les maladies en appliquant le modèle aux données. Nous obtenons un temps d'émergence moyen de 63 ans avec une déviation standard de 13 ans et un temps d'expansion de 8.8 ans, donnant une moyenne de 71.8 ans. De plus, nous comparons PV et TE de façon séparée et trouvons une différence de 1.4 ans dans le temps d'émergence entre les deux maladies. Ce résultat peut être expliqué par la recombinaison mitotique nécessaire pour PV, où la majorité des cellules présentent la mutation dans les deux allèles, contrairement à TE où la majorité des cellules présentent la mutation dans un allèle.

1.4.3 Résultats du Chapitre 4

Ce travail en cours est issu d'une étroite collaboration avec les médecins Stéphane Giraudier et Nabih Maslah de l'Hôpital de Saint-Louis. Ce travail est réalisé en collaboration avec Charlotte Dion-Blanc.

Les SMP ont pour origine une mutation primaire qui induit une accélération de la prolifération, menant à un excès d'un type de cellules sanguines matures. Ce travail se centre autour de la mutation primaire JAK2V617F, la plus fréquente dans les SMP. Les cellules portant la mutation JAK2V617F peuvent aussi porter des mutations secondaires qui sont apparues avant ou après la mutation primaire, et cause possiblement des changements du taux de prolifération, et de la sensibilité de la cellule à certains traitements. Cette étude vise à identifier l'effet que les mutations secondaires ont dans la croissance et la réponse au traitement des SMP à partir de données cliniques. Cette information vise à soutenir la sélection du traitement en fonction des mutations portées par un patient.

Nous étudions exclusivement les trois traitements cytoréducteurs: l'Interféron- α (IFN), l'Hydroxyurée (HU) et le Ruxolitinib (Ruxo). Compte tenu de leurs mécanismes différents, nous supposons que si plus d'un traitement est administré simultanément, l'effet résultant est simplement la somme des effets de tous les traitements en cours.

Les données utilisées consistent en une analyse de séquençage en vrac (*bulk*) de prélèvements de sang de $N = 244$ patients diagnostiqués, avec au moins quatre prélèvements pris à différents instants de temps. À chaque échantillon, nous avons les informations suivantes :

- L'âge et le sexe du patient ;
- Le temps écoulé depuis le premier prélèvement;
- Le diagnostic et les traitements en cours au moment du prélèvement ;
- Le pourcentage de cellules portant la mutation JAK2V617F ;
- Pour chaque mutation additionnelle : le pourcentage de cellules portant la mutation.

Un travail préliminaire a consisté à nettoyer, filtrer et mettre en forme les données. Les données sont disponibles ici: <https://github.com/ana-fernandezbaranda/>

additional-mutations.

Nous définissons un modèle à effets mixtes pour la dynamique du pourcentage des cellules mutées JAK2V617F qui intègre la croissance naturelle, l'effet des traitements et l'influence des mutations additionnelles dans la croissance et la réponse aux traitements. Le logiciel Monolix est utilisé pour l'estimation des paramètres et l'analyse statistique de ce modèle. Cette estimation est basée sur l'algorithme SAEM.

Nous construisons un modèle initial pour la dynamique du pourcentage de cellules portant la mutation JAK2V617F. Ce modèle combine la croissance naturelle de la population à travers une fonction logistique généralisée, et la réduction de cette croissance par chaque traitement, en fonction de son mode d'action. Soit $P(t)$ le pourcentage de cellules mutées JAK2V617F au temps t mesuré en jours. Le modèle pour la dynamique de $P(t)$ est donné par

$$\begin{aligned} \frac{dP}{dt}(t) = & C P(t) \left(1 - \left(\frac{P(t)}{100} \right)^\gamma \right) [1 - HU(t) K_{HU} - IFN(t - 182) K_{IFN}] \\ & - Ruxo(t) K_{Ruxo} P(t), \end{aligned}$$

où $C, \gamma > 0$, $K_l > 0$ pour $l \in \{IFN, HU, Ruxo\}$. Nous prenons la fonction logistique généralisée pour la croissance de la population puisqu'elle regroupe une variété de modèles tels que le modèle logistique si $\gamma = 1$ ou le modèle de Gompertz quand γ approche 0 et permet aussi de borner $P(t)$ qui ne peut pas excéder 100.

Pour tenir compte de la variabilité inter-individuelle qui émerge de caractéristiques individuelles qui peuvent affecter la vitesse de prolifération et des doses et fréquences des traitements, nous considérons les cinq paramètres C , γ , K_{IFN} , K_{HU} , K_{Ruxo} comme des effets aléatoires. Étant donné leur positivité stricte, nous les prenons comme des échantillons de distributions lognormales.

En utilisant ce modèle, nous étudions l'ajout des deux covariable, âge et sexe, au paramètres de croissance C et γ en utilisant l'outil *Stochastic Approximation for Model Building Algorithm* (SAMBA) implémenté dans Monolix. Nous trouvons que le modèle qui s'ajuste le mieux aux données inclut une dépendance en âge pour les deux paramètres de croissance. Ceci indique qu'il n'existe pas une dépendance en sexe significative de ces paramètres. Nous savons que $\log(C^n)$ et $\log(\gamma^n)$ pour $1 \leq n \leq N$ sont distribués selon une loi normale étant donné que C et γ suivent une loi normale.

La dépendance en âge est introduite de la façon suivante

$$\begin{aligned}\log(C^n) &= \log(C_{pop}) + \beta_C \log\left(\frac{age_n}{\widehat{Age}}\right) + \eta_C^n \\ \log(\gamma^n) &= \log(\gamma_{pop}) + \beta_\gamma \log\left(\frac{age_n}{\widehat{Age}}\right) + \eta_\gamma^n,\end{aligned}$$

où $\eta_C^n \sim \mathcal{N}(0, \omega_C^2)$, $\eta_\gamma^n \sim \mathcal{N}(0, \omega_\gamma^2)$; C_{pop} et γ_{pop} sont les médianes des distributions de C^n et γ^n ; β_C et β_γ sont des paramètres à estimer. Dans cette construction, les paramètres de la population C et γ sont définis pour un individu typique de la population d'âge $\widehat{Age}$, la moyenne pondérée de l'âge de la population.

Pour résumer, ce modèle de base intègre:

- Une fonction de croissance logistique généralisée avec des paramètres K et γ qui dépendent de l'âge,
- L'effet des trois traitements administrés: Interféron- α , Hydroxyurée and Ruxolitinib.

Les paramètres du modèle sans mutations additionnelles sont estimés en utilisant les données avec le logiciel Monolix. Nous estimons les paramètres en utilisant un échantillon aléatoire avec 80% des patients, et obtenons un bon ajustement aux données et un petit nombre de valeurs aberrantes (7.56%) (voir Figure 1.7). Nous utilisons ensuite cette estimation pour prédire les paramètres individuels du 20% des individus restant, en obtenant à nouveau une bonne estimation avec seulement 7.41% de valeurs aberrantes. Cette validation croisée vérifie le choix du modèle pour la croissance naturelle.

Après validation du modèle de base, nous adaptons ce modèle pour prendre en compte les effets des mutations additionnelles. Dans cet objectif, nous définissons $\mathcal{M}$ comme l'ensemble de toutes les mutations additionnelles et $P^i(t)$ pour $i \in \mathcal{M}$ comme la population de cellules JAK2V617F portant aussi la mutation i (et possiblement d'autres mutations additionnelles). Quatre paramètres sont associés à chaque mutation additionnelle $i \in \mathcal{M}$ qui nous donnent des informations sur son impact :

- $C_i \in \mathbb{R}$: avantage ou désavantage dans la croissance de la mutation i
- $K_l^i \in \mathbb{R}$ pour chaque traitement $l \in \{HU, IFN, Ruxo\}$: l'effet de la mutation i sur le traitement l .

Si $C_i > 0$, cela signifie que la vitesse de croissance d'une population portant exclusivement la mutation i (et JAK2V617F) est plus rapide que celle de la population

JAK2V617F sans aucune autre mutation. Si $C_i < 0$, cela signifie que la vitesse de croissance d'une population portant exclusivement la mutation i (et JAK2V617F) est plus lente que celle de la population JAK2V617F sans aucune autre mutation.

Si $K_l^i > 0$ pour un traitement l , cela signifie qu'une cellule portant exclusivement la mutation i (et JAK2V617F) a une meilleure réponse au traitement l qu'une cellule portant uniquement la mutation JAK2V617F ; à l'inverse, si $K_l^i < 0$, la mutation i induit une résistance plus forte au traitement l . Ainsi, les paramètres K_l^i , $l \in \{HU, IFN, Ruxo\}$, apportent des informations cruciales qui peuvent être utiles pour le choix du traitement.

Nous supposons que les effets des mutations additionnelles sont indépendants les uns des autres. Ainsi, si une cellule possède les deux mutations $i, j \in \mathcal{M}$, l'effet sur la croissance est $C_i + C_j$ et l'effet sur le traitement $l \in \{HU, IFN, Ruxo\}$ est $K_l^i + K_l^j$. Nous supposons également que les nouvelles mutations apparaissent dans les cellules les plus mutées, ce qui implique que le nombre de cellules portant (non exclusivement) les mutations $i, j \in \mathcal{M}$ est égal à $\min(P^i(t), P^j(t))$. Après calculs, le modèle devient

$$\frac{dP}{dt}(t) = C g(P(t)) \left[\left(P(t) + \sum_{i \in \mathcal{M}} P^i(t) C_i \right) [1 - HU(t) K_{HU} - IFN(t) K_{IFN}] \right. \quad (1.4.4)$$

$$\left. - \sum_{i \in \mathcal{M}} P^i(t) (HU(t) K_{HU}^i + IFN(t) K_{IFN}^i) \right. \quad (1.4.5)$$

$$\left. - \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{M}} C_i (HU(t) K_{HU}^j + IFN(t) K_{IFN}^j) \min(P^i(t), P^j(t)) \right] \quad (1.4.6)$$

$$- Ruxo(t) \left[P(t) K_{Ruxo} + \sum_{i \in \mathcal{M}} P^i(t) K_{Ruxo}^i \right]. \quad (1.4.7)$$

Nous considérons les quatre paramètres associés à chaque mutation additionnelle $i \in \mathcal{M}$, C_i , K_{IFN}^i , K_{HU}^i , K_{Ruxo}^i comme fixes. Un résumé des paramètres du modèle est présenté dans la Table [1.3](#).

Nous étudions l'effet sur la croissance et les traitements IFN et HU pour les trois mutations les plus prévalentes : *TET2* (48.8% des patients), *DNMT3A* (39.3%) et *ASXL1**exon12* (29.9%). Ceci nous permet de comparer les deux traitements quand au moins une de ces mutations est présente. L'effet des mutations sur le traitement Ruxo n'a pas été pris en compte puisque ce n'a pas été possible d'obtenir des bonnes estimations avec des intervalles de confiance finis.

Nous travaillons avec l'ensemble de mutations additionnelles $\mathcal{M}_1 = \{TET2, DNMT3A, ASXL1\}$. Nous incluons l'effet de ces trois mutations sur la croissance et la réponse aux traitements IFN et HU à partir de [\(1.4.4\)](#). Alors, le

Paramètre	Domaine	Distribution	Description
C	$(0, \infty)$	lognormal	Relié à la croissance naturelle
β_C	$(-\infty, \infty)$		Effet de l'âge dans le paramètre C
γ	$(0, \infty)$	lognormal	Relié à la croissance naturelle
β_γ	$(-\infty, \infty)$		Effet de l'âge dans le paramètre γ
K_l	$(0, \infty)$	lognormal	Réponse au traitement $l \in \{HU, IFN, Ruxo\}$
C_i	$(-\infty, \infty)$	Fixe	Effet sur la croissance de la mutation $i \in \mathcal{M}$
K_l^i	$(-\infty, \infty)$	Fixe	Effet sur la réponse au traitement $l \in \{HU, IFN, Ruxo\}$ de la mutation $i \in \mathcal{M}$

Table 1.3: Paramètres du modèle.

modèle est donné par

$$\begin{aligned}
 \frac{dP}{dt}(t) = C g(P(t)) & \left[\left(P(t) + \sum_{i \in \mathcal{M}_1} P^i(t) C_i \right) [1 - HU(t) K_{HU} - IFN(t) K_{IFN}] \right. \\
 & - \sum_{i \in \mathcal{M}_1} P^i(t) (HU(t) K_{HU}^i + IFN(t) K_{IFN}^i) \\
 & \left. - \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{M}} C_i (HU(t) K_{HU}^j + IFN(t) K_{IFN}^j) P^{i,j}(t) \right] \\
 & - Ruxo(t) P(t) K_{Ruxo}.
 \end{aligned}$$

Nous effectuons l'estimation pour le même jeu de données avec 80% des individus. Dans la Figure 1.8, nous pouvons voir un gain important dans la prédiction pour certains individus portant au moins une de ces mutations quand nous incluons les mutations additionnelles par rapport au modèle sans les mutations additionnelles. En effet, nous observons que ce nouveau modèle capture des fluctuations dans les observations qui étaient mal prédites. Nous utilisons cette estimation sur le 20% restant, nous trouvons aussi une amélioration dans les prédictions pour certains patients portant au moins une de ces mutations, comme observé dans la Figure 1.9.

Les valeurs des paramètres obtenues à partir de cette estimation et leurs intervalles de confiance au seuil de 95% sont présentés dans la Table 1.4.

Paramètre	Valeur	Intervalle de confiance
C_{TET2}	12.76	[8.8 , 16.73]
K_{IFN}^{TET2}	-2.08	[-2.28, -1.87]
K_{HU}^{TET2}	-4.45	[-4.73, -4.17]
C_{DNMT3A}	-0.0041	[-0.016, 0.0077]
K_{IFN}^{DNMT3A}	-33.72	[-42.29, -25.16]
K_{HU}^{DNMT3A}	-6.41	[-8.89, -3.92]
C_{ASXL1}	5.93	[5.18, 6.68]
K_{IFN}^{ASXL1}	-37.62	[-43.8, -31.44]
K_{HU}^{ASXL1}	-2.32	[-2.9, -1.73]

Table 1.4: Estimation des paramètres et intervalles de confiance au seuil de 95%.

Ces résultats peuvent être interprétés de la façon suivante :

- $C_{TET2}, C_{ASXL1} > 0$: les mutations $TET2$ et $ASXL1$ favorisent une croissance plus rapide de la population JAK2V617F, l'effet de $TET2$ étant beaucoup plus important que celui de $ASXL1$.
- $C_{DNMT3A} \approx 0$: la mutation $DNMT3A$ n'a pas d'effet significatif sur la croissance de la population JAK2V617F.
- $K_l^i < 0$ for $l \in \{IFN, HU\}$ and $i \in \{TET2, ASXL1, DNMT3A\}$: les trois mutations induisent une pire réponse pour les traitements IFN et HU.
- $K_{HU}^{TET2} < K_{IFN}^{TET2} < 0$: la mutation $TET2$ a un effet plus significatif sur la réponse à HU que celle à IFN.
- $K_{IFN}^i < K_{HU}^i < 0$ for $i \in \{ASXL1, DNMT3A\}$ les mutations $DNMT3A$ et $ASXL1$ ont un effet beaucoup plus significatif sur la réponse à IFN que celle à HU.

Ces résultats suggèrent qu'en présence de $TET2$ ou $ASXL1$, on devrait s'attendre à une évolution plus rapide, surtout pour $TET2$. Quand la mutation $TET2$ est présente seule, le traitement IFN devrait être plus effectif que HU. Au contraire, avec l'existence de $DNMT3A$ ou $ASXL1$, on devrait avoir une meilleure réponse à HU qu'à IFN. Dans le cas où $TET2$ et $DNMT3A$ ou $TET2$ et $ASXL1$ sont présentes, le meilleur

traitement dépendra du pourcentage de cellules avec chaque mutation. Cependant, l'effet de *TET2* sur HU est beaucoup plus petit que ceux de *DNMT3A* et *ASXL1* sur IFN, indiquant que HU probablement donnerai les meilleurs résultats, indépendamment de la proportion de cellules portant chaque mutation.

Perspectives

Ce travail définit le cadre pour identifier l'impact des mutations additionnelles, donnant des informations importantes pour le choix du traitement. Nous caractérisons l'effets des trois mutations les plus présentes sur les traitements IFN et HU. Il reste à réaliser des analyses supplémentaires. Les effets des mutations additionnelles sur le traitement Ruxo n'ont pas pu être inclus, puisqu'il n'a pas été possible d'obtenir des bonnes estimations. Nous visons à résoudre ce problème en discutant s'il faudrait modifier le modèle pour inclure le mécanisme de ce traitement différemment, ou en simplifiant le modèle pour faciliter l'estimation. Nous visons à continuer d'inclure plus de mutations additionnelles dans notre analyse. De plus, dans ce travail nous caractérisons l'effet individuel de chaque mutation, nous voulons étendre cette information à des profils mutationnels, où un individu pourrait avoir plusieurs mutations additionnelles, chacune dans une proportion différente.

CHAPTER 2

Hematopoiesis as a continuum: from stochastic compartmental model to hydrodynamic limit

This chapter consists of the preprint *Hematopoiesis as a continuum: from stochastic compartmental model to hydrodynamic limit*, a joint work with Vincent Bansaye, Stéphane Giraudier and Sylvie Méléard [\[7\]](#). It is reproduced here with minimal modifications.

Abstract

We consider a multiscale stochastic compartmental model with three types of cells (stem cells, immature cells and mature cells) which combines cell proliferation and cell differentiation. We derive a hydrodynamic limit when the number of immature compartments goes to infinity obtaining a partial differential equations system with boundary conditions, modelling hematopoiesis as a continuum. We assume that proliferation and differentiation are regulated and let the corresponding rates depend on the number of mature cells. This leads us to model the dynamics of the population by a Markov process in continuous time and discrete space, which does not satisfy the branching property. We prove the convergence in law of the stem and mature cells population size processes and of the empirical measures of the immature cells dynamics, conveniently rescaled, to the unique triplet involving coupled functions and a measure, which are solutions of a deterministic measure valued equation with boundary dynamics. The cell differentiation induces a transport term in space and the main difficulty comes from the boundary effects coming from stem and mature cells. We also prove that the limiting measure admits at each time a density with respect to Lebesgue measure and can be characterized as solution of a partial differential equation.

2.1 Introduction

Hematopoiesis is the process by which blood cells are produced. This process starts with hematopoietic stem cells (denoted HSCs) that go through differentiation: the cells gradually become more and more specialized until they finally become mature blood cells with a specific function in the body. These immature cells can also self-renew through division, where two daughter cells are created from a mother cell; or die. The most mature myeloid cells (myeloid leucocytes, erythrocytes, platelets) do not divide and can only be created by differentiation or die with different half-life durations. As shown by Smith [71], the growth and differentiation of hematopoietic immature cells are regulated by signals from the microenvironment, mainly through cytokines secreted by mature cells.

The process of hematopoiesis has long been described as a hierarchical step-wise process. Biological observations of hematopoietic cells showed phenotype differences among the progenitor cells which allowed to create distinct groups of cells by phenotype as explained in Cheng et al. [18]. In this framework, HSCs differentiate a fixed number of times, becoming at each step a different more specialized progenitor, until finally becoming mature blood cells. Regardless, this classical modeling oversimplifies the complexity that hematopoietic cells have. Indeed, by using new techniques of observation at single-cell level (single-cell RNA-Seq analysis), this hierarchical discrete structure can be questioned as presented in Quesenberry et al. [62] and [18]. Current hypothesis concerning hematopoiesis development defined types of hematopoietic maturation stages are able to divide into more subtypes, leading to a continuum of hematopoiesis differentiation rather than a discrete model (see Wilson et al. [82] and Pietras et al. [60]). The authors in Zhang et al. [85] and [62] suggest that hematopoiesis is a continuum process where the phenotype of stem cells widely fluctuates over time, giving a heterogeneity among cells rather than clearly defined homogeneous phenotypes that allow to separate into different cell types. There are no obvious boundaries between the different hierarchical levels of hematopoietic cells. Moreover, [62] explained that the phenotype of a cell changed during the cell cycle which would not allow for a clear separation of cells into distinct categories. All of these arguments suggest that rather than being a step-wise differentiation, it is a progressive process. Then, rather than having different maturation cell types, we would be talking about maturation level, where the most immature cells are HSC and the final stage is mature cells.

The emergence of HSC that represent the upper compartment of the hematopoietic tree is also originating during embryogenesis from the AGM region where very few cells (mainly 1 to 3) (see, for example, Kumaravelu et al. [41], Rybtsov et al. [68] and Tober et al. [77]) emerge, migrate to the liver where the cells expand and give rise to HSC. Regardless, the precise mechanisms, number of cells at each step, and death rates at each step of this process are not well known. Then, we firstly hypothesized a

infinite number of steps from HSC to mature cells i.e. a continuum in differentiation process.

Previous modeling of hematopoiesis has been widely based on finite compartmental models where each compartment represents a distinct type of cell corresponding to the different maturation stages. This compartmental approach can be seen, for example, in Bonnet et al. [15], Dingli et al. [22], Knauer et al. [40], Mon Père et al. [51] and Stiehl et al. [73]. In all of these models, differentiation is directly linked to division. The hypothesis of regulation is often included in modeling as in [40], [51] and [15] through a feedback loop depending on the number of mature cells. While all of these models consider at least three compartments corresponding to HSCs, immature cells and mature cells, the number of immature compartments varies from one to the other depending on the phenomenon to study. Indeed, while in [40] they consider that a single immature cell type is enough to properly explain oscillatory dynamics, in [15] they prove that 6 compartments at least are necessary to explain the amplification in the erythroid lineage and in [22] they suggest that up to 31 types are needed for amplification.

In this article, we derive a deterministic model of an hematopoiesis continuum from a stochastic compartmental model. Starting from a continuous in time pure jump Markov process with self-renewal and cell differentiation, we study the limiting behavior when the number of immature compartments goes to infinity. In this scenario, division and differentiation are separate phenomena and follow slow-fast dynamics where division and death rates are much smaller than differentiation rates. Indeed, it is well known that the number of cell divisions from stem to mature cells is finite, whereas differentiation becoming a continuous process, the biological changes in the cell's state become smaller and smaller and happen faster and faster. The renewal and differentiation rates both depend on the number of mature cells (as motivated previously), making the model nonlinear. The stochastic system depends on a scaling parameter N which scales both the (large) order of magnitude of the stem cells population size as well as the number of compartments. We consider separately, on one hand the stem and mature cells dynamics ($X_1^N(t)$, $t \geq 0$) and ($X_N^N(t)$, $t \geq 0$) which are involved in division (for the first ones) and deaths (for the last ones), and on the other hand the immature compartments dynamics in large number ($X_i^N(t)$, $t \geq 0$; $2 \leq i \leq N-1$), which depend on the fast differentiation rates. The latter are gathered in an empirical measure-valued process on spatial sites corresponding to the compartment levels, defined by

$$\mu_t^N = \frac{1}{N} \sum_{i=2}^{N-1} X_i^N(t) \delta_{i/N}.$$

This results in three coupled processes $(X_1^N(t), \mu_t^N, X_N^N(t))_{t \geq 0}$, one describing the

stem cells, another for the immature cells, and a third for the mature cells. When doing so, we find a boundary condition issue: the dependence of the mature cells dynamics on the final immature compartment X_{N-1}^N . We overpass this difficulty by finding an equation only depending on the processes of the triplet $(X_1^N(t), \mu_t^N, X_N^N(t))_{t \geq 0}$. Our main result is Theorem 2.3.6, stating that these dynamics converge, as N tends to infinity, to a deterministic nonlinear system $((a(t), \mu_t, z(t)), t \geq 0)$ composed of two ordinary differential equations satisfied by a and z describing limiting stem and mature dynamics, and a measure-valued equation satisfied by μ describing the immature dynamics for the continuum of compartments. We obtain this convergence through a tightness-identification-uniqueness procedure. We first use the semimartingale decomposition of the processes to prove tightness results. We characterize the limiting values as solutions of the limiting system. Differentiation behaves now as a transport term in the measure-valued equation. The uniqueness in the space of measure-valued functions was a difficult point. To prove it, we take into account the associated mild equation obtained by using the flow of the ordinary differential equation representing the cell maturation dynamics. This mild formulation is also used to prove that when the initial empirical measure admits a density, then this property is preserved over time.

Similar limiting behaviors have been widely studied for interacting particle systems over $\mathbb{Z}^d$ when both the number of particles and the number of sites go to infinity and the number of particles is constant, see for example Kipnis and Landim [39] and De-Masi and Presutti [21]. Let us also mention Blount [13] and Perrut [59] which include birth and deaths in each site, for respectively one and two species with competition, and whose hydrodynamic limit is a reaction diffusion process. Up to our knowledge, much less work has been achieved for hydrodynamic limits combining birth and death processes, transport and boundary effects. We refer to Bahadoran [6] for a model where the birth and death appear in reservoirs located at the boundary. Let us also observe that when the interactions are neglected, our process with N compartments is a N types branching process, as studied in Athreya and Ney [4], Mode [50], Kimmel and Axelrod [38]. We are interested here in the approximation of the process for a large number of types ($N \rightarrow \infty$). Taking into account the regulation, the branching property is broken and the techniques we use involve tightness of the sequence of processes and identification of the limit rather than the branching structure of the process.

Doumic et al. [23] studied the continuous limiting values of a deterministic compartmental model for cellular differentiation, with a feedback loop in the differentiation. Their work provided the framework for limiting continuous models for cellular differentiation. In contrast, we present a stochastic compartmental model that describes the dynamics of the population of cells based on the individual cell behavior and consider regulation in both the differentiation and regulation rates. Besides, we extend the

space of solutions for the limiting system to the set of finite non-negative measures on $[0, 1]$ that do not necessarily have a density. The relation between our work and [23] is done in the particular case where densities exist at each time for the limiting measures μ_t . Uniqueness of the limiting process in measure-valued solutions is proven in [28] in the case where differentiation transitions are discrete and continuous at the same time, whereas we focus in strictly continuous differentiation.

The structure of the paper is as follows. In Section 2.2, we construct the discrete jump process for any fixed number N of compartments including the feedback mechanism on the differentiation and division rates. Section 2.3 is devoted to the proof of the convergence result. In Section 2.4, we show that if the initial condition of the measure gathering the immature compartments admits a density, then, this property propagates through time, allowing us to find an associated PDE system with two boundary ODEs. In Section 2.5, we illustrate our results with simulations for both the stochastic and deterministic models.

Notation: $\mathcal{C}(S)$ and $\mathcal{C}^1(S)$ denote respectively the sets of continuous and continuously differentiable functions on the set S .

The set of finite non-negative measures on $[0, 1]$ endowed with the weak topology is denoted by $(\mathcal{M}_F, w)$. Note that this topology is metrizable with the following metric. Let $BL(S)$ be the set of bounded Lipschitz continuous functions on the set S with the norm

$$\forall g \in BL(S), \quad ||g|| = \sup_{x \in S} |g(x)| + \sup_{x, y \in S, x \neq y} \frac{|g(x) - g(y)|}{|x - y|},$$

and define the unit ball

$$BL_1(S) = \{g \in BL(S), ||g|| \leq 1\}.$$

For two measures $\nu^1, \nu^2 \in \mathcal{M}_F$, we define

$$||\nu^1 - \nu^2||_{BL} = \sup_{g \in BL_1([0,1])} |\langle \nu^1 - \nu^2, g \rangle|.$$

C and C_T will denote respectively a positive constant that can change from line to line and a positive constant depending on T that can change from line to line. Both of these constants can depend on the parameters of the model, but not on the scaling parameter N nor on any chosen test function.

2.2 Discrete model for cellular differentiation

In this model, we characterize a cell by a quantitative trait $x \in [0, 1]$ representing its maturation level. A maturation level of $x = 0$ corresponds to HSC, one of $x = 1$

to mature cells and for $x \in (0, 1)$ we are talking about immature cells. We consider initially that there are N different cell types from HSCs to mature cells. A cell of type $i \in \{1, \dots, N\}$ will have a maturation level of $x = i/N$. In this framework, HSC have a maturation level of $x = 1/N$, approaching $x = 0$ as N grows. The parameter N also defines the speed of differentiation in immature compartments. Indeed, the more maturation stages the cell has to pass by before becoming a mature cell, the faster it has to go through them in order to maintain the time scale of mature cells creation of order one. We define a discrete jump process $(X(t), t \geq 0) = ((X_1(t), \dots, X_N(t)))_{t \geq 0}$ where $X_i(t)$ is the number of cells of type i at time t . The goal is to study the limiting behavior of this system when $N \rightarrow \infty$.

We introduce the functions $(x, z) \rightarrow r(x, z)$, $(x, z) \rightarrow m(x, z)$ defined on $[0, 1] \times [0, \infty)$ that correspond to the division and differentiation rates for a cell of maturation level x when the mature population is of size z . The dependence on the mature population size gathers the regulation factors on the division and differentiation rates. In the whole paper, we work with the following assumption.

Assumption 1. We assume that the functions r and m are continuous and non-negative functions, that m belongs to $\mathcal{C}^1([0, 1] \times [0, \infty))$, and that there exist constants $\hat{r}, \hat{m}, m_{\min} \in \mathbb{R}_+^*$ such that for all $(x, z) \in [0, 1] \times [0, \infty)$,

$$0 \leq r(x, z) \leq \hat{r}, \quad m_{\min} \leq m(x, z) \leq \hat{m}.$$

Furthermore, we assume that there exist $L_m > 0$ and $L_r > 0$ such that for any $x_1, x_2 \in [0, 1]$ and any $z_1, z_2 \in [0, \infty)$,

$$\begin{aligned} |m(x_1, z_1) - m(x_2, z_2)| &\leq L_m (|x_1 - x_2| + |z_1 - z_2|), \\ \left| \frac{\partial}{\partial x} m(x_1, z_1) - \frac{\partial}{\partial x} m(x_2, z_2) \right| &\leq L_m (|x_1 - x_2| + |z_1 - z_2|), \\ |r(x_1, z_1) - r(x_2, z_2)| &\leq L_r (|x_1 - x_2| + |z_1 - z_2|). \end{aligned}$$

The boundedness by above of the division and differentiation rates will ensure, in particular, the control of the first moment and the non-explosion of the stochastic process, while the Lipschitz continuity conditions on m , $\partial m / \partial x$ and r will guarantee uniqueness of the deterministic limit objects. In order to ensure creation of mature cells, the differentiation rate function has to be bounded by below by some $m_{\min} > 0$. Indeed, if this was not the case and $m(x, z) = 0$ for a fixed $z \in [0, \infty)$ and some $x \in [0, 1]$, then the cells may not be able to attain a maturation level greater than x .

For the N subpopulations of cells we have the following transitions:

- Each cell of type i , ($1 \leq i \leq N - 1$) divides at rate $r(i/N, X_N/N)$ creating two cells of same type,
- Each cell of type 1 becomes a cell of type 2 at rate $m(1/N, X_N/N)$,
- Each cell of type i , ($2 \leq i \leq N - 1$) becomes a cell of type $i + 1$ at rate $Nm(i/N, X_N/N)$,
- Each cell of type N dies at rate d ,

where d is a positive constant. For simplicity, we did not consider deaths in immature compartments and the parameter d is not regulated by the mature size population. We believe that these dynamics could be included by directly adapting the proofs presented here.

Since the differentiation rate of the immature compartments is accelerated by the number of compartments, each of these immature compartment both fills and empties very fast. Then, the population size of stem and mature cells are of same order of magnitude. This assumption is not contradictory with the amplification phenomenon observed between HSCs and mature cells compartments (see simulations in Figure 2.1). We focus on the case where the population sizes of stem and mature cells are of order N . Hence, the $N - 2$ immature compartments are expected to have an order of magnitude of 1. Thus, we consider the scaling

$$X_1^N(t) = \frac{X_1(t)}{N}, \quad X_i^N(t) = X_i(t), \quad X_N^N(t) = \frac{X_N(t)}{N}.$$

We are thus working with a multitype density dependent pure jump Markov process $(X^N(t), t \geq 0)$, whose trajectorial representation is given using Poisson point measures. Let $(\mathcal{N}_i^j : 1 \leq i \leq N, j \in \{r, m, d\})$ be independent Poisson point measures with intensity $ds d\theta$ on $\mathbb{R}_+^2$. We introduce $(\mathcal{F}_t)_t$ as the canonical filtration generated by the initial condition $X^N(0)$ and these Poisson point measures. Then, the process $X^N(t)$ is the unique strong solution in $\mathbb{D}\left([0, T], \frac{\mathbb{N}}{N} \times \mathbb{N}^{N-2} \times \frac{\mathbb{N}}{N}\right)$, for all $T > 0$, of

the following system of stochastic differential equations

$$\begin{aligned} X_1^N(t) &= X_1^N(0) + \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N r(1/N, X_N^N(s^-))} X_1^N(s^-) \mathcal{N}_1^r(ds, d\theta) \\ &\quad - \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m(1/N, X_N^N(s^-))} X_1^N(s^-) \mathcal{N}_1^m(ds, d\theta) ; \end{aligned} \quad (2.2.1)$$

$$\begin{aligned} X_i^N(t) &= X_i^N(0) + \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq r(i/N, X_N^N(s^-))} X_i^N(s^-) \mathcal{N}_i^r(ds, d\theta) \\ &\quad + \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m((i-1)/N, X_N^N(s^-))} X_{i-1}^N(s^-) \mathcal{N}_{i-1}^m(ds, d\theta) \\ &\quad - \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m(i/N, X_N^N(s^-))} X_i^N(s^-) \mathcal{N}_i^m(ds, d\theta), \quad 2 \leq i \leq N-1 ; \end{aligned} \quad (2.2.2)$$

$$\begin{aligned} X_N^N(t) &= X_N^N(0) + \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m((N-1)/N, X_N^N(s^-))} X_{N-1}^N(s^-) \mathcal{N}_{N-1}^m(ds, d\theta) \\ &\quad - \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N d} X_N^N(s^-) \mathcal{N}_N^d(ds, d\theta). \end{aligned} \quad (2.2.3)$$

By considering the compensated martingale measures of $(\mathcal{N}_i^j, 1 \leq i \leq N, j \in \{r, m, d\})$, we can write the semimartingale decomposition of these equations for $2 \leq i \leq N-1$ as

$$\begin{cases} X_1^N(t) = X_1^N(0) + \int_0^t \left[r \left(\frac{1}{N}, X_N^N(s) \right) - m \left(\frac{1}{N}, X_N^N(s) \right) \right] X_1^N(s) ds + M_1^N(t) ; & (2.2.4a) \\ X_i^N(t) = X_i^N(0) + \int_0^t N m \left(\frac{i-1}{N}, X_N^N(s) \right) X_{i-1}^N(s) ds \\ \quad + \int_0^t \left[r \left(\frac{i}{N}, X_N^N(s) \right) - N m \left(\frac{i}{N}, X_N^N(s) \right) \right] X_i^N(s) ds + M_i^N(t) ; & (2.2.4b) \\ X_N^N(t) = X_N^N(0) + \int_0^t m \left(\frac{N-1}{N}, X_N^N(s) \right) X_{N-1}^N(s) ds - d \int_0^t X_N^N(s) ds + M_N^N(t), & (2.2.4c) \end{cases}$$

where the processes $(M_j^N(t), 1 \leq j \leq N)_{t \geq 0}$ are square-integrable martingales whose quadratic variation can be easily computed (see Lemma [A.1](#) in Appendix).

In order to study the asymptotic behavior of the immature compartments of type $2 \leq i \leq N-1$ which are in a different scaling, we consider the stochastic measure μ_t^N defined for any $t \geq 0$ as

$$\mu_t^N = \frac{1}{N} \sum_{i=2}^{N-1} X_i^N(t) \delta_{i/N}. \quad (2.2.5)$$

In all that follows, we will work with the triplet $(X_1^N(t), \mu_t^N, X_N^N(t))_{t \geq 0}$ composed of the empirical measure containing all the immature compartments, and the boundaries

that correspond to the cell and mature cells population sizes. From System (2.2.4a), we can write the semimartingale decomposition. The equations are written in such a way that they only depend on elements on the triplet and not on any X_i^N for $2 \leq i \leq N-1$.

Proposition 2.2.1. *Under Assumption 1, for any function $f \in \mathcal{C}([0, 1])$, for any $t \geq 0$, we have*

$$X_1^N(t) = X_1^N(0) + A_1^N + M_1^N(t) \quad (2.2.6a)$$

$$\langle \mu_t^N, f \rangle = \langle \mu_0^N, f \rangle + A^{N,f}(t) + M^{N,f}(t) \quad (2.2.6b)$$

$$X_N^N(t) = X_N^N(0) + A_N^N(t) + M^N(t), \quad (2.2.6c)$$

where

$$A_1^N(t) = \int_0^t \left[r\left(\frac{1}{N}, X_N^N(s)\right) - m\left(\frac{1}{N}, X_N^N(s)\right) \right] X_1^N(s) ds; \quad (2.2.7a)$$

$$\begin{aligned} A^{N,f}(t) &= f\left(\frac{2}{N}\right) \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds \\ &\quad + \int_0^t \langle \mu_s^N, f r(\cdot, X_N^N(s)) \rangle ds + \int_0^t \langle \mu_s^N, (\Delta_{1/N} f) m(\cdot, X_N^N(s)) \rangle ds \\ &\quad - f(1) B^N(t) \end{aligned} \quad (2.2.7b)$$

$$A_N^N(t) = B^N(t) - d \int_0^t X_N^N(s) ds, \quad (2.2.7c)$$

$$\begin{aligned} B^N(t) &= \langle \mu_0^N, 1 \rangle - \langle \mu_t^N, 1 \rangle + \int_0^t \langle \mu_s^N, r(\cdot, X_N^N(s)) \rangle ds \\ &\quad + \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds \end{aligned} \quad (2.2.7d)$$

$\Delta_{1/N} f$ corresponds to the discrete derivative of f defined by

$$\Delta_h f(x) = \frac{f(x+h) - f(x)}{h},$$

and the square-integrable martingales $M^{N,f}$, M^N are given by

$$\begin{aligned} M^{N,f} &= \frac{1}{N} \sum_{i=2}^{N-1} \left(f\left(\frac{i}{N}\right) - f(1) \right) M_i^N \\ M^N &= \frac{1}{N} \sum_{i=2}^{N-1} M_i^N + M_N^N. \end{aligned}$$

Proof. The equation for $X_1^N(t)$ comes directly from (2.2.4a). Let us consider now

$$\langle \mu_t^N, f \rangle = \frac{1}{N} \sum_{i=2}^{N-1} X_i^N(t) f\left(\frac{i}{N}\right)$$

and use (2.2.4b). We deduce

$$\begin{aligned}
\langle \mu_t^N, f \rangle &= \frac{1}{N} \sum_{i=2}^{N-1} f\left(\frac{i}{N}\right) X_i^N(0) + \int_0^t \frac{1}{N} \sum_{i=2}^{N-1} f\left(\frac{i}{N}\right) r\left(\frac{i}{N}, X_N^N(s)\right) X_i^N(s) ds \\
&\quad + f\left(\frac{2}{N}\right) \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds \\
&\quad + \int_0^t \frac{1}{N} \sum_{i=2}^{N-1} \left(f\left(\frac{i+1}{N}\right) - f\left(\frac{i}{N}\right)\right) N m\left(\frac{i}{N}, X_N^N(s)\right) X_i^N(s) ds \\
&\quad - f(1) \int_0^t m\left(\frac{N-1}{N}, X_N^N(s)\right) X_{N-1}^N(s) ds + \frac{1}{N} \sum_{i=2}^{N-1} f\left(\frac{i}{N}\right) M_i^N(t) \\
&= \langle \mu_0^N, f \rangle + \int_0^t \langle \mu_s^N, f r(\cdot, X_N^N(s)) \rangle ds \\
&\quad + f\left(\frac{2}{N}\right) \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds \\
&\quad + \int_0^t \langle \mu_s^N, \Delta_{1/N} f m(\cdot, X_N^N(s)) \rangle ds \\
&\quad - f(1) \int_0^t m\left(\frac{N-1}{N}, X_N^N(s)\right) X_{N-1}^N(s) ds + \frac{1}{N} \sum_{i=2}^{N-1} f\left(\frac{i}{N}\right) M_i^N(t).
\end{aligned} \tag{2.2.8}$$

As we want the equations to only depend on elements of the triplet (X_1^N, μ^N, X_n^N) , we need to get rid of the term involving $X_{N-1}^N(t)$ in $\langle \mu_t^N, f \rangle$. Note that for $2 \leq i \leq N-1$, from (2.2.4b), we have that

$$N \int_0^t m\left(\frac{i}{N}, X_N^N(s)\right) X_i^N(s) ds \tag{2.2.9}$$

$$\begin{aligned}
&= X_i^N(0) - X_i^N(t) + N \int_0^t m\left(\frac{i-1}{N}, X_N^N(s)\right) X_{i-1}^N(s) ds \\
&\quad + \int_0^t r\left(\frac{i}{N}, X_N^N(s)\right) X_i^N(s) ds + M_i^N(t).
\end{aligned} \tag{2.2.10}$$

We apply this formula for $2 \leq i \leq N-1$, we sum the corresponding equations and then divide both sides by N to finally obtain that

$$\begin{aligned}
&\int_0^t m\left(\frac{N-1}{N}, X_N^N(s)\right) X_{N-1}^N(s) ds = \\
&\quad \langle \mu_0^N, 1 \rangle - \langle \mu_t^N, 1 \rangle + \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds \\
&\quad + \int_0^t \langle \mu_s^N, r(\cdot, X_N^N(s)) \rangle ds + \frac{1}{N} \sum_{i=2}^{N-1} M_i^N(t).
\end{aligned} \tag{2.2.11}$$

Plugging (2.2.11) in (2.2.8) yields (2.2.6b) and (2.2.7b). Alternatively, (2.2.11) can be computed from (2.2.8) using the test function $f \equiv 1$. Finally, for $X_N^N(t)$ we replace the term involving $X_{N-1}^N(t)$ in (2.2.4c) using (2.2.11). $\square$

The quadratic variations of the martingales in System (2.2.6) are given below.

Proposition 2.2.2. *The martingales M_1^N , $M^{N,f}$ and M^N in Proposition 2.2.1 are square-integrable and their quadratic variations are*

$$\begin{aligned} \langle M_1^N \rangle_t &= \frac{1}{N} \int_0^t \left(r \left(\frac{1}{N}, X_N^N(s) \right) + m \left(\frac{1}{N}, X_N^N(s) \right) \right) X_1^N(s) ds; \\ \langle M^{N,f} \rangle_t &= \frac{1}{N} \left[\int_0^t \langle \mu_s^N, (f - f(1))^2 r(\cdot, X_N^N(s)) \rangle ds \right. \\ &\quad \left. + \frac{1}{N} \int_0^t \langle \mu_s^N, (\Delta_{1/N} f)^2 m(\cdot, X_N^N(s)) \rangle ds \right. \\ &\quad \left. + \left(f \left(\frac{2}{N} \right) - f(1) \right)^2 \int_0^t m \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds \right]; \\ \langle M^N \rangle_t &= \frac{1}{N} \left[\int_0^t \langle \mu_s^N, r(\cdot, X_N^N(s)) \rangle ds + \int_0^t m \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds \right. \\ &\quad \left. + d \int_0^t X_N^N(s) ds \right]; \\ \langle M_1^N, M^{N,f} \rangle_t &= \frac{1}{N} \left(f(1) - f \left(\frac{2}{N} \right) \right) \int_0^t m \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds; \\ \langle M^{N,f}, M^N \rangle_t &= \frac{1}{N} \left[\int_0^t \langle \mu_s^N, \left(f(1) - f \left(\frac{i}{N} \right) \right) r(\cdot, X_N^N(s)) \rangle ds \right. \\ &\quad \left. + \int_0^t \left(f(1) - f \left(\frac{2}{N} \right) \right) m \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds \right]; \\ \langle M_1^N, M^N \rangle_t &= - \frac{1}{N} \int_0^t m \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds, \end{aligned}$$

for any $t \geq 0$ and any $f \in \mathcal{C}([0, 1])$.

Proof. Recall that the compensated martingale measures $(\mathcal{N}_i^j, 1 \leq i \leq N, j \in \{r, m, d\})$ are all independent. The quadratic variation of M_1^N is computed in the

proof of Lemma A.1 The martingale $M^{N,f}$ is given by

$$\begin{aligned}
 M^{N,f}(t) &= \frac{1}{N} \sum_{i=2}^{N-1} \left(f\left(\frac{i}{N}\right) - f(1) \right) \left[\int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq r(i/N, X_N^N(s^-))} X_i^N(s^-) \widetilde{\mathcal{N}}_i^r(ds, d\theta) \right. \\
 &\quad + \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m((i-1)/N, X_N^N(s^-))} X_{i-1}^N(s^-) \widetilde{\mathcal{N}}_{i-1}^m(ds, d\theta) \\
 &\quad \left. - \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m(i/N, X_N^N(s^-))} X_i^N(s^-) \widetilde{\mathcal{N}}_i^m(ds, d\theta) \right] \\
 &= \frac{1}{N} \sum_{i=2}^{N-1} \left(f\left(\frac{i}{N}\right) - f(1) \right) \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq r(i/N, X_N^N(s^-))} X_i^N(s^-) \widetilde{\mathcal{N}}_i^r(ds, d\theta) \\
 &\quad + \frac{1}{N} \left(f\left(\frac{2}{N}\right) - f(1) \right) \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m((1)/N, X_N^N(s^-))} X_1^N(s^-) \widetilde{\mathcal{N}}_1^m(ds, d\theta) \\
 &\quad + \frac{1}{N^2} \sum_{i=2}^{N-1} \Delta_{1/N} f\left(\frac{i+1}{N}\right) \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m(i/N, X_N^N(s^-))} X_i^N(s^-) \widetilde{\mathcal{N}}_i^m(ds, d\theta)
 \end{aligned}$$

Given the independence of the compensated measures we obtain

$$\begin{aligned}
 \langle M^{N,f} \rangle_t &= \frac{1}{N^2} \sum_{i=2}^{N-1} \left(f\left(\frac{i}{N}\right) - f(1) \right)^2 \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq r(i/N, X_N^N(s^-))} X_i^N(s^-) d\theta ds \\
 &\quad + \frac{1}{N^2} \left(f\left(\frac{2}{N}\right) - f(1) \right)^2 \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m((1)/N, X_N^N(s^-))} X_1^N(s^-) d\theta ds \\
 &\quad + \frac{1}{N^4} \sum_{i=2}^{N-1} \left(\Delta_{1/N} f\left(\frac{i+1}{N}\right) \right)^2 \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m(i/N, X_N^N(s^-))} X_i^N(s^-) d\theta ds \\
 &= \frac{1}{N} \int_0^t \left\langle \mu_s^N, (f - f(1))^2 r(\cdot, X_N^N(s)) \right\rangle ds \\
 &\quad + \frac{1}{N^2} \int_0^t \left\langle \mu_s^N, (\Delta_{1/N} f)^2 m(\cdot, X_N^N(s)) \right\rangle ds \\
 &\quad + \frac{1}{N} \left(f\left(\frac{2}{N}\right) - f(1) \right)^2 \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds.
 \end{aligned}$$

For M^N we have that

$$\begin{aligned}
 M^N(t) &= \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m((1)/N, X_N^N(s^-))} X_1^N(s^-) \widetilde{\mathcal{N}}_1^m(ds, d\theta) \\
 &\quad + \frac{1}{N} \sum_{i=2}^{N-1} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq r(i/N, X_N^N(s^-))} \widetilde{\mathcal{N}}_i^r(ds, d\theta) \\
 &\quad - \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N d X_N^N(s^-)} \widetilde{\mathcal{N}}_N^d(ds, d\theta),
 \end{aligned}$$

which implies, by similar arguments as before

$$\langle M^N \rangle_t = \frac{1}{N} \left[\int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds + \int_0^t \left\langle \mu_s^N, r(\cdot, X_N^N(s)) \right\rangle ds + \int_0^t d X_N^N(s) ds \right].$$

Additionally, we compute

$$\begin{aligned}\langle M_1^N, M^{N,f} \rangle_t &= -\frac{1}{N^2} \left(f\left(\frac{2}{N}\right) - f(1) \right) \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m(\frac{1}{N}, X_N^N(s))} X_1^N(s) d\theta ds \\ &= -\frac{1}{N} \left(f\left(\frac{2}{N}\right) - f(1) \right) \int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds.\end{aligned}$$

And we can find in the same way $\langle M^{N,f}, M^N \rangle_t$ and $\langle M_1^N, M^N \rangle_t$. $\square$

We can estimate the first and second moments of the triplet (X_1^N, μ^N, X_n^N) uniformly on finite time intervals, which will be needed in the following proofs.

Proposition 2.2.3. *Under Assumption [1](#) for any $N > 0$ and $T > 0$ we have that*

$$\mathbb{E} \left[\sup_{t \leq T} \left\{ X_1^N(t) + \langle \mu_t^N, 1 \rangle + X_N^N(t) \right\} \right] \leq \mathbb{E} \left[X_1^N(0) + \langle \mu_0^N, 1 \rangle + X_N^N(0) \right] e^{\hat{r}T}, \quad (2.2.12)$$

and

$$\begin{aligned}\mathbb{E} \left[\sup_{t \leq T} \left(X_1^N(t) + \langle \mu_t^N, 1 \rangle + X_N^N(t) \right)^2 \right] \\ \leq \left[\mathbb{E} \left[\left(X_1^N(0) + \langle \mu_0^N, 1 \rangle + X_N^N(0) \right)^2 \right] + \frac{T}{2N} \right] \exp \left(2\hat{r} \left(1 + \frac{1}{2N} \right) T \right). \quad (2.2.13)\end{aligned}$$

Proof. Let us prove [\(2.2.13\)](#). Similar computations can be done to prove [\(2.2.12\)](#). We define the process $Y^N(t) = X_1^N(t) + \langle \mu_t^N, 1 \rangle + X_N^N(t)$ and from [\(2.2.1\)](#), [\(2.2.2\)](#), [\(2.2.3\)](#) and [\(2.2.5\)](#), it follows that

$$\begin{aligned}Y^N(t) &= Y^N(0) + \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N r(1/N, X_N^N(s^-))} X_1^N(s^-) \mathcal{N}_1^r(ds, d\theta) \\ &\quad + \sum_{i=2}^{N-1} \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq r(i/N, X_N^N(s^-))} X_i^N(s^-) \mathcal{N}_i^r(ds, d\theta) \\ &\quad - \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq dN X_N^N(s^-)} \mathcal{N}_N^d(ds, d\theta).\end{aligned}$$

For any $n \in \mathbb{N}^*$, we define the stopping time $T_n = \{t > 0, Y^N(t) \geq n\}$ with respect

to the filtration $\mathcal{F}_t$. Then, for $s \leq t$ we have

$$\begin{aligned}
& \left(Y^N(s \wedge T_n) \right)^2 \\
&= \left(Y^N(0) \right)^2 \\
&+ \int_0^{s \wedge T_n} \int_{\mathbb{R}^+} \left(\left(Y^N(s^-) + \frac{1}{N} \right)^2 - \left(Y^N(s^-) \right)^2 \right) \mathbb{1}_{\theta \leq N r(1/N, X_N^N(s^-)) X_1^N(s^-)} \mathcal{N}_1^r(ds, d\theta) \\
&+ \sum_{i=2}^{N-1} \int_0^{s \wedge T_n} \int_{\mathbb{R}^+} \left(\left(Y^N(s^-) + \frac{1}{N} \right)^2 - \left(Y^N(s^-) \right)^2 \right) \mathbb{1}_{\theta \leq r(i/N, X_N^N(s^-)) X_i^N(s^-)} \mathcal{N}_i^r(ds, d\theta) \\
&+ \int_0^{s \wedge T_n} \int_{\mathbb{R}^+} \left(\left(Y^N(s^-) - \frac{1}{N} \right)^2 - \left(Y^N(s^-) \right)^2 \right) \mathbb{1}_{\theta \leq N d X_N^N(s^-)} \mathcal{N}_N^d(ds, d\theta).
\end{aligned} \tag{2.2.14}$$

By only considering the positive terms we obtain the upper bound

$$\begin{aligned}
& \sup_{s \leq t} \left(Y^N(s \wedge T_n) \right)^2 \\
&\leq \left(Y^N(0) \right)^2 \\
&+ \int_0^{t \wedge T_n} \int_{\mathbb{R}^+} \left(\left(Y^N(s^-) + \frac{1}{N} \right)^2 - \left(Y^N(s^-) \right)^2 \right) \mathbb{1}_{\theta \leq N r(1/N, X_N^N(s^-)) X_1^N(s^-)} \mathcal{N}_1^r(ds, d\theta) \\
&+ \sum_{i=2}^{N-1} \int_0^{t \wedge T_n} \int_{\mathbb{R}^+} \left(\left(Y^N(s^-) + \frac{1}{N} \right)^2 - \left(Y^N(s^-) \right)^2 \right) \mathbb{1}_{\theta \leq r(i/N, X_N^N(s^-)) X_i^N(s^-)} \mathcal{N}_i^r(ds, d\theta).
\end{aligned}$$

Then, taking expectation and using Assumption [1](#), we obtain

$$\begin{aligned}
\mathbb{E} \left[\sup_{s \leq t} \left(Y^N(s \wedge T_n) \right)^2 \right] &\leq \mathbb{E} \left[\left(Y^N(0) \right)^2 \right] + \mathbb{E} \left[\int_0^{t \wedge T_n} \left(2Y^N(s) + \frac{1}{N} \right) r \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds \right] \\
&\quad + \mathbb{E} \left[\sum_{i=2}^{N-1} \int_0^{t \wedge T_n} \frac{1}{N} \left(2Y^N(s) + \frac{1}{N} \right) r \left(\frac{i}{N}, X_N^N(s) \right) X_i^N(s) ds \right] \\
&\leq \mathbb{E} \left[\left(Y^N(0) \right)^2 \right] + 2\hat{r} \mathbb{E} \left[\int_0^{t \wedge T_n} \left(Y^N(s) + \frac{1}{2N} \right) X_1^N(s) ds \right] \\
&\quad + 2\hat{r} \mathbb{E} \left[\int_0^{t \wedge T_n} \left(Y^N(s) + \frac{1}{2N} \right) \langle \mu_s^N, 1 \rangle ds \right] \\
&\leq \mathbb{E} \left[\left(Y^N(0) \right)^2 \right] + 2\hat{r} \mathbb{E} \left[\int_0^{t \wedge T_n} \left(Y^N(s) + \frac{1}{2N} \right) Y^N(s) ds \right].
\end{aligned}$$

Using the fact that for any $x \in \mathbb{R}$ and $\varepsilon \geq 0$, $x^2 + \varepsilon x \leq (1 + \varepsilon)x^2 + \varepsilon$, we deduce that

$$\begin{aligned} & \mathbb{E} \left[\sup_{s \leq t} \left(Y^N(s \wedge T_n) \right)^2 \right] \\ &= \mathbb{E} \left[\left(Y^N(0) \right)^2 \right] + 2\hat{r} \mathbb{E} \left[\int_0^{t \wedge T_n} \left(\left(1 + \frac{1}{2N} \right) \left(Y^N(s) \right)^2 + \frac{1}{2N} \right) ds \right] \\ &\leq \mathbb{E} \left[\left(Y^N(0) \right)^2 \right] + \frac{t}{2N} + 2\hat{r} \left(1 + \frac{1}{2N} \right) \mathbb{E} \left[\int_0^{t \wedge T_n} \left(Y^N(s) \right)^2 ds \right] \\ &= \mathbb{E} \left[\left(Y^N(0) \right)^2 \right] + \frac{t}{2N} + 2\hat{r} \left(1 + \frac{1}{2N} \right) \mathbb{E} \left[\int_0^t \sup_{u \leq s} \left(Y^N(u \wedge T_n) \right)^2 ds \right]. \end{aligned}$$

Since the process is bounded by $n + 1$ before T_n , Gronwall's lemma tells us that for any $T > 0$

$$\mathbb{E} \left[\sup_{s \leq T \wedge T_n} \left(Y^N(s) \right)^2 \right] \leq \left(\mathbb{E} \left[\left(Y^N(0) \right)^2 \right] + \frac{T}{2N} \right) \exp \left(2\hat{r} \left(1 + \frac{1}{2N} \right) T \right). \quad (2.2.15)$$

We have that $(T_n)_n$ goes to infinity almost surely, since otherwise there would exist $S > 0$ such that $\mathbb{P}(\sup_n T_n < S) > 0$, which would imply $\mathbb{E} \left(\sup_{t \leq S \wedge T_n} (Y^N(t))^2 \right) \geq n^2 \mathbb{P}(\sup_n T_n < S)$ and this is a contradiction with (2.2.15). Then, by Fatou's lemma

$$\mathbb{E} \left[\sup_{s \leq T} \left(Y^N(s) \right)^2 \right] < \left(\mathbb{E} \left[\left(Y^N(0) \right)^2 \right] + \frac{T}{2N} \right) \exp \left(2\hat{r} \left(1 + \frac{1}{2N} \right) T \right).$$

This proves (2.2.13). □

Furthermore we can find a bound on the time average expected value of each immature compartment, which will be useful when studying the limit of System (2.2.6).

Proposition 2.2.4. *For any $N > 0$, $T > 0$ and $2 \leq i \leq N - 1$, we have that*

$$\int_0^T \mathbb{E} \left[X_i^N(s) \right] ds \leq \frac{1}{m_{min}} \left[\mathbb{E} \left[X_1^N(0) + \langle \mu_0^N, 1 \rangle + X_1^N(0) \right] + \frac{\widehat{m}}{\hat{r}} \mathbb{E} \left[X_1^N(0) \right] \right] \exp(\hat{r}T).$$

Proof. We have that

$$\frac{d}{dt} \mathbb{E} \left[X_1^N(t) \right] = \mathbb{E} \left[\left(r \left(\frac{1}{N}, X_1^N(u) \right) - m \left(\frac{1}{N}, X_1^N(u) \right) \right) X_1^N(t) \right] \leq \hat{r} \mathbb{E} \left[X_1^N(t) \right].$$

Using Gronwall's Lemma, we obtain

$$\mathbb{E} \left[X_1^N(t) \right] \leq \mathbb{E} \left[X_1^N(0) \right] \exp(\hat{r}t). \quad (2.2.16)$$

Considering any $2 \leq i \leq N-1$, we can apply formula (2.2.9) to all $2 \leq j \leq i$ and sum the corresponding equations. Then taking the expectations on both side, we infer that

$$\begin{aligned} & \int_0^t \mathbb{E} \left[m \left(\frac{i}{N}, X_N^N(s) \right) X_i^N(s) \right] ds \\ &= \frac{1}{N} \sum_{k=2}^i \left[\mathbb{E} [X_k^N(0)] - \mathbb{E} [X_k^N(t)] + \int_0^t \mathbb{E} \left[r \left(\frac{k}{N}, X_N^N(s) \right) X_k^N(s) \right] ds \right] \\ & \quad + \int_0^t \mathbb{E} \left[m \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) \right] ds. \end{aligned}$$

Since for any $1 \leq i \leq N$, the initial value $X_i^N(0)$ is non-negative, it follows that

$$\frac{1}{N} \sum_{k=2}^i \mathbb{E} [X_k^N(0)] \leq Y^N(0) = X_1^N(0) + \langle \mu_0, 1 \rangle + X_N^N(0).$$

This implies using (2.2.12) and (2.2.16) that

$$\begin{aligned} & \int_0^t \mathbb{E} \left[m \left(\frac{i}{N}, X_N^N(s) \right) X_i^N(s) \right] ds \\ & \leq \mathbb{E} [Y^N(0)] + \widehat{r} \mathbb{E} [Y^N(0)] \int_0^t \exp(\widehat{r}s) ds + \widehat{m} \mathbb{E} [X_1^N(0)] \int_0^t \exp(\widehat{r}s) ds \\ & \leq \left[\mathbb{E} [Y^N(0)] + \frac{\widehat{m}}{\widehat{r}} \mathbb{E} [X_1^N(0)] \right] \exp(\widehat{r}T). \end{aligned}$$

Since by Assumption 1, $m_{\min} \leq m(i/N, X_N^N(s))$, it yields that

$$\int_0^t \mathbb{E} [X_i^N(s)] ds \leq \frac{1}{m_{\min}} \left[\mathbb{E} [Y_1^N(0)] + \frac{\widehat{m}}{\widehat{r}} \mathbb{E} [X_1^N(0)] \right] \exp(\widehat{r}T).$$

This ends the proof. $\square$

2.3 Approximation for a large number of compartments

2.3.1 Tightness

The goal now is to study the triplet $(X_1^N(t), \mu_t^N, X_N^N(t))_{t \geq 0}$ as the number N of compartments goes to infinity. Each empirical measure μ_t^N lives in the space (M_F, w) . Some assumptions on the initial conditions are needed.

Assumption 2. Assume that

1. The initial condition $(X_1^N(0), \mu_0^N, X_N^N(0))$ is such that

$$\sup_N \mathbb{E} \left[\left(X_1^N(0) + \langle \mu_0^N, 1 \rangle + X_N^N(0) \right)^2 \right] < \infty.$$

2. The initial condition $(X_1^N(0), \mu_0^N, X_N^N(0))$ converges in law (and in probability) in $\mathbb{R}^+ \times (M_F, w) \times \mathbb{R}^+$ to the deterministic triplet (a_0, μ_0, z_0) , where $a_0 \in (0, \infty)$, $\mu_0 \in M_F$ and $z_0 \in [0, \infty)$.

Notice that under this assumption, the estimates obtained in Propositions [2.2.3](#) and [2.2.4](#) are now bounded uniformly in N .

This allows us to now show the tightness of the sequence of laws of the triplet (X_1^N, μ^N, X_N^N) on any finite time interval.

Proposition 2.3.1. *Under Assumptions [1](#) and [2](#) for each $T > 0$ the sequence of laws of the triplet $(X_1^N(t), \mu_t^N, X_N^N(t), t \in [0, T])_N$ is tight in $\mathcal{P}(\mathbb{D}([0, T], \mathbb{R}^+) \times \mathbb{D}([0, T], (M_F, w)) \times \mathbb{D}([0, T], \mathbb{R}^+))$.*

Proof. We want to show that

- the sequences of laws of $(X_1^N(t), t \in [0, T])_N$ and of $(X_N^N(t), t \in [0, T])_N$ are tight in $\mathcal{P}(\mathbb{D}([0, T]), \mathbb{R})$,
- the sequence of laws of the processes $(\mu_t^N, t \in [0, T])_N$ is tight in $\mathcal{P}(\mathbb{D}([0, T]), (M_F, w))$,

which would prove the proposition.

We know that $X_1^N(t)$ is a semimartingale given by

$$X_1^N(t) = X_1^N(0) + A_1^N(t) + M_1^N(t).$$

As shown by Joffe and Métivier [\[35\]](#), in order to prove tightness of the sequence $(\mathcal{L}(X_1^N))_N$ it is sufficient to show that the sequence $(\mathcal{L}(\sup_{t \leq T} |X_1^N(t)|))_N$ is tight and that A_1^N and $\langle M_1^N \rangle$ satisfy the Aldous condition found in Aldous [\[2\]](#).

Since $X_1^N(t) \geq 0$, then, for $\eta > 0$, by Markov's inequality

$$\mathbb{P} \left(\sup_{t \leq T} |X_1^N(t)| \geq \eta \right) \leq \frac{1}{\eta} \mathbb{E} \left[\sup_{t \leq T} |X_1^N(t)| \right] = \frac{1}{\eta} \mathbb{E} \left[\sup_{t \leq T} X_1^N(t) \right].$$

Using Proposition 2.2.3 and Assumption 2 this value goes uniformly to 0 when $\eta \rightarrow \infty$. Hence, the sequence of laws of $(\sup_{t \leq T} |X_1^N|)_N$ is tight.

Let $\delta > 0$ and σ, τ be two stopping times such that $\sigma \leq \tau \leq (\sigma + \delta) \wedge T$. From (2.2.7a), by Markov's inequality and using Assumption 2 we have

$$\begin{aligned} \mathbb{P}(|A_1^N(\sigma) - A_1^N(\tau)| > \eta) &\leq \frac{1}{\eta} \mathbb{E}(|A_1^N(\sigma) - A_1^N(\tau)|) \\ &= \frac{1}{\eta} \mathbb{E}\left[\left|\int_{\sigma}^{\tau} \left(r\left(\frac{1}{N}, X_N^N(s)\right) - m\left(\frac{1}{N}, X_N^N(s)\right)\right) X_1^N(s) ds\right|\right] \\ &\leq \frac{1}{\eta} (\hat{r} + \widehat{m}) \mathbb{E}\left[\int_{\sigma}^{\tau} X_1^N(s) ds\right] \leq \frac{\delta}{\eta} (\hat{r} + \widehat{m}) \sup_N \mathbb{E}\left[\sup_{s \leq T} X_1^N(s)\right]. \end{aligned}$$

By Proposition 2.2.3 and Assumption 2 this value tends to 0 when $\delta \rightarrow 0$ uniformly in N . Hence, the process A_1^N satisfy the Aldous condition. It's easy to prove in a similar way that this is also the case for $\langle M_1^N \rangle$. Then $\mathcal{L}(X_1^N)$ is tight in $\mathcal{P}(\mathbb{D}([0, T], \mathbb{R}))$.

On the other hand, for the measure μ_t we have that for any $f \in \mathcal{C}([0, 1])$

$$\langle \mu_t^N, f \rangle = \langle \mu_0^N, f \rangle + A^{N,f}(t) + M^{N,f}(t).$$

In order to prove tightness of the sequence of laws of μ^N in $\mathcal{P}(\mathbb{D}([0, T]), (M_F, w))$, as stated in Roelly-Coppoletta [66], it is sufficient to prove that for any $t \geq 0$ and each $f \in \mathcal{C}([0, 1])$, the sequence of laws of $\langle \mu^N, f \rangle$ is tight in $\mathbb{D}([0, T], \mathbb{R})$, and that $A^{N,f}$ and $\langle M^{N,f} \rangle$ satisfy the Aldous condition. We start by proving that for any continuous function f on $[0, 1]$,

$$\sup_N \mathbb{E}\left[\sup_{t \leq T} |\langle \mu_t^N, f \rangle|\right] < \infty.$$

This is an immediate consequence of Assumption 2 and Proposition 2.2.3 since $X_i^N(t)$ is non-negative for each $2 \leq i \leq N - 1$. It follows that

$$\sup_N \mathbb{E}\left[\sup_{t \leq T} |\langle \mu_t^N, f \rangle|\right] \leq \|f\|_{\infty} \sup_N \mathbb{E}\left[\sup_{t \leq T} \langle \mu_t^N, 1 \rangle\right].$$

Let σ and τ be two stopping times such that $\sigma \leq \tau \leq (\sigma + \delta) \wedge T$, then

$$\begin{aligned} \mathbb{E} \left[\left| \langle M^{N,f} \rangle_\sigma - \langle M^{N,f} \rangle_\tau \right| \right] &= \mathbb{E} \left[\left| \frac{1}{N} \left[\int_\sigma^\tau \langle \mu_s^N, (f - f(1))^2 r(\cdot, X_N^N(s)) \rangle ds \right. \right. \right. \\ &\quad \left. \left. + \frac{1}{N} \int_\sigma^\tau \langle \mu_s^N, (\Delta_{1/N} f) m(\cdot, X_N^N(s)) \rangle ds \right. \right. \\ &\quad \left. \left. + \left(f\left(\frac{2}{N}\right) - f(1) \right)^2 \int_\sigma^\tau m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds \right] \right| \right] \\ &\leq C \|f\|_\infty^2 \left[\mathbb{E} \left[\int_\sigma^\tau \langle \mu_s^N, 1 \rangle ds \right] + \mathbb{E} \left[\int_\sigma^\tau X_1^N(s) ds \right] \right] \\ &\leq \delta C \|f\|_\infty^2 \left[\mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} \langle \mu_s^N, 1 \rangle \right] + \mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} X_1^N(s) \right] \right]. \end{aligned} \tag{2.3.1}$$

Similarly, we find that

$$\mathbb{E} \left[\left| A^{N,f}(\tau) - A^{N,f}(\sigma) \right| \right] \leq \delta C \|f\|_\infty^2 \left[\mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} \langle \mu_s^N, 1 \rangle \right] + \mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} X_1^N(s) \right] \right]. \tag{2.3.2}$$

By Proposition [2.2.3](#), and given that f is bounded, $\langle M^{N,f} \rangle$ and $A^{N,f}$ both satisfy the Aldous condition. Hence, all the conditions for the collection of laws $(\mu_t^N, t \in [0, T])_N$ to be tight in $\mathcal{P}(\mathbb{D}([0, T]), (M_F, w))$ are satisfied.

Finally, we have that $X_N^N(t)$ is a semimartingale given by

$$X_N^N(t) = X_N^N(0) + A_N^N(t) + M^N(t).$$

As for X_1^N , to prove the tightness of the sequence $(\mathcal{L}(X_N^N))_N$, we will prove that the sequence $(\mathcal{L}(\sup_{t \leq T} |X_N^N(t)|))_N$ is tight and that A_N^N and $\langle M^N \rangle$ satisfy the Aldous condition.

The tightness of the sequence of laws of $(\sup_{t \leq T} |X_N^N(t)|)_N$ follows from Markov's inequality and previous estimates.

Similar computations can be done as the ones for $X_1^N(t)$ to find that for $\delta > 0$ and σ, τ two stopping times such that $\sigma \leq \tau \leq (\sigma + \delta) \wedge T$, we have

$$\begin{aligned} \mathbb{P} \left(|A_N^N(\sigma) - A_N^N(\tau)| > \eta \right) &\leq \frac{1}{\eta} \left[\mathbb{E} \left[\left| \int_\sigma^\tau m\left(\frac{N-1}{N}, X_N^N(s)\right) X_{N-1}^N(u) du \right| \right] \right. \\ &\quad \left. + \delta d \mathbb{E} \left[\sup_{\sigma \leq u \leq \tau} X_N^N(u) \right] \right], \end{aligned}$$

and

$$\mathbb{P} \left(\left| \langle M^N \rangle_\tau - \langle M^N \rangle_\sigma \right| > \eta \right) \leq \frac{1}{N} \frac{1}{\eta} \left[\mathbb{E} \left[\int_\sigma^\tau m \left(\frac{N-1}{N}, X_N^N(s) \right) X_{N-1}^N(u) \right] + \delta d \mathbb{E} \left[\sup_{\sigma \leq u \leq \tau} X_N^N(u) \right] \right].$$

We use equation (2.2.11) to deal with the term involving X_{N-1}^N , by studying the expectation of the equation. We observe that

$$\begin{aligned} & \mathbb{E} \left[\int_\sigma^\tau m \left(\frac{N-1}{N}, X_N^N(s) \right) X_{N-1}^N(s) ds \right] \\ & \leq \mathbb{E} \left[\left| \langle \mu_\sigma^N, 1 \rangle - \langle \mu_\tau^N, 1 \rangle \right| \right] + \mathbb{E} \left[\left| M^{N,1}(\tau) - M^{N,1}(\sigma) \right| \right] \\ & \quad + \mathbb{E} \left[\left| \int_\sigma^\tau \langle \mu_s^N, r(\cdot, X_N^N(s)) \rangle ds + \int_\sigma^\tau m \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds \right| \right]. \end{aligned} \quad (2.3.3)$$

We study each term in the right hand side individually. By (2.3.1) it yields that

$$\begin{aligned} \mathbb{E} \left[\left| M^{N,1}(\tau) - M^{N,1}(\sigma) \right| \right] & \leq \left(\mathbb{E} \left[\left| \langle M^{N,1} \rangle_\tau - \langle M^{N,1} \rangle_\sigma \right| \right] \right)^{1/2} \\ & \leq \delta^{1/2} C \left[\mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} \langle \mu_s^N, 1 \rangle \right] + \mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} X_1^N(s) \right] \right]^{1/2}. \end{aligned}$$

Using this argument and (2.3.2) it follows that

$$\begin{aligned} \mathbb{E} \left[\left| \langle \mu_\sigma^N, 1 \rangle - \langle \mu_\tau^N, 1 \rangle \right| \right] & \leq \mathbb{E} \left[\left| A^{N,1}(\tau) - A^{N,1}(\sigma) \right| \right] + \left(\mathbb{E} \left[\left| \langle M^{N,1} \rangle_\sigma - \langle M^{N,1} \rangle_\tau \right| \right] \right)^{1/2} \\ & \leq \delta C \left[\mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} \langle \mu_s^N, 1 \rangle \right] + \mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} X_1^N(s) \right] \right] \\ & \quad + \left(\delta C \left[\mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} \langle \mu_s^N, 1 \rangle \right] + \mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} X_1^N(s) \right] \right] \right)^{1/2}. \end{aligned}$$

Finally, by Assumption 1 we get that

$$\begin{aligned} & \mathbb{E} \left(\int_\sigma^\tau \langle \mu_s^N, r(\cdot, X_N^N(s)) \rangle ds + \int_\sigma^\tau m \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds \right) \\ & \leq \delta C \left[\mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} \langle \mu_s^N, 1 \rangle \right] + \mathbb{E} \left[\sup_{\sigma \leq s \leq \tau} X_1^N(s) \right] \right]. \end{aligned}$$

Hence, by gathering these results and using Proposition 2.2.3, we obtain that (2.3.3) tends to 0 when $\delta \rightarrow 0$ uniformly in N , which implies the same for $\mathbb{P} \left(\left| A_N^N(\tau) - A_N^N(\sigma) \right| > \eta \right)$ and $\mathbb{P} \left(\left| \langle M^N \rangle_\tau - \langle M^N \rangle_\sigma \right| > \eta \right)$. Hence, the processes A_N^N and $\langle M^N \rangle$ satisfy the Aldous condition. Then, $\mathcal{L}(X_N^N)$ is tight in $\mathcal{P}(\mathbb{D}([0, T], \mathbb{R}))$. This ends the proof. $\square$

2.3.2 Identification of the limit

The tightness of the sequence of laws of $(X_1^N, \mu_t^N, X_N^N)_N$ guarantees the existence of converging subsequences. Let us now identify the system satisfied by these limiting values. Recall that the initial condition (a_0, μ_0, z_0) defined in Assumption 2 is deterministic.

Proposition 2.3.2. *Under Assumptions 1-2, let $T > 0$ and consider a subsequence $(X_1^{N_k}, \mu^{N_k}, X_{N_k}^{N_k})$ that converges in law in $\mathbb{D}([0, T], \mathbb{R}^+) \times \mathbb{D}([0, T], (M_f, w)) \times \mathbb{D}([0, T], \mathbb{R}^+)$ to a limit (a, μ, z) as $N_k \rightarrow \infty$. Then, this limit is deterministic, belongs to $\mathcal{C}([0, T], \mathbb{R}^+) \times \mathcal{C}([0, T], M_F) \times \mathcal{C}([0, T], \mathbb{R}^+)$, and for any function $f \in \mathcal{C}^1([0, 1])$ and $t \in [0, T]$, it satisfies the system*

$$\left\{ \begin{array}{l} a(t) = a_0 + \int_0^t [r(0, z(s)) - m(0, z(s))] a(s) ds; \\ \langle \mu_t, f \rangle = f(0) \int_0^t m(0, z(s)) a(s) ds + \langle \mu_0, f \rangle + \int_0^t \langle \mu_s, f r(\cdot, z(s)) \rangle ds \\ \quad + \int_0^t \langle \mu_s, f' m(\cdot, z(s)) \rangle ds - f(1) b(t); \\ z(t) = z_0 + b(t) - d \int_0^t z(s) ds; \\ b(t) = \langle \mu_0, 1 \rangle - \langle \mu_t, 1 \rangle + \int_0^t \langle \mu_s, r(\cdot, z(s)) \rangle ds + \int_0^t m(0, z(s)) a(s) ds \\ \mu_t(\{0\}) = \mu_t(\{1\}) = 0 \quad \text{for almost every } t \in [0, T]. \end{array} \right. \quad (2.3.4)$$

Proof. Let $\mathbb{D}^3 = \mathbb{D}([0, T], \mathbb{R}^+) \times \mathbb{D}([0, T], (M_f, w)) \times \mathbb{D}([0, T], \mathbb{R}^+)$. Let us consider a limiting value (a, μ, z) in $\mathbb{D}^3$, which is the limit in law of the subsequence $(X_1^{N_k}, \mu^{N_k}, X_{N_k}^{N_k})$ when $N_k \rightarrow \infty$. By simplicity, we will denote this subsequence by (X_1^N, μ^N, X_N^N) . Let us first show that (a, μ, z) has continuous paths almost surely.

For $\nu \in \mathbb{D}([0, T], (M_F, w))$, $t \geq 0$ and $x \in \mathbb{D}([0, T], \mathbb{R}^+)$, we define $|\Delta \nu_t| = \|\nu_t - \nu_{t-}\|_{BL}$ and $|\Delta x(t)| = |x(t) - x(t-)|$. Notice that the application

$$(x, \nu, z) \in \mathbb{D}^3 \mapsto \sup_{t \leq T} (|\Delta x(t)|, |\Delta \nu_t|, |\Delta z(t)|) \in (\mathbb{R}^+)^3$$

is continuous. Moreover, we easily observe that

$$|\Delta X_1^N(t)| \leq \frac{1}{N}, \quad |\Delta \mu_t^N| \leq \frac{1}{N}, \quad |\Delta X_N^N(t)| \leq \frac{1}{N},$$

and then (a, μ, z) has continuous paths almost surely.

To prove that the triplet (a, μ, z) is a solution of System (2.3.4), let us consider $f \in \mathcal{C}^1([0, 1])$ and $T \geq 0$. We define for any $t \in [0, T]$, the functions $\Psi_t^i(x, \nu, y)$:

$\mathbb{D}^3 \rightarrow \mathbb{R}$, for $i = 1, 2, 3$ by

$$\begin{aligned}\Psi_t^1(x, \nu, y) &= x(t) - x_0 - \int_0^t (r(0, z(s)) - m(0, y)) x(s) ds ; \\ \Psi_t^2(x, \nu, y) &= \langle \nu_t, f \rangle - \langle \nu_0, f \rangle - \int_0^t \langle \nu_s, f r(\cdot, y) \rangle ds + \int_0^t \langle \nu_s, f' m(\cdot, y) \rangle ds \\ &\quad + f(0) \int_0^t m(0, y) x(s) ds \\ &\quad + f(1) \left[\langle \nu_0, 1 \rangle - \langle \nu_t, 1 \rangle + \int_0^t \langle \nu_s, r(\cdot, y) \rangle ds + \int_0^t m(0, y) x(s) ds \right] ; \\ \Psi_t^3(x, \nu, y) &= y(t) - y_0 - \langle \nu_0, 1 \rangle + \langle \nu_t, 1 \rangle - \int_0^t \langle \nu_s, r(\cdot, y) \rangle ds - \int_0^t m(0, y) x(s) ds \\ &\quad + d \int_0^t y(s) ds .\end{aligned}$$

We will prove that $\mathbb{E} [|\Psi_t^i(a, \mu, z)|] = 0$ for $i \in \{1, 2, 3\}$ and $t \in [0, T]$, which will give the desired result.

Since the sequence of triplets $(X_1^N, \mu^N, X_N^N)_N$ is $\mathcal{C}$ -tight, and since the functions Ψ^i , $i = 1, 2, 3$ are continuous on $\mathcal{C}([0, T], \mathbb{R}^+) \times \mathcal{C}([0, T], (M_F, w)) \times \mathcal{C}([0, T], \mathbb{R}^+)$, we can deduce that $\Psi_t(X_1^N, \mu^N, X_N^N)$ converges in law to $\Psi_t(a, \mu, z)$ as $N \rightarrow \infty$. Let us check that the sequences $(\Psi_t^i(X_1^N, \mu^N, X_N^N))_N$ are uniformly integrable to get convergence in L^1 .

By Assumption [2](#) and Proposition [2.3.1](#), and since $f \in \mathcal{C}^1([0, 1])$, it follows that for any N

$$\begin{aligned}|\Psi_t^1(X_1^N, \mu^N, X_N^N)| &\leq X_1^N(t) + X_1^N(0) + (\hat{r} + \widehat{m}) \int_0^t X_1^N(s) ds \\ &\leq C \sup_{s \leq T} X_1^N(s) ; \\ |\Psi_t^2(X_1^N, \mu^N, X_N^N)| &\leq C \|f\|_\infty \left[\langle \mu_t^N, 1 \rangle + \langle \mu_0^N, 1 \rangle + \int_0^t \langle \mu_s^N, 1 \rangle ds + \int_0^t X_1^N(s) ds \right] \\ &\leq C \|f\|_\infty \sup_{s \leq T} (X_1^N(s) + \langle \mu_s^N, 1 \rangle) ; \\ |\Psi_t^3(X_1^N, \mu^N, X_N^N)| &\leq X_N^N(t) + X_N^N(0) + \langle \mu_0, 1 \rangle + \langle \mu_t, 1 \rangle + \hat{r} \int_0^t \langle \mu_s, 1 \rangle ds \\ &\quad + \widehat{m} \int_0^t X_1^N(s) ds + d \int_0^t X_N^N(s) ds \\ &\leq C \sup_{s \leq T} (X_1^N(s) + \langle \mu_s^N, 1 \rangle + X_N^N(s)) .\end{aligned}$$

Then,

$$\begin{aligned}
\left| \Psi_t^1(X_1^N, \mu^N, X_N^N) \right| \mathbb{1}_{|\Psi_t^1(X_1^N, \mu^N, X_N^N)| > \eta} &\leq \frac{1}{\eta} \left| \Psi_t^1(X_1^N, \mu^N, X_N^N) \right|^2 \\
&\leq \frac{C}{\eta} \sup_{s \leq T} (X_1^N(s))^2 ; \\
\left| \Psi_t^2(X_1^N, \mu^N, X_N^N) \right| \mathbb{1}_{|\Psi_t^2(X_1^N, \mu^N, X_N^N)| > \eta} &\leq \frac{1}{\eta} \left| \Psi_t^2(X_1^N, \mu^N, X_N^N) \right|^2 \\
&\leq \frac{C}{\eta} \|f\|_\infty \sup_{s \leq T} (X_1^N(s) + \langle \mu_s^N, 1 \rangle)^2 ; \\
\left| \Psi_t^3(X_1^N, \mu^N, X_N^N) \right| \mathbb{1}_{|\Psi_t^3(X_1^N, \mu^N, X_N^N)| > \eta} &\leq \frac{1}{\eta} \left| \Psi_t^3(X_1^N, \mu^N, X_N^N) \right|^2 \\
&\leq \frac{C}{\eta} \sup_{s \leq T} (X_1^N(s) + \langle \mu_s^N, 1 \rangle + X_N^N(s))^2 .
\end{aligned}$$

From Proposition 2.2.3 and Assumption 2 and since f is bounded, the sequences $(\Psi_t^1(X_1^N, \mu^N, X_N^N))_N$, $(\Psi_t^2(X_1^N, \mu^N, X_N^N))_N$ and $(\Psi_t^3(X_1^N, \mu^N, X_N^N))_N$ are uniformly integrable for any $t \in [0, T]$.

Combining convergence in law and uniform integrability of the subsequences, we obtain that

$$\mathbb{E} \left[\left| \Psi_t^i(a, \mu, z) \right| \right] = \lim_{N \rightarrow \infty} \mathbb{E} \left[\left| \Psi_t^i(X_1^N, \mu^N, X_N^N) \right| \right] \leq \lim_{N \rightarrow \infty} \left(\mathbb{E} \left[\Psi_t^i(X_1^N, \mu^N, X_N^N)^2 \right] \right)^{1/2} .$$

Recognizing the martingales

$$\begin{aligned}
\Psi_t^1(X_1^N, \mu^N, X_N^N) &= M_1^N(t), \quad \Psi_t^2(X_1^N, \mu^N, X_N^N) = M^{N,f}(t), \\
\Psi_t^3(X_1^N, \mu^N, X_N^N) &= M^N(t),
\end{aligned}$$

from System (2.2.6), Proposition 2.2.2 shows that these martingales are of order $1/N$ and ensures the convergence of the expectations of each Ψ^i to 0 when $N \rightarrow \infty$. Then, we can finally conclude that any limiting value of the sequence $(X_1^N, \mu_t^N, X_N^N)_N$ is deterministic and is a solution of the system (2.3.4).

Recall that the subsequence $(X_1^N, \mu^N, X_N^N)_N$ converges to a limit $(a, \mu, z)_t$. We will now proceed to prove that for any limiting value, we have almost surely, almost everywhere in $t \in [0, T]$, $\mu_t(\{0\}) = 0$ and $\mu_t(\{1\}) = 0$.

Let $\varepsilon, \delta \in (0, 1)$ and $f_\varepsilon, f_\delta \in \mathcal{C}([0, 1])$ be two continuous positive functions such that

$$\begin{aligned}
f_\varepsilon(0) &= 1, \quad f_\varepsilon(x) = 0 \text{ for } x \geq \varepsilon, \quad \|f_\varepsilon\|_\infty \leq 1 ; \\
f_\delta(1) &= 1, \quad f_\delta(x) = 0 \text{ for } x \leq 1 - \delta, \quad \|f_\delta\|_\infty \leq 1 .
\end{aligned}$$

We study $(\langle \mu_t, f_\varepsilon \rangle)_{t \geq 0}$. Since by construction for any $2 \leq i \leq N-1$ such that $\varepsilon \leq i/N$ we have $f_\varepsilon(i/N) = 0$, then

$$\int_0^T \mathbb{E} [\langle \mu_s^N, f_\varepsilon \rangle] ds = \int_0^T \mathbb{E} \left[\frac{1}{N} \sum_{i=2}^{\lceil N\varepsilon \rceil} X_i^N(s) f_\varepsilon \left(\frac{i}{N} \right) \right] ds = \frac{1}{N} \sum_{i=2}^{\lceil N\varepsilon \rceil} \int_0^T \mathbb{E} [X_i^N(s)] ds.$$

Using Proposition 2.2.4 and Assumption 2 we obtain

$$\int_0^T \mathbb{E} [\langle \mu_s^N, f_\varepsilon \rangle] ds \leq \frac{C_T}{N} (\lceil N\varepsilon \rceil - 1) \leq \varepsilon C_T. \quad (2.3.5)$$

Since $\|f_\varepsilon\|_\infty \leq 1$, we have that

$$0 \leq \langle \mu_t^N, f_\varepsilon \rangle \leq \langle \mu_t^N, 1 \rangle,$$

which implies that

$$\int_0^t \langle \mu_s^N, f_\varepsilon \rangle ds \mathbb{1}_{\int_0^t \langle \mu_s^N, f_\varepsilon \rangle ds > \eta} \leq \frac{1}{\eta} \left(\int_0^t \langle \mu_s^N, f_\varepsilon \rangle ds \right)^2 \leq \frac{1}{\eta} \left(\int_0^t \langle \mu_s^N, 1 \rangle ds \right)^2.$$

As the right hand side has a bounded expectation with respect to N for all $t \in [0, T]$ by Proposition 2.2.3 and Assumption 2, we obtain the uniform integrability of the sequence $\left(\int_0^T \langle \mu_s^N, f_\varepsilon \rangle ds \right)_N$.

Using that μ^N converges in law to μ in $\mathbb{D}([0, T], (M_F, w))$ as N tends to infinity, then $\int_0^T \langle \mu_s^N, f_\varepsilon \rangle ds$ converges in law to $\int_0^T \langle \mu_s, f_\varepsilon \rangle ds$. By uniform integrability we get

$$\mathbb{E} \left[\int_0^T \langle \mu_s, f_\varepsilon \rangle ds \right] = \lim_{N \rightarrow \infty} \mathbb{E} \left[\int_0^T \langle \mu_s^N, f_\varepsilon \rangle ds \right].$$

By (2.3.5), it follows that

$$\mathbb{E} \left[\int_0^T \langle \mu_s, f_\varepsilon \rangle ds \right] \leq \varepsilon C_T.$$

Finally, when $\varepsilon \rightarrow 0$ we get by monotone limit

$$\int_0^T \mu_s(\{0\}) ds = \lim_{\varepsilon \rightarrow 0} \int_0^T \langle \mu_s, f_\varepsilon \rangle ds = 0 \quad a.s.$$

Hence, almost everywhere in t , we have

$$\mu_t(\{0\}) = 0.$$

By applying similar computations to $\langle \mu_u^N, f_\delta \rangle$, we find that

$$\int_0^T \mathbb{E} [\langle \mu_s^N, f_\delta \rangle] ds \leq \delta C_T.$$

Therefore we get that almost everywhere in t ,

$$\mu_t(\{1\}) = 0.$$

This proves the proposition. □

2.3.3 Uniqueness

Having found the characterization of the limiting system (2.3.4), we are interested in studying the uniqueness of its solution. We start by introducing a classical flow construction to adapt the characteristics approach which will allow us to prove uniqueness.

We consider $z \in \mathcal{C}^1(\mathbb{R}_+)$ and extend this function to $z \in \mathcal{C}^1(\mathbb{R})$ in such a way that it remains bounded with bounded derivatives. Then for any $(t, x) \in \mathbb{R}^2$, we consider the solution $M^z(\cdot, t, x)$ of the (non-autonomous) Cauchy problem defined for $s \in \mathbb{R}$ by

$$\frac{\partial M^z}{\partial s}(s, t, x) = m(M^z(s, t, x), z(s)), \quad M^z(t, t, x) = x, \quad (2.3.6)$$

where m has been extended to a function of $\mathcal{C}^1(\mathbb{R} \times \mathbb{R})$ in such a way that it remains bounded, with bounded derivative. The function $M^z(\cdot, t, x)$ represents the dynamics of the maturation of a cell whose level at time t is given by x . The function m is Lipschitz continuous in $x \in \mathbb{R}$ uniformly with respect to the other variable. Then we can use the classical Cauchy Lipschitz theory (see Hirsch and Smale [32] Chapter 15) and obtain existence and uniqueness of the Cauchy problem (2.3.6) on a maximal time interval. Adding that m is bounded, the solution cannot explode in finite time. Thus, M^z is well defined for any time and generates a flow in time and space on $\mathbb{R}^3$. This flow is also a diffeomorphism in space, which can be seen by reversing the flow in time. Observe that in our model, m is bounded on $\mathbb{R} \times \mathbb{R}$ and the flow is also a diffeomorphism in time. More precisely, let us gather these useful and classical properties in the following statement.

Proposition 2.3.3. *The following properties hold :*

i) For all $(t, x) \in \mathbb{R}^2$, System (2.3.6) admits a unique solution $M^z(\cdot, t, x)$ on $\mathbb{R}$. It defines an application $M^z : (s, t, x) \in \mathbb{R} \times \mathbb{R} \times \mathbb{R} \mapsto M^z(s, t, x) \in \mathbb{R}$.

ii) For all $t_1, t_2, t_3 \in \mathbb{R}$ and $x \in \mathbb{R}$,

$$M^z(t_1, t_3, x) = M^z(t_2, t_3, M^z(t_1, t_2, x)), \quad (2.3.7)$$

iii) The function M^z belongs to $\mathcal{C}^1(\mathbb{R}^3, \mathbb{R})$.

iv) For all $s, t \in \mathbb{R}$, the function $x \in \mathbb{R} \mapsto M^z(s, t, x) \in \mathbb{R}$ is a $\mathcal{C}^1$ -diffeomorphism.

v) For all $s \in \mathbb{R}$ and $x \in \mathbb{R}$, the function $t \in \mathbb{R} \mapsto M^z(s, t, x) \in \mathbb{R}$ is a $\mathcal{C}^1$ -diffeomorphism. The inverse function $(s, y, x) \in \mathbb{R}^3 \mapsto \kappa(s, y, x) \in \mathbb{R}$ satisfying $M^z(s, \kappa(s, y, x), x) = y$ on $\mathbb{R}^3$ belongs to $\mathcal{C}^1(\mathbb{R}^3, \mathbb{R})$.

vi) For any $(s, t, x) \in \mathbb{R}^3$,

$$\partial_s M^z(s, t, x) + m(x, z(s)) \partial_x M^z(s, t, x) = 0. \quad (2.3.8)$$

Proof. The existence and uniqueness of point i) for non-autonomous systems can be found in Chapter 15 of [32]. The composition property ii) can be found in Chapter 8 of [32] for autonomous systems and extends directly to the non-autonomous case. For global existence and uniqueness and for points iii) and iv) in non-autonomous systems, we refer to Chapter 4, Theorem 3.3 of Mouhot [53] which covers our framework. We prove that the function $t \in \mathbb{R} \mapsto M^z(s, t, x)$ is a diffeomorphism (first part of v)) by differentiating (2.3.7) with respect to t_2 and taking $t_2 = t_3$:

$$\frac{\partial M^z}{\partial t}(s, t, x) = -m(M^z(s, t, x), z(s)), \quad (2.3.9)$$

and using the strict positivity of the function m . Continuity of κ on $\mathbb{R}^3$ in point v) follows from the implicit function theorem since $M^z(t, s, x) - y$ belongs to $\mathcal{C}^1(\mathbb{R}^4, \mathbb{R})$ and $M^z(t, \kappa(s, y, x), x) - y = 0$ and the differential of $M^z(t, s, x) - y$ with respect to s is non zero due to the positivity of m . Finally, Theorem 3.6 in [53] proves vi). $\square$

We complement the above properties on the flow M by the following estimation, which will be useful for the proof of the uniqueness of the solution of (2.3.4).

Proposition 2.3.4. For any $z, \hat{z} \in \mathcal{C}(\mathbb{R}_+)$, we have

$$|M^z(s, t, x) - M^{\hat{z}}(s, t, x)| \leq \exp(L_m T) \int_s^t |z(u) - \hat{z}(u)| du,$$

for all $s, t \in [0, T]$ and $x \in [0, 1]$.

Proof. Using (2.3.6) and the Lipschitz continuity of the function m , we have

$$\begin{aligned} |M^z(s, t, x) - M^{\hat{z}}(s, t, x)| &= \left| \int_s^t \left(m(M^z(u, t, x), z(u)) - m(M^{\hat{z}}(u, t, x), \hat{z}(u)) \right) du \right| \\ &\leq L_m \int_s^t \left[|M^z(u, t, x) - M^{\hat{z}}(u, t, x)| + |z(u) - \hat{z}(u)| \right] du. \end{aligned}$$

Gronwall's Lemma allows to conclude. $\square$

We can then prove the uniqueness of the solution of the system (2.3.4).

Proposition 2.3.5. *Under Assumption 1, for any $T > 0$ and any $(a_0, \mu_0, z_0) \in \mathbb{R}_+ \times M_F \times \mathbb{R}_+$, for any function $f \in \mathcal{C}^1([0, 1])$, the system (2.3.4) has a unique solution $(a, \mu, z) \in \mathcal{C}([0, T], \mathbb{R}) \times \mathcal{C}([0, T], (M_F, w)) \times \mathcal{C}([0, T], \mathbb{R})$.*

Proof. Consider the measure that gathers the three elements of the triplet (a, μ, z) given for each $t \geq 0$ by

$$\nu_t = a(t)\delta_0 + \mu_t + z(t)\delta_1, \quad (2.3.10)$$

where $a(t)$, μ_t , $z(t)$ are defined by System (2.3.4).

Then, for any $f \in \mathcal{C}^1([0, 1])$ and any $t \geq 0$

$$\begin{aligned} \langle \nu_t, f \rangle &= f(0) a(t) + \langle \mu_t, f \rangle + f(1) z(t) \\ &= \langle \nu_0, f \rangle + \int_0^t f(0) r(0, z(s)) a(s) ds + \int_0^t \langle \mu_s, f r(\cdot, z(s)) \rangle ds \\ &\quad + \int_0^t \langle \mu_s, f' m(\cdot, z(s)) \rangle ds - d \int_0^t f(1) z(s) ds. \end{aligned}$$

For every $t \in \mathbb{R}_+$, we can extend the last equation by classical arguments to a test function $\phi_t(s, x) \in \mathcal{C}^1([0, t] \times [0, 1], \mathbb{R})$ as follows:

$$\begin{aligned} \langle \nu_t, \phi_t(t, \cdot) \rangle &= \langle \nu_0, \phi_t(0, \cdot) \rangle + \int_0^t \phi_t(s, 0) r(0, z(s)) a(s) ds + \int_0^t \langle \mu_s, \phi_t(s, \cdot) r(\cdot, z(s)) \rangle ds \\ &\quad + \int_0^t \langle \mu_s, m(\cdot, z(s)) \partial_x \phi_t(s, \cdot) + \partial_s \phi_t(s, \cdot) \rangle ds - d \int_0^t \phi_t(s, 1) z(s) ds. \end{aligned}$$

Let $f \in \mathcal{C}^1([0, 1])$. We extend f to a function $\tilde{f} \in \mathcal{C}_b^1(\mathbb{R})$. We define the test function ϕ for $t \geq 0$ and $x \in [0, 1]$ by $\phi_t(s, x) = \tilde{f}(M^z(t, s, x))$. Using point vi) of Proposition 2.3.3, we can observe that for any $s, t \in \mathbb{R}$ and $x \in [0, 1]$,

$$\phi_t(t, x) = f(x), \quad m(x, z(s)) \partial_x \phi_t(s, x) + \partial_s \phi_t(s, x) = 0.$$

Therefore, we obtain that $(\nu_t)_t$ is solution of the mild equation given by

$$\begin{aligned} \langle \nu_t, f \rangle &= \langle \nu_t, \tilde{f} \rangle = \langle \nu_0, \tilde{f}(M^z(t, 0, \cdot)) \rangle + \int_0^t \tilde{f}(M^z(t, s, 0)) r(0, z(s)) a(s) ds \\ &\quad + \int_0^t \langle \mu_s, \tilde{f}(M^z(t, s, \cdot)) r(\cdot, z(s)) \rangle ds - d \int_0^t \tilde{f}(M^z(t, s, 1)) z(s) ds. \end{aligned} \quad (2.3.11)$$

Let us now consider $g \in BL_1([0, 1])$. Then g can be extended on $\mathbb{R}$ to a bounded Lipschitz continuous function $\tilde{g} \in BL_1(\mathbb{R})$. Since any function in $BL_1(\mathbb{R})$ is the limit of a sequence of $\mathcal{C}_b^1(\mathbb{R})$ -functions for the bounded pointwise convergence, (take for example convoluted functions of f by smooth mollifiers), the mild equation above is also satisfied by any function $g \in BL_1$ and its bounded Lipschitz continuous extension $\tilde{g}$. Let us prove that the mild equation (2.3.11) has a unique solution (a, μ, z) in $\mathcal{C}([0, T], \mathbb{R}) \times \mathcal{C}([0, T], (M_F, w)) \times \mathcal{C}([0, T], \mathbb{R})$ which satisfies that a.e. in t , $\mu_t(\{0\}) = \mu_t(\{1\}) = 0$.

Let (a, μ, z) and $(\hat{a}, \hat{\mu}, \hat{z})$ be two solutions of System (2.3.4) and $\nu_t, \hat{\nu}_t$ as defined in (2.3.10) with $\nu_0 = \hat{\nu}_0$. Using (2.3.11), it follows that

$$\begin{aligned} |\langle \nu_t - \hat{\nu}_t, \tilde{g} \rangle| &\leq \left| \langle \nu_0, \tilde{g}(M^z(t, 0, \cdot)) - \tilde{g}(M^{\hat{z}}(t, 0, \cdot)) \rangle \right| \\ &\quad + \int_0^t \left| \tilde{g}(M^z(t, s, 0)) r(0, z(s)) a(s) - \tilde{g}(M^{\hat{z}}(t, s, 0)) r(0, \hat{z}(s)) \hat{a}(s) \right| ds \\ &\quad + \int_0^t \left| \langle \mu_s, \tilde{g}(M^z(t, s, \cdot)) r(\cdot, z(s)) \rangle - \langle \hat{\mu}_s, \tilde{g}(M^{\hat{z}}(t, s, \cdot)) r(\cdot, \hat{z}(s)) \rangle \right| ds \\ &\quad + d \int_0^t \left| \tilde{g}(M^z(t, s, 1)) z(s) - \tilde{g}(M^{\hat{z}}(t, s, 1)) \hat{z}(s) \right| ds. \end{aligned}$$

We study individually the different terms on the right hand side. For the first term, given Assumption 2, the Lipschitz continuity of g and the estimation given in Proposition 2.3.4 we derive that

$$\begin{aligned} \left| \langle \nu_0, \tilde{g}(M^z(t, 0, \cdot)) - \tilde{g}(M^{\hat{z}}(t, 0, \cdot)) \rangle \right| &\leq |\langle \nu_0, 1 \rangle| \sup_{x \in [0, 1]} \left| \tilde{g}(M^z(t, 0, x)) - \tilde{g}(M^{\hat{z}}(t, 0, x)) \right| \\ &\leq C_T \int_0^t |z(s) - \hat{z}(s)| ds. \end{aligned}$$

Using the Lipschitz continuity of r and m (Assumption 1), the Lipschitz continuity of

$\tilde{g}$, and Proposition 2.3.4, we prove that the second term is bounded as follows

$$\begin{aligned} & \left| \int_0^t \tilde{g}(M^z(t, s, 0)) r(0, z(s)) a(s) - \tilde{g}(M^{\hat{z}}(t, s, 0)) r(0, \hat{z}(s)) \hat{a}(s) ds \right| \\ &= \int_0^t \left| \tilde{g}(M^z(t, s, 0)) r(0, z(s)) (a(s) - \hat{a}(s)) \right. \\ &\quad \left. + \hat{a}(s) r(0, z(s)) (\tilde{g}(M^z(t, s, 0)) - \tilde{g}(M^{\hat{z}}(t, s, 0))) \right. \\ &\quad \left. + \hat{a}(s) \tilde{g}(M^{\hat{z}}(t, s, 0)) (r(0, z(s)) - r(0, \hat{z}(s))) \right| ds \\ &\leq C_T \|\tilde{g}\|_\infty \left[\int_0^t |a(s) - \hat{a}(s)| ds + \int_0^t |z(s) - \hat{z}(s)| ds \right]. \end{aligned}$$

By similar arguments, we obtain

$$\begin{aligned} & \left| \int_0^t \left(\langle \mu_s, \tilde{g}(M^z(t, s, \cdot)) r(\cdot, z(s)) \rangle - \langle \hat{\mu}_s, \tilde{g}(M^{\hat{z}}(t, s, \cdot)) r(\cdot, \hat{z}(s)) \rangle \right) ds \right| \\ &\leq \int_0^t \left| \left(\langle \mu_s - \hat{\mu}_s, \tilde{g}(M^z(t, s, \cdot)) r(\cdot, z(s)) \rangle \right. \right. \\ &\quad \left. + \langle \hat{\mu}_s, (\tilde{g}(M^z(t, s, \cdot)) - \tilde{g}(M^{\hat{z}}(t, s, \cdot))) r(\cdot, z(s)) \rangle \right. \\ &\quad \left. + \langle \hat{\mu}_s, \tilde{g}(M^{\hat{z}}(t, s, \cdot)) (r(\cdot, z(s)) - r(\cdot, \hat{z}(s))) \rangle \right) ds \\ &\leq C_T \|\tilde{g}\|_\infty \int_0^t (|\langle \mu_s - \hat{\mu}_s, 1 \rangle| + |z(s) - \hat{z}(s)|) ds. \end{aligned}$$

Finally, by Lipschitz continuity of $\tilde{g}$ and using Proposition 2.3.4, the last term satisfies

$$\left| \int_0^t (\tilde{g}(M^z(t, s, 1)) z(s) - \tilde{g}(M^{\hat{z}}(t, s, 1)) \hat{z}(s)) ds \right| \leq C_T \int_0^t |z(s) - \hat{z}(s)| ds.$$

Then, by gathering all these results, we conclude that

$$\|\nu_t - \hat{\nu}_t\|_{BL} \leq C_T \|\tilde{g}\|_\infty \int_0^t \|\nu_s - \hat{\nu}_s\|_{BL} ds.$$

Hence, by boundedness of $\tilde{g}$, Gronwall's Lemma ensures that $\|\nu_t - \hat{\nu}_t\|_{BL} = 0$. Since for almost all $t \in [0, T]$, the measures μ_t and $\hat{\mu}_t$ don't charge the two singletons $\{0\}$ and $\{1\}$, we have $a(t) = \hat{a}(t)$ and $z(t) = \hat{z}(t)$ for all t (invoking also the continuity of the functions) and thus $\mu_t = \hat{\mu}_t$ for all $t \in [0, T]$. Uniqueness of the solution of (2.3.4) is proved. $\square$

We can then conclude to the convergence of the system $(X_1^N, \mu^N, X_N^N)_N$ as stated in the following theorem.

Theorem 2.3.6. *Under Assumptions 1-2, for each $T > 0$, the sequence of processes $(X_1^N, \mu^N, X_N^N)_N$ converges in probability in $\mathbb{D}([0, T], \mathbb{R}_+) \times \mathbb{D}([0, T], (M_F, w)) \times \mathbb{D}([0, T], \mathbb{R}_+)$ to the unique deterministic triplet $(a, \mu, z) \in \mathcal{C}([0, T], \mathbb{R}) \times \mathcal{C}([0, T], (M_F, w)) \times \mathcal{C}([0, T], \mathbb{R})$ satisfying (2.3.4).*

2.4 Existence of a density

In this section, we assume that the initial condition μ_0 admits a density with respect to the Lebesgue measure on $[0, 1]$. We will prove that this property propagates over time. We introduce the following assumption.

Assumption 3. Assume that the limiting initial measure μ_0 is absolutely continuous with respect to the Lebesgue measure on $[0, 1]$.

Then, the following result holds.

Proposition 2.4.1. Under Assumptions 1, 2, 3, for any $t \in [0, T]$, the measure μ_t defined in System (2.3.4) admits a density $u(t, \cdot)$ with respect to the Lebesgue measure.

Proof. Consider the measure that gathers the immature and mature cells given by

$$\omega_t = \mu_t + z(t)\delta_1.$$

Then, using (2.3.4) we have that for any function $f \in \mathcal{C}^1([0, 1])$

$$\begin{aligned} \langle \omega_t, f \rangle &= \langle \omega_0, f \rangle + \int_0^t f(0) m(0, z(s)) a(s) ds + \int_0^t \langle \mu_s, f r(\cdot, z(s)) \rangle ds \\ &\quad + \int_0^t \langle \mu_s, f' m(\cdot, z(s)) \rangle ds - d \int_0^t f(1) z(s) ds. \end{aligned}$$

Similarly as for (2.3.11), we consider the flow (2.3.6) and study the measure ω_t with the test function $\phi(s, x) = \tilde{f}(M^z(s, t, x))$ where $\tilde{f}$ is an extension of f in $\mathcal{C}_b^1(\mathbb{R})$. Then, we obtain that ω_t is a solution of

$$\begin{aligned} \langle \omega_t, f \rangle &= \langle \omega_t, \tilde{f} \rangle \tag{2.4.1} \\ &= \langle \omega_0, \tilde{f}(M^z(t, 0, \cdot)) \rangle + \int_0^t \tilde{f}(M^z(t, s, 0)) m(0, z(s)) a(s) ds \\ &\quad + \int_0^t \langle \mu_s, \tilde{f}(M^z(t, s, \cdot)) r(\cdot, z(s)) \rangle ds - d \int_0^t \tilde{f}(M^z(t, s, 1)) z(s) ds. \tag{2.4.2} \end{aligned}$$

We study individually the terms in the right hand side.

Assumption 3 implies the existence of a measurable function u_0 such that $\mu_0(dx) = u_0(x)dx$. We use now that $x \in (0, 1) \rightarrow M^z(t, 0, x) \in (M^z(t, 0, 0), M^z(t, 0, 1))$ is a C^1 diffeomorphism (for any fixed t) with inverse function denoted by $h(t, \cdot)$, see Proposition 2.3.3. We deduce that for each $t \in [0, T]$,

$$\begin{aligned} \langle \mu_0, \tilde{f}(M^z(t, 0, \cdot)) \rangle &= \int_{(0,1)} u_0(x) \tilde{f}(M^z(t, 0, x)) dx \\ &= \int_{(M^z(t,0,0), M^z(t,0,1))} u_0(h(t, y)) \tilde{f}(y) \frac{\partial}{\partial y} h(t, y) dy. \end{aligned}$$

Moreover

$$\begin{aligned} \int_0^t \tilde{f}(M^z(t, s, 0)) m(0, z(s)) a(s) ds = \\ - \int_0^{M^z(t, 0, 0)} \tilde{f}(y) m(0, z(\tau(t, y))) a(\tau(t, y)) \frac{\partial}{\partial y} \tau(t, y) dy \end{aligned}$$

where $s \rightarrow M^z(t, s, 0)$ is a C^1 -diffeomorphism from $(0, t)$ into its image by Proposition 2.3.3 and $\tau(t, \cdot)$ denotes its inverse function. Notice that the bounds of the integration interval have been exchanged since for any $t, x \in \mathbb{R}$, the application $s \mapsto M^z(t, s, x)$ is decreasing (by (2.3.9) since m is positive).

We now justify a change of variables for the term $\int_0^t \langle \mu_s, \tilde{f}(M^z(t, s, \cdot)) r(\cdot, z(s)) \rangle ds$. We have defined in Proposition 2.3.3 the inverse function $y \rightarrow \kappa(t, x, y)$ of $s \rightarrow M^z(t, s, x)$, and κ belongs to $\mathcal{C}^1(\mathbb{R}^3, \mathbb{R})$. Note also that for any $(t, x) \in [0, T] \times [0, 1]$ and any $y \in [x, M^z(t, s, x)]$, we have $\kappa(t, y, x) \in [0, t]$.

We will use indifferently the notations $\mu_s(dx)$ or $\mu(s, dx)$. For a fixed $t \in [0, T]$, we define $\tilde{\mu}$ by

$$\tilde{\mu}(y, dx) = \mu(\kappa(t, y, x), dx).$$

We will now show that

$$\begin{aligned} \int_0^t \int_0^1 \tilde{f}(M^z(t, s, x)) r(x, z(s)) \mu(s, dx) ds \\ = - \int_{\mathbb{R}} \int_0^1 \mathbb{1}_{x \leq y \leq M^z(t, 0, x)} \tilde{f}(y) r(x, z(\kappa(t, y, x))) \frac{\partial}{\partial y} \kappa(t, y, x) \tilde{\mu}(y, dx) dy. \end{aligned} \quad (2.4.3)$$

In that aim, we will use approximating discrete measures. By Lemma A.2 in Appendix, given that the functions f , M^z , r , z are all continuous and the composition of continuous functions remains continuous, it follows that

$$\begin{aligned} \int_0^t \int_0^1 \tilde{f}(M^z(t, s, x)) r(x, z(s)) \mu(s, dx) ds = \\ \lim_{k \rightarrow \infty} \int_0^t \int_0^1 \tilde{f}(M^z(t, s, x)) r(x, z(s)) \mu^k(s, dx) ds \end{aligned}$$

where μ^k is defined as in Appendix (2.6.1) by

$$\mu^k(s, dx) = \sum_{i \in \mathbb{Z}} \mu \left(s, \left(\frac{i-1}{k}, \frac{i}{k} \right] \right) \delta_{\frac{i}{k}}.$$

For a fixed $k \in \mathbb{N}$, for each $0 \leq i \leq k$ and $x = i/k$, we consider the change of variable $y = M^z(t, s, x)$ ($s = \kappa(t, x, y)$) to obtain

$$\begin{aligned} & \int_0^t \tilde{f}(M^z(t, s, x)) r(x, z(s)) \mu\left(s, \left(x - \frac{1}{k}, x\right]\right) ds \\ &= - \int_{\mathbb{R}} \mathbb{1}_{x \leq y \leq M^z(t, 0, x)} \tilde{f}(y) r(x, z(\kappa(t, y, x))) \frac{\partial}{\partial y} \kappa(t, y, x) \mu\left(\kappa(t, y, x), \left(x - \frac{1}{k}, x\right]\right) dy. \end{aligned}$$

Then, if we sum over all $0 \leq i \leq k$, it follows that

$$\begin{aligned} & \int_0^t \int_0^1 \tilde{f}(M^z(t, s, x)) r(x, z(s)) \mu^k(s, dx) ds \\ &= - \int_{\mathbb{R}} \int_0^1 \mathbb{1}_{x \leq y \leq M^z(t, 0, x)} \tilde{f}(y) r(x, z(\kappa(t, y, x))) \frac{\partial}{\partial y} \kappa(t, y, x) \tilde{\mu}^k(y, dx) dy. \end{aligned}$$

Then, by continuity of $\kappa(t, y, x)$ with respect to (y, x) (Proposition 2.3.3 v), Lemma A.2 in Appendix yields that

$$\begin{aligned} & \int_0^t \int_0^1 \tilde{f}(M^z(t, s, x)) r(x, z(s)) \mu(s, dx) ds \\ &= \lim_{k \rightarrow \infty} \int_0^t \int_0^1 \tilde{f}(M^z(t, s, x)) r(x, z(s)) \mu^k(s, dx) ds \\ &= - \lim_{k \rightarrow \infty} \int_{\mathbb{R}} \int_0^1 \mathbb{1}_{x \leq y \leq M^z(t, 0, x)} \tilde{f}(y) r(x, z(\kappa(t, y, x))) \frac{\partial}{\partial y} \kappa(t, y, x) \tilde{\mu}^k(y, dx) dy \\ &= - \int_{\mathbb{R}} \int_0^1 \mathbb{1}_{x \leq y \leq M^z(t, 0, x)} \tilde{f}(y) r(x, z(\kappa(t, y, x))) \frac{\partial}{\partial y} \kappa(t, y, x) \tilde{\mu}(y, dx) dy. \end{aligned}$$

We gather the results of each term of (2.4.1) to conclude that

$$\int_0^1 \tilde{f}(x) \omega_t(dx) = f(M^z(t, 0, 1)) z_0 + \int_{\mathbb{R}} \tilde{f}(y) G(y, t) dy,$$

where

$$\begin{aligned} G(y, t) &= \mathbb{1}_{M^z(t, 0, 0) < y < M^z(t, 0, 1)} u_0(h(t, y)) \frac{\partial}{\partial y} h(t, y) \\ &\quad - \mathbb{1}_{0 \leq y \leq M^z(t, 0, 0)} m(0, z(\tau(t, y))) a(\tau(t, y)) \frac{\partial}{\partial y} \tau(t, y) \\ &\quad - \int_0^1 \mathbb{1}_{x \leq y \leq M^z(t, 0, x)} r(x, z(\kappa(t, y, x))) \frac{\partial}{\partial y} \kappa(t, y, x) \tilde{\mu}(y, dx). \end{aligned}$$

Let $f \in \mathcal{C}([0, 1])$ be a function such that $f(1) = 0$. We extend it to a function $\tilde{f} \in \mathcal{C}_b^1(\mathbb{R})$ such that $\tilde{f}(x) = 0$ for any $x \in [1, \infty)$. Notice that $M^z(0, t, 1) \geq 1$ since

$M^z(0, 0, 1) = 1$ and by definition in (2.3.6), it is strictly increasing with respect to t since m is positive by Assumption 1. Then, we obtain

$$\int_0^1 \tilde{f}(x) \nu_t(dx) = \int_0^1 \tilde{f}(x) \mu_t(dx) = \int_{\mathbb{R}} \tilde{f}(y) G(y, t) dy.$$

Thus, we conclude that for every $t \in [0, T]$, $\mu_t(dx)$ admits on $[0, 1]$ the density $u(t, x) = G(x, t)$. $\square$

Let us now prove that the triplet (a, u, z) is weak solution of a partial differential equations system (PDES).

Proposition 2.4.2. *Under Assumptions 1, 2, 3, the function (a, u, z) is weak solution of the PDES given by*

$$\frac{d}{dt}a(t) = [r(0, z(t)) - m(0, z(t))]a(t) \quad (2.4.4a)$$

$$\partial_t u(x, t) + \partial_x [m(x, z(t)) u(x, t)] = r(x, z(t)) u(x, t) \quad (2.4.4b)$$

$$\frac{d}{dt}z(t) = m(1, z(t)) u(1, t) - dz(t), \quad (2.4.4c)$$

for $t \geq 0$ and $x \in [0, 1]$, with boundary conditions

$$\begin{aligned} u(0, t) &= a(t) \\ u(1, t) &= \frac{1}{m(1, z(t))} \left[\int_0^1 \frac{d}{dt} u(x, t) dx - \int_0^1 r(x, z(t)) u(x, t) dx - m(0, z(t)) a(t) \right], \end{aligned}$$

and initial conditions $(a_0, z_0) \in (\mathbb{R}_+)^2$.

Proof. By writing $\mu_t(dx) = u(x, t) dx$ in System (2.3.4) and applying integration by

parts to the term involving f' , it follows that

$$\begin{aligned}
& \int_0^1 f(x) u(x, t) dx \\
&= f(0) \int_0^t m(0, z(s)) a(s) ds + \int_0^1 f(x) u(x, 0) dx + \int_0^t \int_0^1 f(x) r(x, z(s)) u(x, s) ds \\
&+ \int_0^t \int_0^1 f'(x) m(x, z(s)) u(x, s) ds \\
&- f(1) \left[\int_0^1 u(x, 0) dx - \int_0^1 u(x, t) dx + \int_0^t \int_0^1 r(x, z(s)) u(x, s) ds \right. \\
&\quad \left. + \int_0^t m(0, z(s)) a(s) ds \right] \\
&= \int_0^1 f(x) u(x, 0) dx + \int_0^t \int_0^1 f(x) r(x, z(s)) u(x, s) ds \\
&- \int_0^t \int_0^1 f(x) \partial_x (m(x, z(s)) u(x, s)) ds \\
&+ f(0) \left[\int_0^t m(0, z(s)) a(s) ds - \int_0^t m(0, z(s)) u(0, s) ds \right] \\
&- f(1) \left[\int_0^1 u(x, 0) dx - \int_0^1 u(x, t) dx + \int_0^t \int_0^1 r(x, z(s)) u(x, s) ds \right. \\
&\quad \left. + \int_0^t m(0, z(s)) a(s) ds + \int_0^t m(1, z(s)) u(1, s) ds \right].
\end{aligned}$$

In order to obtain a PDE we need to set the following boundary conditions for any $t \in [0, T]$

$$\begin{aligned}
\int_0^t m(0, z(s)) a(s) ds &= \int_0^t m(0, z(s)) u(0, s) ds \\
\int_0^t m(1, z(s)) u(1, s) ds &= - \int_0^1 u(x, 0) dx + \int_0^1 u(x, t) dx \\
&\quad - \int_0^t \int_0^1 r(x, z(s)) u(x, s) ds - \int_0^t m(0, z(s)) a(s) ds.
\end{aligned}$$

Then, we obtain the equation for $u(x, t)$ given by

$$\partial_t u(x, t) + \partial_x [m(x, z(t)) u(x, t)] = r(x, z(t)) u(x, t).$$

Finally, using the fact that by assumption $0 < m_{\min} \leq m(x, s)$ for any $(x, s) \in$

$[0, 1] \times \mathbb{R}^+$, we have that

$$\begin{aligned} \int_0^t m(0, z(s)) a(s) ds &= \int_0^t m(0, z(s)) u(0, s) ds \\ \implies \int_0^t m(0, z(s)) (a(s) - u(0, s)) ds &= 0 \\ \implies m(0, z(s)) (a(s) - u(0, s)) &= 0, \quad \forall s \leq t, \\ \implies a(s) - u(0, s) &= 0, \quad \forall s \leq t. \end{aligned}$$

This ends the proof. □

Remark: This system is equivalent to the one found by [23], and our result proves as corollary the uniqueness of the system obtained in [23], and subsequently to conclude a convergence result. Indeed, their limiting system writes

$$\begin{aligned} \frac{d}{dt} a(t) &= [2\alpha(z(t)) - 1] r(0, z(t)) a(t) \\ \partial_t u(x, t) + \partial_x [m(x, z(t)) u(x, t)] &= r(x, z(t)) u(x, t) \\ m(0, z(t)) u(0, t) &= 2[1 - \alpha(z(t))] r(0, z(t)) a(t) \\ \frac{d}{dt} z(t) &= m(1, z(t)) u(1, t) - d z(t), \end{aligned} \tag{2.4.5}$$

where $u(x, t)$ corresponds to the density of cells of maturation stage x at time t and $\alpha(z(t))$ is the fraction of progeny that stays at stem cell level. The term $[2\alpha(z(t)) - 1] r(0, z(t))$ gathers all the changes in size in the stem cell population, which is equivalent in our case to $r(0, z(t)) - m(0, z(t))$. The term $2[1 - \alpha(z(t))] r(0, z(t))$ corresponds to the differentiation rate of the stem cell population, which we consider as $m(0, z(t))$. Finally, the weak formulation of the PDE (2.4.5) in the system allows us to find the boundary condition for $m(1, z(t)) u(1, t)$. For a function $f \in \mathcal{C}^1([0, 1])$ we have that

$$\begin{aligned} \int_0^1 u(x, t) f(x) dx &= \int_0^1 u(x, 0) dx - \int_0^t \int_0^1 \partial_x [m(x, z(s)) u(x, s)] f(x) dx ds \\ &\quad + \int_0^t \int_0^1 r(x, z(s)) u(x, s) f(x) dx ds \\ &= \int_0^t f(0) m(0, z(s)) u(0, s) - f(1) m(1, z(s)) u(1, s) ds \\ &\quad + \int_0^t \int_0^1 m(x, z(s)) u(x, s) f'(x) dx ds \\ &\quad + \int_0^t \int_0^1 r(x, z(s)) u(x, s) f(x) dx ds. \end{aligned}$$

Then,

$$\begin{aligned} \int_0^1 f(1) m(1, z(s)) u(1, s) ds = \\ \int_0^1 u(x, 0) dx - \int_0^1 u(x, t) f(x) dx + \int_0^t f(0) m(0, z(s)) u(0, s) ds \\ + \int_0^t \int_0^1 m(x, z(s)) u(x, s) f'(x) dx ds + \int_0^t \int_0^1 r(x, z(s)) u(x, s) f(x) dx ds. \end{aligned}$$

For $f \equiv 1$, we obtain

$$\begin{aligned} \int_0^1 m(1, z(s)) u(1, s) ds = \int_0^1 u(x, 0) dx - \int_0^1 u(x, t) dx + \int_0^t m(0, z(s)) u(0, s) ds \\ + \int_0^t \int_0^1 r(x, z(s)) u(x, s) dx ds, \end{aligned}$$

Remark: When μ_t admits a density, the equality of System (2.3.4) given by

$$\mu_t(\{0\}) = \mu_t(\{1\}) = 0 \quad \text{for almost every } t \in [0, T]$$

is valid for all $t \in [0, T]$ as the measure μ_t is dominated by the Lebesgue measure.

2.5 Numerical simulation

We performed numerical simulations for both the initial System (2.2.1) with different numbers of compartments ($N = 50, 200$) and the limiting System (2.4.4) under the existence of a density. The stochastic system is simulated by directly drawing event counts from Poisson processes following the stochastic differential system (2.2.1) where time has been discretized. For the PDE part of the deterministic system, the numerical scheme is based on the following rules: the derivatives are approximated using finite backward differences, the PDE is approximated using an upwind method as seen in page 73 of [43] and the integrals are approximated using Riemann sums.

Our aim in these simulations is to illustrate the dynamics of the amplification through time as the maturity level increases. To do so, we started with an initial condition with only very few HSCs. Given the lack of available data on the differentiation and division rates, we considered arbitrary constant differentiation and proliferation functions m and r . We then set $m = 0.02$, $r = 0.015$ and $d = 0.005$ for the simulations and consider an initial condition of 50 stem cells.

In Figure 2.1 and Figure 2.2 we observe the distribution through time of immature cells depending on the maturation level for both systems. Figure 2.2 also shows the dynamics of the mature cells population. Figure 2.1 illustrates the average of 50 simulations of the discrete system with $N = 50$ and $N = 200$ compartments while Figure

[2.2](#) illustrates the deterministic system. In both figures, we observe a propagation front that progressively fills all maturation stages and we observe an amplification in the number of cells as the maturation level grows, which is coherent with biological observations. The dynamics of the mature cells population in Figure [2.2](#) further exemplifies the amplification as the number of mature cells is much higher than the one of stem cells.

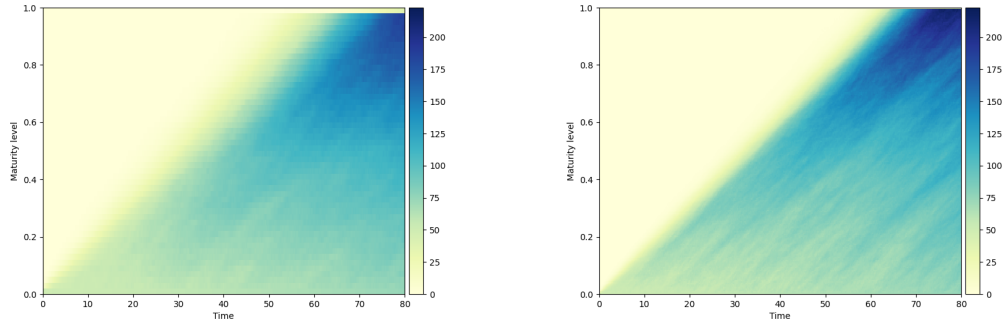


Figure 2.1: Dynamics of immature cells populations by their maturity level for System [\(2.2.4a\)](#) for 50 (left) and 200 (right) compartments.

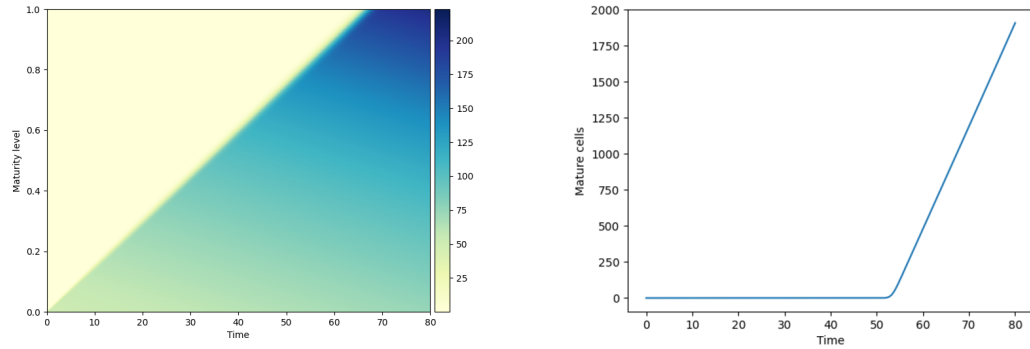


Figure 2.2: Left: dynamics of immature cells populations by their maturity level for System [\(2.4.4\)](#). Right: dynamics of the mature cells population size for System [\(2.4.4\)](#).

2.6 Discussion

In this paper, we presented a stochastic compartmental differential system with continuous time to model hematopoiesis at steady-state with a regulation loop depending

only on the number of mature cells. As the goal was to create a continuum in maturity level, we proposed a model where differentiation and division are two separate events, which differs with standard modeling (see, for example, [15], [22], [40], [51] or [73]) where cells can only differentiate during a division. Note that in our model, cells differentiate to the immediate next maturation stage which corresponds to the biological context of steady-state hematopoiesis. Our model cannot represent two situations where jumps in differentiation stages have been observed: 1) megakaryocytes can emerge directly from LT-HSCs with a very few number of intermediary steps; 2) in stress erythropoiesis, ST-HSC can directly jump into stress erythroid progenitors Harandi et al. [29].

We proposed a rigorous derivation of a deterministic measure-valued system with two boundary ODEs dynamics and an intermediary PDE (defined in a weak sense) from the rescaled stochastic compartmental system. The scaling parameter captures both the large order of magnitude of the number of HSCs and the number of compartments. The limiting system is a first step to approximate the dynamics of hematopoiesis amplification in a continuum. This model could be extended to include more accurately different biological considerations such as stress or study the lineages separately.

As shown by simulations in Figure 2.1 and Figure 2.2, this modeling approach allows to capture the amplification between HSCs and mature cells. Characterizing this mechanism has been the topic of study of [15], with a 3-compartmental model where the scaling parameter represents the size of the HSC population. The characterization of this amplification in the present framework remains to be addressed.

The convergence result we obtained arises from a slow-fast dynamics, with slow division events, slow death events for the last compartment, and fast differentiation. That explains the structure of the limit where the extremal compartments play a specific role: the first one is only filled by (slow) self-renewal and the last one is only emptied by (slow) death. On the other hand, the fast differentiation dynamics in the intermediary compartments allowed to derive the limiting PDE. The topic of a future work will be to study the associated fluctuations and deduce the speed of convergence.

Appendix

Lemma A.1. For any $N > 0$, the martingales $(M_j^N(t), 1 \leq j \leq N)_{t \geq 0}$ are square-integrable and their quadratic variations are given by

$$\begin{aligned}
 \langle M_1^N \rangle_t &= \frac{1}{N} \int_0^t \left[r \left(\frac{1}{N}, X_N^N(s) \right) + m \left(\frac{1}{N}, X_N^N(s) \right) \right] X_1^N(s) ds \\
 \langle M_i^N \rangle_t &= \int_0^t \left[r \left(\frac{i}{N}, X_N^N(s) \right) + N m \left(\frac{i}{N}, X_N^N(s) \right) \right] X_i^N(s) ds \\
 &\quad + \int_0^t N m \left(\frac{i-1}{N}, X_N^N(s) \right) X_{i-1}^N(s) ds, \quad 2 \leq i \leq N-1, \\
 \langle M_N^N \rangle_t &= \frac{1}{N} \int_0^t \left[m \left(\frac{N-1}{N}, X_N^N(s) \right) X_{N-1}^N(s) + d X_N^N(s) \right] ds, \\
 \langle M_1^N, M_2^N \rangle_t &= - \int_0^t m \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds, \\
 \langle M_i^N, M_{i+1}^N \rangle_t &= - \int_0^t N m \left(\frac{i}{N}, X_N^N(s) \right) X_i^N(s) ds, \quad 2 \leq i \leq N-2, \\
 \langle M_{N-1}^N, M_N^N \rangle_t &= - \int_0^t m \left(\frac{N-1}{N}, X_N^N(s) \right) X_{N-1}^N(s) ds, \\
 \langle M_i^N, M_j^N \rangle_t &= 0 \quad \text{otherwise.}
 \end{aligned}$$

Proof. Recall that the compensated measures $(\mathcal{N}_i^j, 1 \leq i \leq N, j \in \{r, m, d\})$ are all independent. The martingale $M_1(t)$ is given by

$$\begin{aligned}
 M_1^N(t) &= \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N r(1/N, X_N^N(s^-)) X_1^N(s^-)} \widetilde{\mathcal{N}}_1^r(ds, d\theta) \\
 &\quad - \frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m(1/N, X_N^N(s^-)) X_1^N(s^-)} \widetilde{\mathcal{N}}_1^m(ds, d\theta).
 \end{aligned}$$

By independence of $\widetilde{\mathcal{N}}_1^r(ds, d\theta)$ and $\widetilde{\mathcal{N}}_1^m(ds, d\theta)$ we can derive

$$\begin{aligned}
 \langle M_1^N \rangle_t &= \frac{1}{N^2} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N r(1/N, X_N^N(s^-)) X_1^N(s^-)} d\theta ds \\
 &\quad + \frac{1}{N^2} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m(1/N, X_N^N(s^-)) X_1^N(s^-)} d\theta ds \\
 &= \frac{1}{N} \int_0^t r \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds + \frac{1}{N} \int_0^t m \left(\frac{1}{N}, X_N^N(s) \right) X_1^N(s) ds.
 \end{aligned}$$

With similar arguments, for $2 \leq i \leq N - 1$ we obtain

$$\begin{aligned} \langle M_i^N \rangle_t &= \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq r(i/N, X_N^N(s^-))} X_i^N(s^-) d\theta ds \\ &\quad + \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m((i-1)/N, X_N^N(s^-))} X_{i-1}^N(s^-) d\theta ds \\ &\quad + \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m(i/N, X_N^N(s^-))} X_i^N(s^-) d\theta ds \\ &= \int_0^t r\left(\frac{i}{N}, X_N^N(s)\right) X_i^N(s) ds + \int_0^t N m\left(\frac{i-1}{N}, X_N^N(s)\right) X_{i-1}^N(s) ds \\ &\quad + \int_0^t N m\left(\frac{i}{N}, X_N^N(s)\right) X_i^N(s) ds, \end{aligned}$$

and for $M_N(t)$ we deduce

$$\begin{aligned} \langle M_N^N \rangle_t &= \frac{1}{N^2} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m((N-1)/N, X_N^N(s^-))} X_{N-1}^N(s^-) d\theta ds + \frac{1}{N^2} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N d X_N^N(s^-)} d\theta ds \\ &= \frac{1}{N} \int_0^t m\left(\frac{N-1}{N}, X_N^N(s)\right) X_{N-1}^N(s) ds + \frac{1}{N} \int_0^t d X_N^N(s) ds. \end{aligned}$$

On the other hand, we have that

$$\begin{aligned} \langle M_1^N, M_2^N \rangle_t &= -\frac{1}{N} \int_0^t \int_{\mathbb{R}^+} \mathbb{1}_{\theta \leq N m(1/N, X_N^N(s^-))} X_1^N(s^-) d\theta ds \\ &= -\int_0^t m\left(\frac{1}{N}, X_N^N(s)\right) X_1^N(s) ds. \end{aligned}$$

Similarly, we can find $\langle M_i, M_{i+1} \rangle_t$ for $2 \leq i \leq N - 2$ and $\langle M_{N-1}, M_N \rangle_t$. Lastly, for any other i, j , given the independence of their compensated measures, it follows that

$$\langle M_i, M_j \rangle_t = 0.$$

□

Lemma A.2. Let us consider M_F , the set of finite measures on $\mathbb{R}$. We also consider a measure-valued function $\mu \in \mathbb{D}(\mathbb{R}_+, M_F)$ and a compact domain D of $\mathbb{R}_+ \times \mathbb{R}$ such that

$$\int_D \mu(s, dx) ds < \infty.$$

Then, for any function g continuous on D , we have that

$$\int_D g(s, x) \mu^k(s, dx) ds \rightarrow \int_D g(s, x) \mu(s, dx) ds$$

as $k \rightarrow \infty$, where

$$\mu^k(s, dx) = \sum_{i \in \mathbb{Z}} \mu\left(s, \left(\frac{i-1}{k}, \frac{i}{k}\right]\right) \delta_{\frac{i}{k}}. \quad (2.6.1)$$

Proof. By Tietze Theorem, we extend the function g in a continuous way from the set D to a rectangle $\mathcal{R} = [t_1, t_2] \times [n_1, n_2] \subset \mathbb{R}^2$ which contains D with $n_1, n_2 \in \mathbb{Z}$.

For any $k \in \mathbb{N}$, we divide the set $\mathcal{R}$ in disjoint subsets of the form $\mathcal{R}_i^k = \{(s, x) \in \mathcal{R} : x \in ((i-1)/k, i/k]\}$ for $i \in \mathbb{Z}$. Notice that $\mathcal{R}_i^k = \emptyset$ for $|i|$ large enough by construction. Then, for any $i \in \mathbb{Z}$ and any $(s, x) \in \mathcal{R}_i^k$ we have that

$$\left| g(s, x) - g\left(s, \frac{i}{k}\right) \right| \leq \eta^k,$$

where

$$\eta^k = \sup_{\substack{(s,x), (s,x') \in \mathcal{R}, \\ |x-x'| \leq 1/k}} |g(s, x) - g(s, x')|. \quad (2.6.2)$$

By integrating over $\mathcal{R}_i^k$ we obtain

$$\left| \int_{\mathcal{R}_i^k} g(s, x) \mu(s, dx) ds - \int_{\mathcal{R}_i^k} g\left(s, \frac{i}{k}\right) \mu(s, dx) ds \right| \leq \eta^k \int_{\mathcal{R}_i^k} \mu(s, dx) ds.$$

By summing over all $i \in \mathbb{Z}$ in we derive

$$\left| \int_{\mathcal{R}} g(s, x) \mu(s, dx) ds - \int_{\mathcal{R}} g\left(s, \frac{i}{k}\right) \mu(s, dx) ds \right| \leq \eta \int_D \mu(s, dx) ds, \quad (2.6.3)$$

since

$$\begin{aligned} \sum_{i \in \mathbb{Z}} \int_{\mathcal{R}_i^k} g\left(s, \frac{i}{k}\right) \mu(s, dx) ds &= \int_{\mathcal{R}} \sum_{i \in \mathbb{Z}} g\left(s, \frac{i}{k}\right) \mu\left(s, \left(\frac{i-1}{k}, \frac{i}{k}\right]\right) ds \\ &= \int_{\mathcal{R}} g(s, x) \mu(s, dx) ds. \end{aligned}$$

Notice that $\eta^k \rightarrow 0$ when $k \rightarrow \infty$ by construction in (2.6.2) and by assumption $\int_D \mu(s, dx) ds < \infty$. Then, using (2.6.3) we prove the Lemma. $\square$

CHAPTER 3

Classical JAK2V617F+ Myeloproliferative Neoplasms emergence and development based on real life incidence and mathematical modeling

This chapter corresponds to the preprint *Classical JAK2V617F+ Myeloproliferative Neoplasms emergence and development based on real life incidence and mathematical modeling*, a joint work with Vincent Bansaye, Evelyne Lauret, Morgane Mounier, Valérie Ugo, Sylvie Méléard and Stéphane Giraudier [25]. It is currently under revision at *Mathematical Biosciences* and is reproduced here with minimal modifications.

Abstract

Mathematical modeling allows us to better understand myeloproliferative neoplasms (MPN), a group of blood cancers, emergence and development. We test different mathematical models on an initial cohort to determine the emergence and evolution times before diagnosis of JAK2V617F+ classical MPN (Polycythemia Vera (PV) and Essential Thrombocythemia (ET)). We consider the time before diagnosis as the sum of two independent periods: the time (from embryonic development) for the JAK2V617F mutation to occur, not disappear and enter proliferation, and a second time corresponding to the expansion of the clonal population until diagnosis. We prove that the rate of active mutation occurrence increases exponentially with age following the Gompertz model rather than being constant. We find that the first tumorous cell takes an average time of 63.1 ± 13 years to appear and start proliferation. On the other hand, the expansion time is constant: 8.8 years once the mutation has emerged. These results are validated in an external cohort. Using this model, we analyze JAK2V617F ET versus PV, and obtain that the time of active mutation occurrence for PV takes approximately 1.5 years more than for ET to develop, while the expansion time was similar. In conclusion, our age-dependent approach for the emergence and development of MPN demonstrates that the emergence of a JAKV617F mutation should be linked to an aging mechanism, and indicates a 8 – 9 years period of time to develop a full MPN.

3.1 Introduction

Myeloproliferative Neoplasms (MPN) are blood cancers issued from the transformation of hematopoietic stem cell and related to the acquisition of JAK2 signaling deregulation. The most frequent driver mutation in MPN is JAK2V617F, followed by JAK2 exon12, CALR and MPL [78]. MPN are categorized into three main diseases that present the JAK2V617F mutation in the majority of cases: Essential Thrombocythemia (ET), Polycythemia Vera (PV) and Primary Myelofibrosis (PMF) [74]. ET is characterized by an increased platelet number, PV by erythroid compartment enlargement and PMF by bone marrow fibrosis. When the JAK2 mutation is responsible for the pathology development, the mutation is present in one allele for most cells for ET, but often undergoes an homologous recombination of JAK2V617F for PV resulting in the mutation being present in both alleles [70]. In PMF, the JAK2 mutation is often coupled with secondary events such as TET2 mutation or ASXL1 [78]. We center our study on ET and PV, given the reduced number of reported cases for PMF. In order to develop one of these pathologies, two steps are needed: the acquisition of a mutation at stem cell level and the development of the cancerous population. The diagnosis of MPN just like Polycythemia Vera is performed when the total tumor mass (erythrocytes) represents at least 25% of the erythroid hematopoietic mass [74]. This suggests a total cancer mass of at least 750 grams in the blood and probably the same mass in the bone marrow. Based on the cell mass, we can speculate that a minimum of one million cell divisions are needed to create such a MPN. We also can speculate that the same order or tumor mass is present in all myeloid chronic myeloproliferative malignancies such as ET.

Mathematical modelling and statistical analysis based on patients data can help us better understand the dynamics of mutation emergence and expansion. This modelling could yield important information for prevention strategies and early diagnosis. Using real life registries of MPN and age of pathology detection in patients, we study the emergence and expansion of MPN for patients presenting the JAK2V617F mutation.

We define the time to diagnosis as composed of two independent elements: the time T_1 of active mutation emergence, namely the time for a first cell harboring the JAK2V617F mutation to emerge and enter the cell cycle, and the time T_2 between emergence and detection, namely the time for cancer population to grow until diagnosis. An important parameter in our model is the active mutation rate defined as the product of the mutation rate at stem cell level and the probability of each mutation to develop into a MPN. Based on a first cohort of diagnosis data, we prove here by a rigorous statistical analysis that this active mutation rate is age-dependent and increases exponentially following the Gompertz model (see (3.3.1)). We further prove that a constant MPN growing time is sufficient to properly explain the data. The statistical analysis of the model on a second cohort of diagnosis data validates the model and its

robustness. The estimation of the parameters of the model allows to obtain information on characteristics of the diseases: an average time of 63.1 ± 13 years for the first tumorous cell to appear and start proliferation and a constant expansion time of 8.8 years once the mutation has emerged. We also analyze JAK2V617F ET versus PV, and obtain that the time of active mutation occurrence for PV takes approximately 1.5 years more than for ET to develop, while the expansion time is similar.

Mathematical modelling of MPN emergence and development based on patients data has already been introduced in different approaches. In [31] the authors use age of detection data to study the behaviour of the size of the mutant population. In other works, sequencing data is used to back-track mutations to define the timeline to driver mutation acquisition, (see for example [57], [79], [49]). In all of these works around MPN emergence, an assumption of a constant mutation occurrence rate throughout the life of an individual is made. Such an assumption would imply an exponential distribution for the emergence time (see subsection 3.2.2). However, the available data (Figure 3.1) shows a growing number of diagnosed patients for older ages (see also [67] and [54]) whose trend is not consistent with an exponential (see Figure 3.4).

In section 3.2 we describe the data used as well as the construction of the model. Section 3.3 is dedicated to introducing, selecting and validating our age-dependent model. Finally, in section 3.4 we apply the selected model to the data to obtain information on the mutation emergence and expansion of MPN and the difference in them for PV and ET.

3.2 Framework

3.2.1 Incidence and frequencies of MPN according to age

In this study, we work with data obtained from the cancer registry of the Côte d'Or in France from 1990 to 2020, which compiles the ages of detection for 264 patients with different types of MPN (Table 3.2 in Appendix). In this region, life expectancy (and then relative incidence of pathology) can be analyzed. We center our analysis to patients diagnosed with either PV or TE and presenting the JAK2V617F mutation. The mean detection age in this data set is of 71.9 years with a standard deviation of 13.8 years. To further validate the models, a second set from FIMBANK was used, containing the ages of detection for ET and PV of 1111 individuals as seen in Table 3.3 in Appendix. In this data set, we have a mean of 64.4 years with a standard deviation of 14.1 years.

The two sets of data used in our analysis are adjusted to take into account the distribution of the population into each age group and consider the proportion of individuals diagnosed within each group. This considers that some individuals could have died before being diagnosed and are thus not present in the data sets. Each

frequency is then divided by the probability of survival up to that age group as follows. Let T_D be the age of death and T_M the age of detection. Then the frequencies given by the data of the probability of being diagnosed at age t is given by

$$\mathbb{P}(T_M = t, T_D \geq t).$$

If we suppose independence between the two variables T_D and T_M , this is equal to

$$\mathbb{P}(T_M = t)\mathbb{P}(T_D \geq t).$$

Using the death rates for each age group, one can obtain $\mathbb{P}(T_D \geq t)$ and then calculate

$$\mathbb{P}(T_M = t) = \frac{\mathbb{P}(T_M = t, T_D \geq t)}{\mathbb{P}(T_D \geq t)}.$$

We apply this adjustment based on the French and European survival data by age and sex of 2021 [34]. The resulting corrected data gives us the age distribution of JAK2V617F for patients who have not died of anything else before. This data indicates that the frequencies of JAK2V617F MPN development increases dramatically in older ages as shown in Figure 3.1.

3.2.2 Model construction and notation

In order to perform a diagnosis of PV or ET, the total cancer mass needs to surpass a certain threshold of around $M = 10^{12}$ cells. We assume that diagnosis is performed when this size is reached. We consider that the age of detection, T_M , is the sum of two independent time periods, T_1 and T_2 : the first one being the time from embryonic development for the mutation to occur, not disappear, and enter in proliferation (that we will name “mutation occurrence”) and a second time corresponding to the expansion time, once the first cancer cell begins to proliferate until the time of MPN diagnosis. The times T_1 and T_2 are taken as independent since the time taken for a cell to become proliferative should not be related to the time for its population to grow to a certain size. In order to consider that the mutation could have appeared at an embryonic level, as suggested in [52], T_M is considered to start 9 months before the birth of an individual.

To define the first period of time T_1 , let τ be rate of mutation or activation of an existing mutation at stem cell level. Then, the time for a mutation to appear follows an exponential distribution with parameter τ . We consider that each activated mutant cell has a probability p to proliferate and then be detected. The number $(1-p)$ corresponds to the frequency of non proliferating mutant cells (just like clonal hematopoiesis does for a long period of time) or of mutant cells whose lineages disappear before reaching detection size of MPN. To find the law of T_1 under these assumptions, let L be a

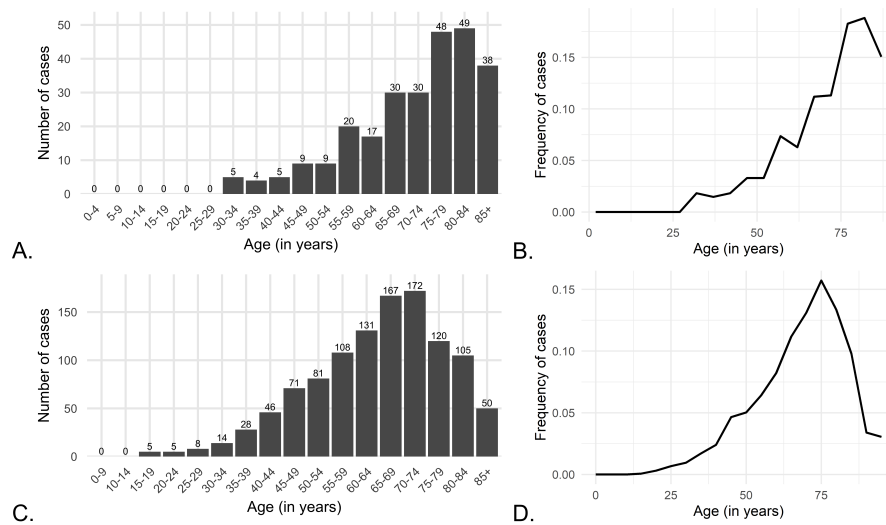


Figure 3.1: **A:** Incidence by age of JAK2V617F Classical MPN (ET and PV, excluding myelofibrosis) in the Cote d'Or Regional Registry Database. **B:** Incidence by age of JAK2V617F classical MPN (ET and PV, excluding myelofibrosis) in the Cote d'Or Regional Registry Database after adjustment excluding other causes of death before the MPN diagnosis. **C:** Incidence by age of JAK2V617F Classical MPN (ET and PV, excluding myelofibrosis) in the National BCB-FIMBANK Cohort. **D:** Incidence by age of JAK2V617F classical MPN (ET and PV, excluding myelofibrosis) in the National BCB-FIMBANK Cohort after adjustment excluding other causes of death before the MPN diagnosis.

random variable of law $Geom(p)$ that corresponds to the number of mutation events before the first active cancer cell appears and let $X_1, X_2, \dots$ be independent and identically distributed random variables of law $Exp(\tau)$ that define the time elapsed between two mutation attempts. Then, T_1 will simply be the sum of these times:

$$T_1 = \sum_{i=1}^L X_i.$$

The law of this variable can be found using the generating function

$$\begin{aligned} \mathbb{E} [e^{-\lambda T_1}] &= \mathbb{E} [\mathbb{E} [e^{-\lambda T_1} | L]] = \mathbb{E} \left[\left(\frac{\tau}{\tau - \lambda} \right)^L \right] \\ &= \sum_{k \geq 1} \left(\frac{\tau}{\tau - \lambda} \right)^k p(1 - p)^{k-1} = \frac{\tau p}{\tau p - \lambda}. \end{aligned}$$

Then, T_1 is an exponential random variable with parameter $\delta = \tau p$, which we will refer to as active mutation rate. If we assume that δ depends on the age $t \geq 0$ of an individual, the density of T_1 is given by

$$f_1(t) = \delta(t) \exp \left(- \int_0^t \delta(s) ds \right). \quad (3.2.1)$$

On the other hand, the time T_2 corresponds to the time taken for a population of cells to reach detection size of 10^{12} cells. Given the high number of divisions needed to reach this size (10^6 divisions), variability in division time is averaged and it makes sense to consider that the expansion time should be similar from one individual to the other. The law of T_2 then needs to be unimodal, centering around a certain value and positive as, for example, a lognormal law whose density for an age $t \geq 0$ is

$$f_2(t) = \frac{1}{t\sigma\sqrt{2\pi}} \exp \left(- \frac{(\log(t) - \mu)^2}{2\sigma^2} \right), \quad (3.2.2)$$

with $\mu \in \mathbb{R}$, $\sigma > 0$. The simplest case is to consider a null variance with $T_2 = \alpha$ for $\alpha \geq 0$, where all individuals have the same expansion time, implying that environmental modifications do not have an effect on the growth of the tumor.

By independence of T_1 and T_2 , the law of T_M is computed as the convolution of the densities f_1 and f_2 of T_1 and T_2 , respectively. That is, for $t \geq 0$, it is given by

$$f_M(t) = \int_0^t f_1(s) f_2(t - s) ds. \quad (3.2.3)$$

The variables and parameters are summarized in the following table.

Variable	Description
T_M	Time to detection. $T_M = T_1 + T_2$
T_1	Mutation emergence time.
T_2	Expansion time.
Parameter	Description
δ	Active mutation rate. $\delta = \tau p$
τ	Mutation rate.
p	Probability of proliferation and detection of a mutant cell.

Table 3.1: Variables and parameters of the model.

3.3 Mathematical model and results

3.3.1 Mathematical age-dependent model

Data in [Figure 3.1](#) shows a clear age structure, with an increasing number of cases as age increases. Then, we hypothesize an age-dependent active mutation occurrence rate that grows as a person gets older. Biological motivation behind considering an age-dependent mutation occurrence rate can be justified by at least three age-dependent modifications of the environment: i- JAK2V617F mutation appearance (from embryo to adulthood), ii- JAK2V617F activation (ie stem cell quiescence exit) or iii- the success probability of the expansion after activation. We consider the Gompertz model (see [\[42\]](#), pages 90-92) for the active mutation rate δ , which is given by

$$\delta_G(t) = A \exp(kt), \quad (3.3.1)$$

where $A > 0$, $k > 0$, and t is the age of an individual. The Gompertz model has been widely used to model different biological age-dependent rates such as mortality [\[26\]](#) or cancer death rate [\[65\]](#). In our case, it can be explained as follows: we assume that individuals have an initial mutation occurrence rate A , and an age-related increase in mutation occurrence represented through a parameter k . Then, for an age t , the active mutation occurrence takes the form $\delta_G(t)$ and the density of T_1 in [\(3.2.1\)](#) is expressed as

$$f_1(t) = A \exp\left(\frac{A}{k}\right) \exp(kt) \exp\left(-\frac{A}{k} \exp(kt)\right)$$

We consider two models for the expansion time. The first one being a fixed T_2 assuming that environmental modifications do not have an effect on the growth of the tumor. The second model takes T_2 as a lognormal random variable defined by [\(3.2.2\)](#). This setting can be explained by invasion process modifications (ie variation time between two divisions), environment modifications or randomness in the diagnostic process (some patients are diagnosed after vascular events (late detection) when others

are diagnosed after a systematic blood cell analysis performed for another pathology (early detection)) or some subjects die (of other unrelated pathologies) before T_2 was finalized. Then, we have two age-dependent models characterized by

$$\text{Model A.1: } \begin{cases} \delta_G(t) = A \exp(k t) \\ T_2 = \alpha \text{ fixed,} \end{cases} \quad (3.3.2)$$

$$\text{Model A.2: } \begin{cases} \delta_G(t) = A \exp(k t) \\ T_2 \sim \text{lognormal}(\mu, \sigma^2). \end{cases} \quad (3.3.3)$$

The respective laws, f_1^M and f_2^M , of the age of detection T_M under each model come from (3.2.3) and are given for an age t by

$$f_1^M(t) = A \exp\left(\frac{A}{k}\right) \exp(k(t - \alpha)) \exp\left(-\frac{A}{k} \exp(k(t - \alpha))\right)$$

$$f_2^M(t) = \int_0^t A \exp\left(\frac{A}{k}\right) \exp(k s) \exp\left(-\frac{A}{k} \exp(k s)\right) \frac{1}{t\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log(t - s) - \mu)^2}{2\sigma^2}\right) ds.$$

3.3.2 Parameter estimation

We estimate the parameters using the generalized Expectation-Maximization (EM) algorithm (see for example [48]) as presented in [64]. Let us recall the method. Let T_M be the vector that corresponds to the observed data of detection ages and has distribution $f_{T_M}(t_M; \theta)$ for an age t_M , where θ is the vector of parameters ($\theta = (A, k, \alpha)$ for Model A.1 (3.3.2) and $\theta = (A, k, \mu, \sigma^2)$ for Model A.2 (3.3.3)). Consider the missing data t_2 , corresponding to the values of T_2 with distribution $f_{T_2}(t_2; \theta)$. Then the complete data is (t_M, t_2) , and it is distributed according to $f_{T_M, T_2}(t_M, t_2; \theta)$. If the complete data was available, the log-likelihood could be calculated by

$$\log L_{T_M, T_2}(\theta) = \log f_{T_M, T_2}(t_M, t_2; \theta).$$

The goal is to find the values of the parameters that maximize this function. This algorithm is iterative and consists of two steps: the E-step and the M-step. Starting from an initial value θ_0 , then, on iteration $k + 1$, the E-step involves computing

$$Q(\theta; \theta_k) = \mathbb{E}_{\theta_k}(\log L_{T_M, T_2}(\theta) | t_M).$$

Then, on the M-step choose θ_{k+1} such that

$$Q(\theta_{k+1}; \theta_k) \geq Q(\theta_k; \theta_k).$$

For the M-step, as suggested in [64], we take θ_{k+1} as one Newton-Raphson step from θ_k over the function $Q(\theta_{k+1}; \theta_k)$, that is

$$\theta_{k+1} = \theta_k + a_k \delta_k,$$

where

$$\delta_k = - \left[\frac{\partial^2(Q(\theta; \theta_k))}{\partial \theta \partial \theta^T} \right]^{-1} \bigg|_{\theta=\theta_k} \left[\frac{\partial(Q(\theta; \theta_k))}{\partial \theta} \right] \bigg|_{\theta=\theta_k}$$

and $0 < a_k \leq 1$. The choice of a_k needs to be such that it stays in the parameters space and the likelihood is non-decreasing. To do this, take $a_k^0 = 1$ and divide by half its value until the parameters satisfy the desired conditions. Then, we apply backtracking line search: while $Q(\theta_k + a_k \delta_k; \theta_k) > Q(\theta_k; \theta_k) + 10^{-4} a_k \nabla Q(\theta_k; \theta_k)^T \delta_k$, set $a_k = 0.8 a_k$. Lastly, we take a_k as the final value of these steps.

3.3.3 Model selection

We want to compare and select a model amongst Model A.1 (3.3.2) and Model A.2 (3.3.3). To do so, we estimate the parameters for both models using the EM-algorithm described in subsection 3.3.2 using the Côte d'Or cohort, and we proceed to perform a Chi-square goodness-of-fit test as follows. We consider the $k = 13$ bins corresponding to ages $0 - 30$, 5 year increments from age 30 to 85 and 85+. For each bin i ($i = 1, \dots, k$), let O_i and E_i be the number and the expected number of observations on bin i , respectively. Then, we construct the statistic as

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}.$$

As the sum of squared standardized random deviations, it follows approximately a Chi-square distribution with $k - c$ degrees of freedom, where c is the number of estimated parameters ($c = 3$ for Model A.1 and $c = 4$ for Model A.2).

These tests resulted in not rejecting Model A.1 with a power of 0.952 and not rejecting Model A.2 with a power of 0.85. We refer to Figure 3.2 for visual comparison between the estimations for fixed T_2 in **A-B** and random T_2 in **C-D**.

Since both models are not rejected, we use the Bayesian information criterion (BIC) presented in [69] to select the most appropriate model amongst these two to explain the data. This criterion is defined by

$$\text{BIC} = c \log N - 2 \log \mathcal{L},$$

where c is the number of parameters of the model, N is the size of the data sample and $\mathcal{L}$ is the likelihood function. The BIC uses the likelihood function to measure how well a model can explain the data, while also penalizing the number of parameters in the model to avoid overfitting. A lower BIC is preferred. We obtain that the Model A.1 with constant MPN expansion time (T_2 as a fixed value) has a BIC of 2079 and Model A.2 with lognormal MPN expansion time has one of 2085.3. This indicates that the Model A.1 with fixed expansion time should be chosen. While incorporating variability

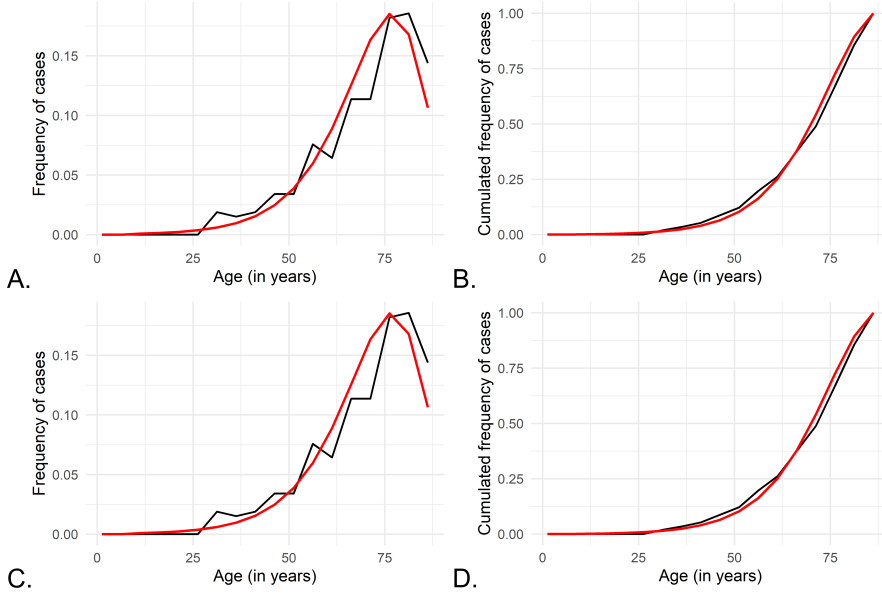


Figure 3.2: **A:** Frequency of cases for the data (in black) and probability density function of the estimation (in red) for the model with aging (3.3.2). **B:** Cumulated frequencies of cases for the data (black) and cumulative distribution of the estimation (in red) for the model with aging (3.3.2). **C:** Frequency of cases for the data (in black) and probability density function of the estimation (in red) for the model with aging and lognormal MPN growing time (3.3.3) **D:** Cumulated frequencies of cases for the data (black) and cumulative distribution of the estimation (in red) for the model with aging and lognormal MPN growing time (3.3.3).

on the expansion time T_2 gives a better fit to the data in terms of the likelihood, the gain is small and not compensated by the cost of adding additional parameter to the model. This would imply that, given our data, the Gompertz model with constant MPN growing time is enough to explain the data. This suggests that environmental modifications (or other sources of variability) do not have a huge impact on the growth phase of the tumor.

3.3.4 Model validation

Validation on a second cohort

The French group of MPN recently developed a national database on ET and PV (BCB-FIMBANK: French National Cancer Institute (INCa) BCB 2013 and 2022, CHU Angers) (Table 2). We considered that this database is representative of the general population given the data size (1111 JAK2V617F MPN patients) (see Figure 3.1 C-D

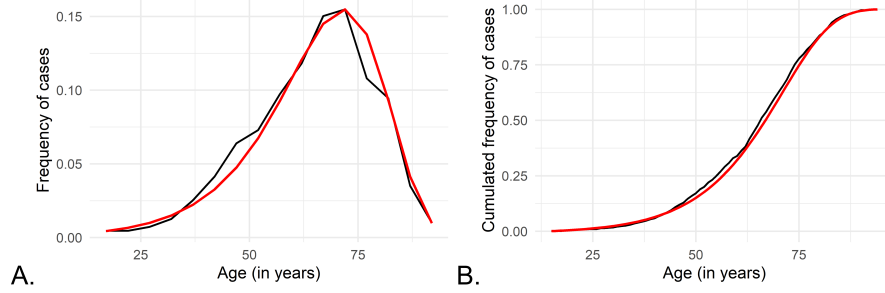


Figure 3.3: **A:** Frequency of cases for the the National BCB-FIMBANK data (in black) and probability density function of the estimation (in red) for the age-dependent model (Model A.1 (3.3.2)). **B:** Cumulated frequencies of cases for the National BCB-FIMBANK data from the (black) and cumulative distribution of the estimation (in red) for the age-dependent model (Model A.1 (3.3.2)).

for the histogram before and after death adjustment), even if it is not as exhaustive as the previous database that reported all the cases in the region. We used this second data set to validate the Gompertz model with fixed expansion time given by Model A.1 (3.3.2). We performed an estimation of the parameters through the same EM algorithm as previously described and then applied a Chi-Square goodness-of-fit test, which was not rejected with a power of 0.99 (see Figure 3.3). The two data sets provide different parameter estimation, yet the functional form of the model seems to properly explain MPN emergence and proliferation.

Age-independent models and Weibull age-dependent model

The proposed age-dependent model was selected among others given its performance with the data. Indeed, when testing different models for the emergence and expansion time (T_1 and T_2), the models fail to fit the data. We study three models with a constant active mutation rate δ , which implies $T_1 \sim \text{Exp}(\delta)$ (see subsection 3.2.2). This assumption of a constant mutation rate has been taken in the majority of genetic analysis backtracking reported to date [57], [79], [49]. The first and most simple model considers a fixed expansion time T_2 . A second model considers that external factors may influence the expansion time, T_2 and is hence taken as a lognormal random variable as defined in (3.2.2). The third model assumes a fixed expansion time, but allows the time of mutation occurrence to vary from one patient to another or the mutation to develop at different times from one patient to another (quiescence/cycling). This takes into consideration that the biological hypothesis of stress (or environment) influences cancer cell emergence. This model considers T_1 as an exponential random variable, but its rate is different from one patient to another. Then, for $1 \leq i \leq N$, where N is the number of patients, we considered that each age of emergence T_1^i is drawn from an

exponential distribution with parameter δ_i . The δ_i 's are considered as an independent sample of a $\text{lognormal}(m, s^2)$. The three age-independent models can be summarized as

$$\text{Model B.1 : } \begin{cases} T_1 \sim \text{Exp}(\delta) \\ T_2 = \alpha \text{ fixed,} \end{cases} \quad (3.3.4)$$

$$\text{Model B.2 : } \begin{cases} T_1 \sim \text{Exp}(\delta) \\ T_2 \sim \text{lognormal}(\mu, \sigma^2), \end{cases} \quad (3.3.5)$$

$$\text{Model B.3 : } \begin{cases} T_1^i \text{ sampled from } \text{Exp}(\delta_i) \\ \delta_i \text{ sampled from } \text{lognormal}(m, s^2), \quad i = 1, \dots, N \\ T_2 = \alpha \text{ fixed.} \end{cases} \quad (3.3.6)$$

By simplicity of Model B.1 and Model B.2 compared to Model A.1 and A.2, we estimate the parameters of both models through a least-squared estimation which minimizes the sum of the square difference between the data and the model for each age. Given the individual variability, the parameters of Model B.3 are estimated using the software Monolix [42]. This software is well adapted for this mixed effect modeling as it allows to easily estimate the parameters when considering the survival analysis approach. The estimation is based of the SAEM algorithm. As illustrated in Figure 3.4 A, B and E, the approximation for Model B.1 and Model B.3 presents such a big discrepancy with the data that it is sufficient to reject this model. While Model B.2 seems closer to the data, we perform a Chi-square goodness-of-fit test which results in rejecting this model at a 99% confidence level with a p-value smaller than 10^{-17} and a very large effect size of $w = 2.5$. The rejection of all three of these models imply that even when including individual or environmental variations, models with constant active mutation occurrence δ cannot explain the data. The estimations of the parameters for each model is given by

$$\text{Model B.1 : } \begin{cases} \delta = 0.0256 \\ \alpha = 32, \end{cases}$$

$$\text{Model B.2 : } \begin{cases} \delta = 0.0011 \\ \mu = 4.11 \\ \sigma = 0.15, \end{cases}$$

$$\text{Model B.3 : } \begin{cases} m = 0.022 \\ s = 0.16. \end{cases}$$

We can then think of other age-dependent models, such as a Weibull model for the active mutation rate (see [42], pages 90-92). For an age t , the Weibull mutation

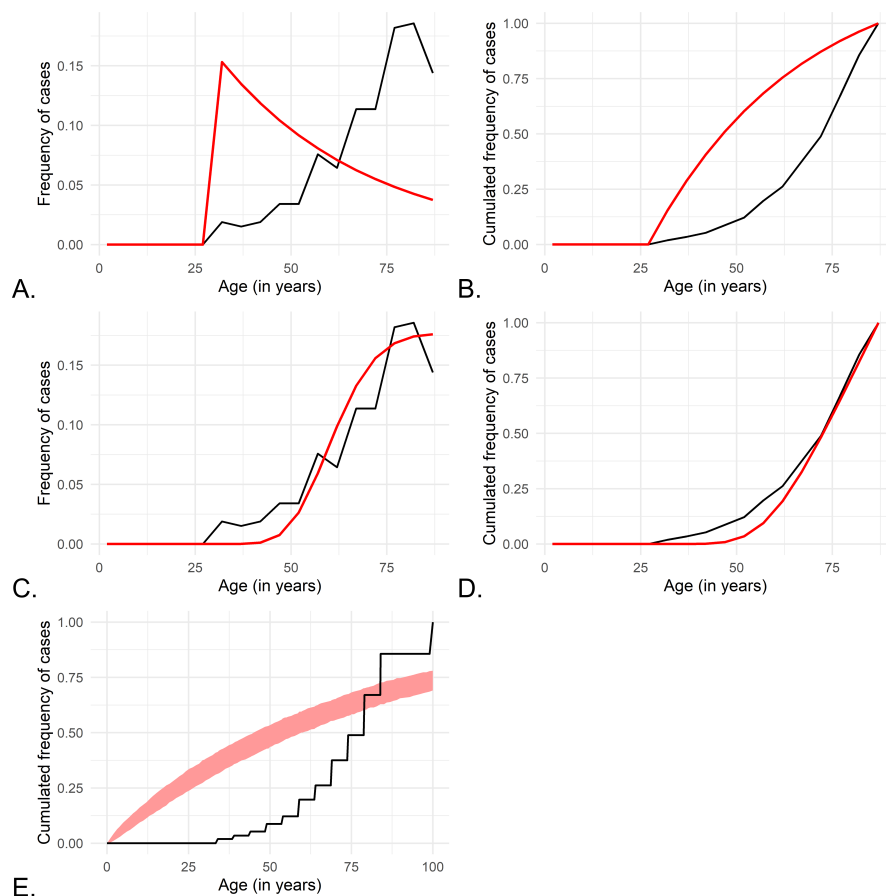


Figure 3.4: **A**: Frequency of cases for the data (in black) and probability density function of the estimation (in red) for the age-independent model (Model B.1 (3.3.4)). **B**: Cumulated frequencies of cases for the data from the (black) and cumulative distribution of the estimation (in red) for the age-independent model (Model B.1 (3.3.4)). **C**: Frequency of cases for the data (in black) and probability density function of the estimation (in red) for the age-independent model with lognormal MPN growing time (Model B.2 (3.3.5)). **D**: Cumulated frequencies of cases for the data (in black) and cumulative density of the estimation (in red) for the age-independent model with lognormal MPN growing time (Model B.2 (3.3.5)). **E**: Cumulated frequencies of cases for the data (in black) and 90% confidence interval of the estimation (in red) for the age-independent model with individual variability (Model B.3 (3.3.6))

occurrence rate function is given by

$$\delta_W(t) = \frac{\gamma}{\lambda} \left(\frac{t}{\lambda} \right)^{\gamma-1},$$

where $\gamma > 0$, $\lambda > 0$. We consider the broader model with T_2 as a lognormal random variable with density function (3.2.2). The addition of variability to the expansion time T_2 should give the same fit or better than a fixed value. The parameters are estimated using the EM-algorithm in subsection 3.3.2. Then, we perform a Chi-square goodness-of fit test and find that the Weibull model is rejected with a p-value smaller than 10^{-17} and a large effect size of $w = 0.69$. As seen in Figure 3.5, this model gives a much better estimation; yet, the initial growth of the distribution is not fast enough to properly estimate the data.

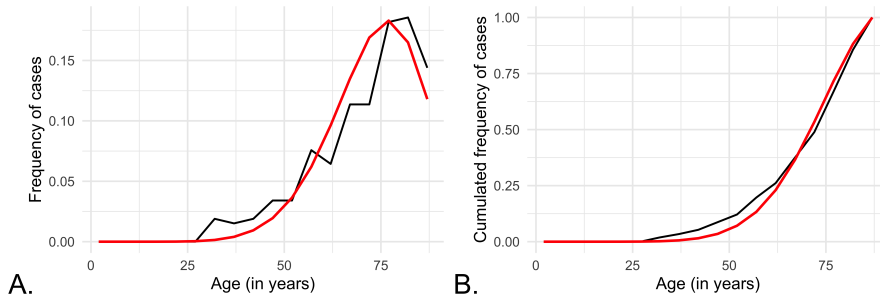


Figure 3.5: **A**: Frequency of cases for the data (in black) and probability density function of the estimation (in red) for the Weibull age-dependent model. **B**: Cumulated frequencies of cases for the data from the (black) and cumulative distribution of the estimation (in red) for the Weibull age-dependent model.

Hence, we obtain a good fit of Model A.1 with Gompertz age-dependent active mutation rate and fixed expansion time in both data sets. Coupled with the rejection of all three age-independent mutation rate models and the Weibull age-dependent model, these results suggest that not only the mutation rate cannot be constant through life, but increases exponentially with age following the Gompertz model. Given these positive results to fit the data, we apply Model A.1 (3.3.2) to our data to obtain information on the emergence and expansion of the disease and to compare the differences in mutation acquisition and expansion between the two pathologies PV and ET.

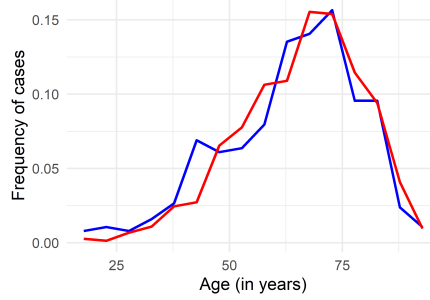


Figure 3.6: Frequencies of JAK2V617F ET (red line) and JAK2V617F PV (blue line) by age at diagnosis in the National BCB-FIMBANK Cohort

3.4 Applications of the model

3.4.1 Mutation emergence and expansion times

By estimating the parameters of Model A.1 (3.3.2) to the Côte d'Or data set, using the EM-algorithm (subsection 3.3.2), we obtain the following values

$$A = 0.000124, \quad k = 0.096577, \quad \alpha = 8.76.$$

This implies that the expected occurrence time T_1 is of 63 years with a standard deviation of 13 years and the pathology development when the first JAK2V617F stem cell is present, fixed and ready to proliferate is 8.8 years (T_2), with a mean detection age of 71.3 years.

3.4.2 Differential analysis of Essential Thrombocythemia and Polycythemia Vera

We hypothesize that ET and PV have a different kinetic because of the double hit in PV compared to single hit in ET: homologous recombination or development of secondary mutations that induces the polycythemia instead of the thrombocytosis only. Given the high number of reported MPN in the FIMBANK cohort, we use this data set to analyze the difference between ET and PV patients.

There is not a clear difference between the two diseases data (see Figure 3.6). Yet, when looking at their means and standard deviations (see Table 3.4), PV presents a bigger mean by 2 years than ET. The standard deviation of ET is bigger by 1.6 years, meaning that the ages of detection are spread across a larger range. We start by testing if ET and PV corrected data come from the same distribution through a Chi-square test, which rejects this hypothesis with a p-value of 10^{-11} . Hence, we confirm that the diagnostic age distributions are different for ET and PV (at a 99% confidence level).

We proceed to estimate the parameters of each pathology under Model A.1 (3.3.2) and find the following values for the parameters, the mean (m_1) and standard deviation (s_1) of T_1 for each disease

$$\begin{aligned} A^{PV} &= 0.00037, & k^{PV} &= 0.08575, & T_2^{PV} &= 8.8, & m_1^{PV} &= 56.9, & s_1^{PV} &= 14.2 \\ A^{ET} &= 0.00056, & k^{ET} &= 0.07935, & T_2^{ET} &= 8.3, & m_1^{ET} &= 55.5, & s_1^{ET} &= 15.1. \end{aligned}$$

The estimation under our model indicates that the average mutation occurrence time (T_1) is approximately 1.5 years higher for PV which can be explained by the time taken for the mitotic recombination to happen.

3.5 Discussion

Our present work provides mathematical modeling of JAK2V617F MPN emergence and expansion based on real life incidence and age at diagnosis, which highlights the role of an exponential age dependence increase of mutation occurrence in MPN. Based on the regional exhaustive registry of Côte d'Or (France), and lastly validated in a MPN national registry (BCB-FIMBANK registry), we model the occurrence and expansion times of JAK2V617F MPN. Most of the previous works performed to decipher the time of JAK2V617F “first” cell appearance, are based on genetic hierarchy and back-tracking performed on ancient blood samples or bone marrow samples (see [57], [79], [49]). These genetic approaches rely upon premises of a rate (a frequency) of mutation occurrence as yet unknown in MPN. It necessitates old blood or bone marrow samples for genetic analysis (to recapitulate the genetic tree evolution of the initial cancer cell) which is obviously very rare and probably biased [33]. Genetic analysis inference performed in such approaches are based on mutation appearance but not necessary proliferative/oncogenic. Indeed, it is now well documented that such JAK2V617F mutations could stay for long time without development of pathologies, as seen in clonal hematopoiesis of indeterminate potential (CHIP). Moreover, these genetic back-tracking approaches were built on stable mutation frequencies that overestimate the time of emergence if mutation rate during life is not constant. Our approach is based on registry analysis, differing from classical genetic back-tracking of mutation emergence and we show that not only the mutation rate cannot be constant, but needs to grow exponentially with age, following the well-known Gompertz model. We assume that two periods of time are necessary to develop and diagnose MPN: a first time period (T_1) corresponding to the mutation acquisition (genetic event from the conception of the embryo) but also its activation (ie ensuring to develop). This first period of time can correspond biologically to a cell cycle entry of a long-lasting quiescent cell for example, but also includes the possibility of a MPN cancer cell to disappear due to the cancer cell extinction probability. The second time period (T_2) corresponds to

the growth and development of such cancer stem cell until diagnosis. This type of modeling has been applied to invasion-fixation of mutants in eco-evolutionary models (see [17], [12]) and to the infectious diseases propagation (see [10], [9]) but less frequently to cancer emergence and development. Most reported modeling assumes that mutation and invasion rate are aging independent (ie a fixed rate of mutation and activation and a fixed probability of emergence of the pathology for a given patient), which would imply that T_1 follows an exponential law. However, models under this hypothesis show a peak of cases in younger ages, which is not coherent with the data, where most cases are present for older ages. We then speculate that myeloproliferative neoplasm emergence could need an age-dependent modeling for T_1 . Age dependence does not necessarily mean that the JAK2V617F mutation acquisition is higher in aged stem cells than in young ones; it also could be interpreted as a mutation appearance during childhood or even before birth, but an activation due to external stimulus long time after. The second time (expansion time) is easier to characterize. A model considering the expansion time as a fixed value properly fits the data and is chosen over one as a random variable. This suggests that external changes (stress, food, toxic exposure, ...) could not significantly modify the behavior of MPN development and that the proliferation time from one "active" cancer cell to the total mass of cancer at diagnosis is relatively fixed. This could mean that the main determining factor is the proliferative speed related to the mutation acquisition. We then can speculate about the possibility (and risks and benefits) to treat subjects with JAK2V617F clonal hematopoiesis to delay the emergence of the MPN or prevent the classical thrombotic complications related to these pathologies. Our mathematical and statistical study allows us to assess the time of emergence and development of MPN. We find that the mean time for cancer cell to emerge is 63 years and time from tumor emergence to diagnosis is approximately 8.8 years in accordance with previous hypothesis, (see [63] and [47]). Based on the Variant Allele Frequency of JAK2V617F in the "FIMBANK" cohort, we observe that the mean JAK2V617F VAF in MPN is around 30% (19.9% in ET and 46.1% in PV respectively) and an increased ratio of 1.4% per year (as reported previously in untreated [72] or Hydroxyurea only treated patients [36]). This should mean a 10-year period of time from undetectable VAF to the 30% allele ratio observed at diagnosis in our MPN cohort. This is very close (and a little bit higher) to the 8.8 years of tumor expansion time to MPN diagnosis that we calculate in our model. Furthermore, our modeling fits well with the JAK2V617F CHIP that has been described to take approximately a decade to transform to MPN [55], [56], [80]. In conclusion, we demonstrate using mathematical modelling that the emergence and fixation of JAK2V617F mutation is strongly linked to aging mechanisms and that the expansion time of MPN can be considered as non random. Our numerical results show a mean of 63.1 years for the emergence and fixation of the mutation, and an expansion time of approximately 8.8 years. Altogether, these results highlight the interest to

test all 50-60 years-old inhabitants for JAK2V617F CHIP and the place of preventive therapies for such subjects.

Appendix

Tables

Age	PMF	PV JAK2+	ET JAK2+	Total JAK2+ cases
0 to 4	0	0	0	0
5 to 9	0	0	0	0
10 to 14	0	0	0	0
15 to 19	0	0	0	0
20 to 24	0	0	0	0
25 to 29	0	0	0	0
30 to 34	0	1	4	5
35 to 39	0	0	4	4
40 to 44	0	1	4	5
45 to 49	0	5	4	9
50 to 54	0	2	7	9
55 to 59	1	8	11	20
60 to 64	1	10	6	17
65 to 69	2	13	15	30
70 to 74	4	12	14	30
75 to 79	5	14	29	48
80 to 85	4	15	30	49
More than 85	4	11	23	38
Total number	21	92	151	264

Table 3.2: First data set (Côte d'Or).

Age	PV JAK2+	ET JAK2+	Total JAK2+ cases
0 to 4	0	0	0
5 to 9	0	0	0
10 to 14	0	0	0
15 to 19	2	3	5
20 to 24	1	4	5
25 to 29	5	3	8
30 to 34	8	6	14
35 to 39	18	10	28
40 to 44	20	26	46
45 to 49	48	23	71
50 to 54	57	24	81
55 to 59	78	30	108
60 to 64	80	51	131
65 to 69	114	53	167
70 to 74	113	59	172
75 to 79	84	36	120
80 to 84	69	36	105
85 to 90	30	9	39
90 to 94	7	4	11
Total number	734	377	1111

Table 3.3: Second data set (FIMBANK).

		Number of cases	Mean age at diagnosis	Standard deviation of age	VAF JAK2 at diagnosis	Standard deviation of VAF
PV	Female	314	66.7	13.7	45.3	22.9
	Male	421	64.0	13.3	46.7	30.3
ET	Female	229	63.0	15.9	19.2	5.7
	Male	147	63.2	13.8	20.9	15.8

Table 3.4: Information concerning the pathologies PV versus ET by sex.

CHAPTER 4

Effects of additional mutations on growth and treatment response in JAK2V617F MPNs through a mixed effect model

This work in progress emerges from a tight collaboration with the medical doctors from Hôpital Saint-Louis Stéphane Giraudier and Nabih Maslah. Using longitudinal sequencing data collected from their patients, this analysis aims to gain valuable information for treatment selection. This work is done in collaboration with Charlotte Dion-Blanc.

4.1 Introduction

4.1.1 Biological framework

Myeloproliferative Neoplasms (MPN) are a group of blood cancers originated by a driver mutation at stem cell level which induces deregulation of proliferation. The mutation of a stem cell is carried forward in its descendancy and eventually produces an excess of one type of mature myeloid cells. The most common driver mutation is JAK2V617F, present in the majority of MPN cases. Additional mutations can arrive before or after JAK2V617 acquisition on the same cell, and while these mutations are not sufficient to develop a MPN, they can increase the probability of the evolution towards more serious diseases and decrease the survival probability (see [75] and [81]). New mutations are frequently acquired within pre-existing mutant clones (see [45]). Additional mutation on JAK2V617F cells can make cells more resistant to treatment as presented in [37]. Studies such as [75] reflect that more than 50% of diagnosed patients carry at least one additional mutation, the most frequent ones being *TET2* and *ASXL1*. Hence, understanding the impact of additional mutations gives important knowledge that allows to optimize the treatment plan (see [27]).

There are three main cytoreductive treatments for MPN, each reducing the number of malignant cells through a different mechanism: Interferon- α (IFN), Hydroxyurea (HU) and Ruxolitinib (Ruxo). HU works by inhibition of proliferation, affecting directly the growth of the mutant population. On the other hand, Ruxo kills JAK2 cells by inhibition of signalisation. Lastly, IFN creates deaths in JAK2 cells by promoting differentiation at stem cell level and exhausting the cell (see [24]), as the mechanism starts in stem cell level, there is a delay between the start of the treatment and the reduction of the population of mature JAK2V617F cells.

In this work, we construct a non-linear mixed-effect model and use longitudinal sequencing data from diagnosed patients to gain insight on the impact of each additional mutations on evolution and treatment response. This analysis aims to obtain information on the optimal treatment depending on the mutational profile of an individual.

4.1.2 Data

We work with longitudinal sequencing data of $N = 244$ patients diagnosed with a JAKV617F MPN¹. For each patient, multiple blood samples were drawn over time. Bulk sequencing analysis was applied to these blood samples, allowing to obtain infor-

¹Data is available here: <https://github.com/ana-fernandezbaranda/additional-mutations>

mation on the proportion of alleles carrying the JAK2V617F mutation as well as 53 additional mutations. For each patient at each sample we have access to the following informations:

- Age and sex of the patient;
- Time elapsed since the first sample;
- Diagnosis and undergoing treatment(s) at the time of the sample;
- The percentage of alleles carrying the JAK2V617F mutation;
- For each additional mutation: the percentage of alleles carrying the mutation.

These patients were not treated with any treatment besides IFN, HU or Ruxo.

4.1.3 Mathematical framework

Following the framework developed by [44], non-linear mixed-effect models have been widely used for describing longitudinal data and individual variability (see also [42]). In the case of cancer, they arise as a powerful tool that allows to capture the dynamics of the cancerous population, account for treatment impact, and also include the variability between individuals that can originate from known information taken as covariates, such as sex or age, or from an unknown source considered as random. A variety of different tumor dynamics can be considered, some examples can be seen in [84]. Different natural growth functions have been developed including linear and exponential growth, as well as models with saturation such as the logistic or Gompertz functions. This approach to modeling cancer allows to better understand treatment response as done in [16] and compare different treatment alternatives as done in [5] and [46]. In [16] they use longitudinal data to study treatment response in chronic lymphocytic leukemia, a type of blood cancer. The authors in [5] compute the probability of success for different treatment options and compare combinations of treatments with monotherapies. The effect of mutational profiles in treatment response is studied in [46] where mutations are taken as covariates and characterize tumor dynamics by distinguishing between proliferating cells and quiescent cells.

4.1.4 Results

We construct a non-linear mixed-effect model for the natural growth dynamics of the percentage of JAK2V617 alleles combining a generalized logistic growth function and the growth reduction by each treatment. We use this model as a base to include the effects of each additional mutation on growth speed and treatment response. Through cross-validation using the data, we validate the initial growth model. Then, we use the

model including additional mutations to obtain quantitative results of the effect of the three most prevalent mutations, *TET2*, *ASXL1* and *DNMT3A*, on treatments IFN and HU. The estimation indicates that:

- Mutations *TET2* and *ASXL1* have an important effect of accelerating the speed of growth of the JAK2V617F population, while mutation *DNMT3A* does not have a significant effect on growth,
- The three mutations make cells more resistant to treatments IFN and HU, at different intensities,
- Mutation *TET2* has a more significant negative impact on the response to HU than the one to IFN,
- Mutations *ASXL1* and *DNMT3A* have a much more significant negative impact on the response to IFN than the one to HU.

These results suggest that under mutation *TET2* cells become more resistant to HU than to IFN. On the other hand, mutations *DNMT3A* and *ASXL1* have a strong negative impact on the response to IFN, indicating that HU should give the best outcome.

The structure of this work is as follows. In Section 2 we define the mixed-effects model combining natural growth, the treatment response and the effect of additional mutations. Section 3 is devoted to applying this model to the data and presenting the results found.

4.2 Statistical method

This section is dedicated to constructing a non-linear mixed-effect model for the dynamics of the percentage of JAK2V617F mutated alleles based on the available information of each individual. We begin by presenting a base model that integrates natural growth and the administration of treatments, which will directly affect this growth. We later use this model as a starting point to include the effects of additional mutations on growth and treatment response. Then, we define the fixed and random effects of the model, and the corresponding distribution of the random effects. Lastly, we discuss the estimation of the parameters using the data.

4.2.1 Base model: model without additional mutations

Let $P(t)$ be the percentage of JAK2V617F mutated alleles at time t which is measured in days. To model natural growth of the mutated population without additional

mutations it is important to consider that we are working with a percentage of mutated alleles and thus the total percentage of JAK2 alleles cannot exceed 100%. With this in mind, we consider a Generalized logistic growth function that gives sigmoidal dynamics and presents a saturation when the population size is close to 100. Then, the dynamics of the percentage of JAK2 alleles without any additional mutation and without treatment is modeled as

$$\frac{dP}{dt}(t) = C P(t) \left(1 - \left(\frac{P(t)}{100} \right)^\gamma \right),$$

where $C, \gamma > 0$. This type of function allows to capture different types of models, such as the logistic if $\gamma = 1$ or the Gompertz when γ approaches 0.

Given the different mechanisms of each treatment, we include them in different ways and we assume that the administration of more than one type of treatment acts as the sum of all administered treatments. As a delay in treatment effect is observed for IFN ([24]), we include it in our model as beginning six months after administration. HU directly impacts the growth function by reducing division of cells. Finally, we assume that Ruxo kills a constant fraction of the cells population. Let $HU, IFN, Ruxo : t \in \mathbb{R}^+ \mapsto \{0, 1\}$ be binary functions that represent the administration of each treatment such that they are equal to 1 if patient is undergoing the treatment at time t and 0 otherwise. To take into account the six month period (equivalent to 182 days) for the mechanism of IFN, we work with the term $IFN(t - 182)$ that is equal to 1 if IFN was administered 182 days prior to time t . We introduce the three treatments to our model defining it as

$$\begin{aligned} \frac{dP}{dt}(t) = & C P(t) \left(1 - \left(\frac{P(t)}{100} \right)^\gamma \right) [1 - HU(t) K_{HU} - IFN(t - 182) K_{IFN}] \\ & - Ruxo(t) K_{Ruxo} P(t), \end{aligned} \quad (4.2.1)$$

where $C, \gamma > 0$ are growth related parameters that will be estimated and $K_l > 0$ for $l \in \{IFN, HU, Ruxo\}$ represents the growth reduction induced by treatment l , parameters that will also be estimated. We use this model as a base to further include the influence of additional mutations on the dynamics of the JAK2V617F population.

4.2.2 Full model: model including additional mutations

Let $\mathcal{M}$ be the set of all additional mutations. For $i \in \mathcal{M}$, $P^i(t)$ denotes the subpopulation of all JAK2V617F alleles also carrying mutation i (and possibly other additional mutations) and for $M \in \mathcal{P}(\mathcal{M})$, we denote by $P_M(t)$ the subpopulation of alleles

carrying exactly all mutations in the set M (and not any other mutation). It follows that

$$P(t) = \sum_{M \in \mathcal{P}(\mathcal{M})} P_M(t). \quad (4.2.2)$$

For $i \in \mathcal{M}$, let $C_i \in \mathbb{R}$ represent the growth advantage or disadvantage of mutation i and for any treatment $l \in \{HU, IFN, Ruxo\}$, let $K_l^i \in \mathbb{R}$ represent the effect of mutation i on treatment l . We assume that the effects of an additional mutation in a cell are independent of the effects of other additional mutations. That is, a cell with mutations $i, j \in \mathcal{M}$ will have a difference in growth given by $C_i + C_j$ and an effect on treatment $l \in \{HU, IFN, Ruxo\}$ given by $K_l^i + K_l^j$.

For each additional mutation $i \in \mathcal{M}$, we consider four parameters associated to this mutation:

- $C_i \in \mathbb{R}$: the growth advantage or disadvantage of mutation i . If $C_i > 0$, it means that the speed of growth of a JAK2V617F population carrying exclusively mutation i is faster than the one of a JAK2V617F population without any mutations. If $C_i < 0$ it means that the speed of growth of a JAK2V617F population carrying exclusively mutation i is slower than the one of a JAK2V617F population without any mutations.
- $K_l^i \in \mathbb{R}$ for $l \in \{HU, IFN, Ruxo\}$: effect of mutation i on treatment l . If $K_l^i > 0$, then a JAK2V617F cell carrying exclusively mutation i has a better response to treatment l than a JAK2V617F cell without any mutation, while $K_l^i < 0$ implies that mutation i makes cells more resistant to treatment l .

The parameters K_l^i for $i \in \mathcal{M}$ and $l \in \{HU, IFN, Ruxo\}$ hence give us important information about the best choice of treatment in the presence of mutation i . Indeed, if cells carry mutation i , then a very positive K_l^i could indicate a preference for treatment l , while a very negative K_l^i could suggest to consider other treatment options.

The parameters of the model are summarized in the Table [4.1](#)

Parameter	Domain	Description
C	$(0, \infty)$	Related to the natural growth
γ	$(0, \infty)$	Related to the natural growth
K_l	$(0, \infty)$	Response to treatment $l \in \{HU, IFN, Ruxo\}$
C_i	$(-\infty, \infty)$	Effect on growth of mutation $i \in \mathcal{M}$
K_l^i	$(-\infty, \infty)$	Effect on response to treatment $l \in \{HU, IFN, Ruxo\}$ of mutation $i \in \mathcal{M}$

Table 4.1: Parameters of the full model ([4.2.1](#)).

With this in mind, the population of alleles carrying exclusively the mutations in $M \in \mathcal{P}(\mathcal{M})$ has growth dynamics given by

$$\begin{aligned} \frac{dP_M}{dt}(t) = & C P_M(t) (1 + C_M) g(P(t)) \\ & \times \left[1 - HU(t)(K_{HU} + K_{HU}^M) - IFN(t - 182)(K_{IFN} + K_{IFN}^M) \right] \\ & - Ruxo(t)(K_{Ruxo} + K_{Ruxo}^M) P_M(t), \end{aligned} \quad (4.2.3)$$

where

$$g(P(t)) = \left(1 - \left(\frac{P(t)}{100} \right)^\gamma \right), \quad C_M = \sum_{i \in M} C_i,$$

and for $l \in \{IFN, Ruxo, HU\}$,

$$K_l^M = \sum_{i \in M} K_l^i.$$

The parameters C_M and K_l^M for $l \in \{HU, IFN, Ruxo\}$ represent the effect of a mutational profile $M \in \mathcal{P}(\mathcal{M})$ on growth and on the response to treatment l .

We compute the complete dynamics of the JAK2V617F with additional mutations. To begin, combining (4.2.2) and (4.2.3) we have that

$$\begin{aligned} \frac{dP}{dt}(t) = & \sum_{M \in \mathcal{P}(\mathcal{M})} C P_M(t) (1 + C_M) g(P(t)) \\ & \times \left[1 - HU(t)(K_{HU} + K_{HU}^M) - IFN(t - 182)(K_{IFN} + K_{IFN}^M) \right] \\ & - Ruxo(t)(K_{Ruxo} + K_{Ruxo}^M) P_M(t) \\ = & C g(P(t)) \\ & \times \left[(1 - HU(t) K_{HU} - IFN(t - 182) K_{IFN}) \sum_{M \in \mathcal{P}(\mathcal{M})} P_M(t) \right. \\ & + (1 - HU(t) K_{HU} - IFN(t - 182) K_{IFN}) \sum_{M \in \mathcal{P}(\mathcal{M})} C_M P_M(t) \\ & - HU(t) \sum_{M \in \mathcal{P}(\mathcal{M})} K_{HU}^M P_M(t) - IFN(t - 182) \sum_{M \in \mathcal{P}(\mathcal{M})} K_{IFN}^M P_M(t) \\ & \left. - HU(t) \sum_{M \in \mathcal{P}(\mathcal{M})} C_M K_{HU}^M P_M(t) - IFN(t - 182) \sum_{M \in \mathcal{P}(\mathcal{M})} C_M K_{IFN}^M P_M(t) \right] \\ & - Ruxo(t) \left[K_{Ruxo} \sum_{M \in \mathcal{P}(\mathcal{M})} P_M(t) + \sum_{M \in \mathcal{P}(\mathcal{M})} K_{Ruxo}^M P_M(t) \right]. \end{aligned} \quad (4.2.4)$$

We rewrite this equation in terms of the quantities $P^i(t)$ for $i \in \mathcal{M}$, which correspond to the available data.

We begin by noticing that

$$\sum_{M \in \mathcal{P}(\mathcal{M})} C_M P_M(t) = \sum_{M \in \mathcal{P}(\mathcal{M})} P_M(t) \sum_{i \in M} C_i = \sum_{i \in \mathcal{M}} C_i \sum_{\substack{M \in \mathcal{P}(\mathcal{M}) \\ i \in M}} P_M(t) = \sum_{i \in \mathcal{M}} C_i P^i(t),$$

and similarly we find that for any $l \in \{HU, IFN, Ruxo\}$

$$\sum_{M \in \mathcal{P}(\mathcal{M})} K_l^M P_M(t) = \sum_{i \in \mathcal{M}} K_l^i P^i(t).$$

Using similar arguments, we compute for $l \in \{HU, IFN\}$

$$\begin{aligned} \sum_{M \in \mathcal{P}(\mathcal{M})} C_M K_l^M P_M(t) &= \sum_{M \in \mathcal{P}(\mathcal{M})} P_M(t) \sum_{i \in M} C_i \sum_{j \in M} K_l^j \\ &= \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{M}} C_i K_l^j \sum_{\substack{M \in \mathcal{P}(\mathcal{M}) \\ i, j \in M}} P_M(t) \\ &= \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{M}} C_i K_l^j P^{i,j}(t), \end{aligned}$$

where $P^{i,j}(t)$ is the subpopulation of JAK2V617F alleles having mutations i and j . We replace these results in (4.2.4) and obtain that

$$\begin{aligned} \frac{dP}{dt}(t) &= C g(P(t)) \left[\left(P(t) + \sum_{i \in \mathcal{M}} P^i(t) C_i \right) [1 - HU(t) K_{HU} - IFN(t - 182) K_{IFN}] \right. \\ &\quad - \sum_{i \in \mathcal{M}} P^i(t) (HU(t) K_{HU}^i + IFN(t - 182) K_{IFN}^i) \\ &\quad \left. - \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{M}} C_i (HU(t) K_{HU}^j + IFN(t - 182) K_{IFN}^j) P^{i,j}(t) \right] \\ &\quad - Ruxo(t) \left[P(t) K_{Ruxo} + \sum_{i \in \mathcal{M}} P^i(t) K_{Ruxo}^i \right]. \end{aligned} \quad (4.2.5)$$

In the following we explain how to deal with the term $P^{i,j}(t)$ given our data.

Computation of mutation proportion

In order to implement this model to the data, notice that for any mutations $i, j \in \mathcal{M}$ the only information we have is the percentage of alleles carrying (non-exclusively) mutation i and the percentage for mutation j . Yet, model (4.2.5) depends on $P^{i,j}(t)$,

the percentage of alleles having mutations i and j . We then need to characterize this quantity using the available data.

As new mutations often appear in already mutated cells (see [45]), we assume that they always occur in cells carrying the most mutations. It follows that for any $i, j \in \mathcal{M}$, the percentage of alleles carrying both mutations is equal to $\min(P^i(t), P^j(t))$. We replace this results in (4.2.5) and obtain the considered dynamics given by

$$\begin{aligned} \frac{dP}{dt}(t) = & C g(P(t)) \\ & \times \left[\left(P(t) + \sum_{i \in \mathcal{M}} P^i(t) C_i \right) [1 - HU(t) K_{HU} - IFN(t - 182) K_{IFN}] \right. \\ & - \sum_{i \in \mathcal{M}} P^i(t) \left(HU(t) K_{HU}^i + IFN(t - 182) K_{IFN}^i \right) \\ & \left. - \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{M}} C_i \left(HU(t) K_{HU}^j + IFN(t - 182) K_{IFN}^j \right) \min(P^i(t), P^j(t)) \right] \\ & - Ru xo(t) \left[P(t) K_{Ru xo} + \sum_{i \in \mathcal{M}} P^i(t) K_{Ru xo}^i \right]. \end{aligned} \quad (4.2.6)$$

Recall that the description of the parameters and their domains are summarized in Table 4.1.

4.2.3 Mixed-effects model

Before applying the models to our data, we define the fixed and random effects of the models and determine the distributions of the random ones.

We consider the two growth related parameters C and γ as random effects. This implies that each individual can have its own growth function. This choice factors the variability arising from personal intrinsic and extrinsic characteristics that are not considered in our model, but can affect the growth speed of the mutated population in an individual. Given the strict positivity of both parameters, we consider them as a sample from a lognormal distribution, each with its own set of parameters.

Furthermore, as the information we have on treatment is binary, only knowing whether a treatment was being administered or not, the response to treatment for each individual can highly vary depending on dosage and frequency of treatment. Then, the three treatment related parameters K_{IFN} , K_{HU} and $K_{Ru xo}$ are considered as random. Assuming a strict negative effect, ie, treatment cannot promote growth, these parameters are also taken as a sample from lognormal distributions.

The lognormal is a unimodal distribution that depends on two parameters $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ and has a density given for $x > 0$ by

$$f(x) = \frac{1}{x \sigma \sqrt{2\pi}} \exp \left(-\frac{(\log(x) - \mu)^2}{2\sigma^2} \right).$$

For any parameter of our model $\theta \in \{C, \gamma, K_{IFN}, K_{HU}, K_{Ru xo}\}$ as they are lognormally distributed, we have that the individual parameters θ_n for $1 \leq n \leq N$, with N the number of patients, are such that $\log(\theta_n)$ are normally distributed with parameters $(\log(\theta_n^{pop}), \omega_\theta^2)$, where θ_n^{pop} is called the population parameter and represents the typical value of θ_n while ω_θ^2 accounts for the inter-individual variability. As the lognormal distribution is defined on $(0, \infty)$, it is not symmetric but right skewed, with a median given by θ_n^{pop} .

The data has two covariates for each patient: age and sex. We perform a covariate analysis to analyze whether the growth parameters C and γ depend on the age and/or the sex of an individual. In this aim, we use the Stochastic Approximation for Model Building Algorithm (SAMBA) tool implemented in Monolix. Presented in [61], this powerful tool allows to select the best covariate model without having to fit the complete model to the data. The model selection is based on minimization of a penalized likelihood. To obtain a good selection we consider all possible covariate models as a initial models. For each initial covariate model, we perform several runs and compare the different choices for each run. We obtain that the minimal penalized likelihood corresponds to the covariate model such that

- The parameter C depends on age,
- The parameter γ depends on age.

These results indicate no significant dependence of these parameters on sex.

Both parameters are assumed to be lognormally distributed and hence $\log(K_n)$ and $\log(\gamma_n)$ for $1 \leq n \leq N$ are normally distributed. We then introduce the dependence on age in the model as follows

$$\begin{aligned} \log(C^n) &= \log(C_{pop}) + \beta_C \log \left(\frac{age_n}{\overline{Age}} \right) + \eta_C^n \\ \log(\gamma_n) &= \log(\gamma_{pop}) + \beta_\gamma \log \left(\frac{age_n}{\overline{Age}} \right) + \eta_\gamma^n, \end{aligned}$$

where $\eta_C^n \sim \mathcal{N}(0, \omega_C^2)$, $\eta_\gamma^n \sim \mathcal{N}(0, \omega_\gamma^2)$; C_{pop} and γ_{pop} are the medians of the distribution of K_n and γ_n ; β_C and β_γ are parameters to be estimated representing the effect

of age in C and γ . Additionally, $\widehat{Age}$ is the weighted age average of our population data given by

$$\widehat{Age} = \exp \left(\sum_{a \in Ages} \frac{\#Obs_a}{\#Obs} \log(a) \right),$$

with $\#Obs_a$ being the number of observations for an age a and $\#Obs$ the total number of observations in the data. This implies that the population parameters are defined for a typical individual of the population with age $\widehat{Age}$.

To summarize, the following table presents the parameters of the model, the distribution taken for each one and their description.

Parameter	Domain	Distribution	Description
C	$(0, \infty)$	$lognormal(\mu_C, \sigma_C^2)$	Related to the natural growth
β_C	$(-\infty, \infty)$		Age effect on parameter C
γ	$(0, \infty)$	$lognormal(\mu_\gamma, \sigma_\gamma^2)$	Related to the natural growth
β_γ	$(-\infty, \infty)$		Age effect on parameter γ
K_l	$(0, \infty)$	$lognormal(\mu_l, \sigma_l^2)$	Response to treatment $l \in \{HU, IFN, Ruxo\}$
C_i	$(-\infty, \infty)$	Fixed	Effect on growth of mutation $i \in \mathcal{M}$
K_l^i	$(-\infty, \infty)$	Fixed	Effect on response to treatment $l \in \{HU, IFN, Ruxo\}$ of mutation $i \in \mathcal{M}$

Table 4.2: Parameters and distributions of the full model.

For patient $1 \leq n \leq N$ with N_n samples, the observation $1 \leq m \leq N_n$, denoted $o_{n,m} \in [0, 100]$ follows the model given by

$$o_{n,m} = f(t_{n,m}, \Theta_n) + g(t_{n,m}, \Theta_m, \xi) \varepsilon_{n,m},$$

where Θ_n are the individual parameters; $t_{n,m}$ corresponds to the time of observation m ; the structural model f is a solution of the ordinary differential equation (4.2.6) with initial condition given by the first observation $f(0, \Theta_n) = o_{n,1}$; $\xi = (a, b)$ are parameters to be estimated of the error model g which is a function of the structural model such that

$$g(t_{n,m}, \Theta_m, \xi_n) = a + bf(t_{n,m}, \Theta_n),$$

and the residual errors satisfy $\varepsilon_{n,m} \sim \mathcal{N}(0, 1)$.

Additionally, to apply the model we consider the following restriction based on biological considerations:

- While the impact of an additional mutation on treatment can be positive or negative, the resulting treatment response cannot be positive (cannot make the population grow). Hence, the sum of the elements multiplying each treatment function $HU(t)$, $IFN(t - 182)$, $Ruxo(t)$ must remain non-negative.

4.2.4 Estimation of parameters

The estimation of the parameters using the data is done using the software Monolix. We estimate both population parameters Θ_{pop} and the individual parameters Θ_n , $1 \leq n \leq N$. The estimation of the population parameters is done based on the Stochastic Approximation Expectation-Maximization (SAEM) algorithm developed in [20]. This approximation is done in three steps: a burn-in phase of 5 iterations to initialize SAEM, an exploratory phase with high stochasticity to explore the parameter space, and a smoothing phase. Both the exploratory and smoothing phases are stopped once convergence has been detected.

Once the population parameters are found, the individual parameters are estimated using Empirical Bayes Estimates, giving the most probable value of the parameters for each individual given its observations. To obtain the standard errors of the estimation, the Fisher Information Matrix (FIM) is calculated by stochastic approximation.

We begin estimations with uninformative initial conditions for the parameters, and run iteratively the algorithm, each time using the previous estimates as a new initial conditions. For the first runs, we enable simulated annealing to encourage broader exploration of the parameter space and reduces the risk of convergence to local solutions. Once a good estimation has been found where estimations stabilize across iterations, we disable simulated annealing to reduce stochastic variability and obtain more precise convergence with smaller standard errors of the approximation. Our final choice is made by minimizing the relative standard error.

4.3 Results

4.3.1 Validation of the model

As a first step, we verify that the growth model (4.2.1) is appropriate given using data. To do this, we perform a cross validation by estimating the population and individual parameters on a randomly selected sample consisting of 80% of the individuals. We

use this estimation of the population parameters to predict the individual parameters of the 20% remaining individuals. We study the fit of the prediction to the data for both samples.

In Figure 4.1, we can visually compare the observations with respect to their corresponding predictions for both the initial 80% of the data and the remaining 20%. The plots show not only a good fit for the initial 80%, given the small number of outliers values in the predictions compared to the observations (7.56%), but the estimation also performs well with the remaining 20% of the data with only 7.41% of outliers. This implies that in the majority of observations, the model properly fits the data. Hence, the proposed model (4.2.1) is validated for this data.

The resulting estimation of the median of the population parameters and the coefficient of variation (CV) are presented below in Table 4.3. The CV represents the ratio between the standard deviation and the mean, giving information on the inter-individual variability. All parameters except γ have an important CV, indicating a strong variability between individuals, and validating the choice of these parameters as random variables. The important inter-individual variability in the growth parameter C implies that intrinsic or extrinsic individual characteristics that are not considered give rise to highly variable growth speed between individuals. On the other hand, the inter-individual variability in treatment parameters K_{IFN} , K_{HU} and K_{RUXO} most likely arises from the binary modeling of the treatments (is being administered/is not being administered) that does not take into account factors like dosage or frequency, which can lead to a very different response between individuals. The values of the age related parameters β_C and β_γ indicate an increase in speed of growth of the JAK2V617F population with age.

Commonly in the data, an individual is only administered at most one treatment at a time, and this treatment can vary over time. Furthermore, the treatment approaches are very mixed between individuals. This high variety of the data coupled with the positive results in the cross-validation suggest good identifiability of these parameters, even with a high CV.

Parameter	Median	CV(%)
C	0.0072	334.28
β_C	4.29	
γ	0.023	0.22
β_γ	-2.68	
K_{IFN}	0.79	695.71
K_{HU}	0.34	408.57
K_{RUXO}	0.00014	506.07

Table 4.3: Estimation results for the base model (4.2.1).

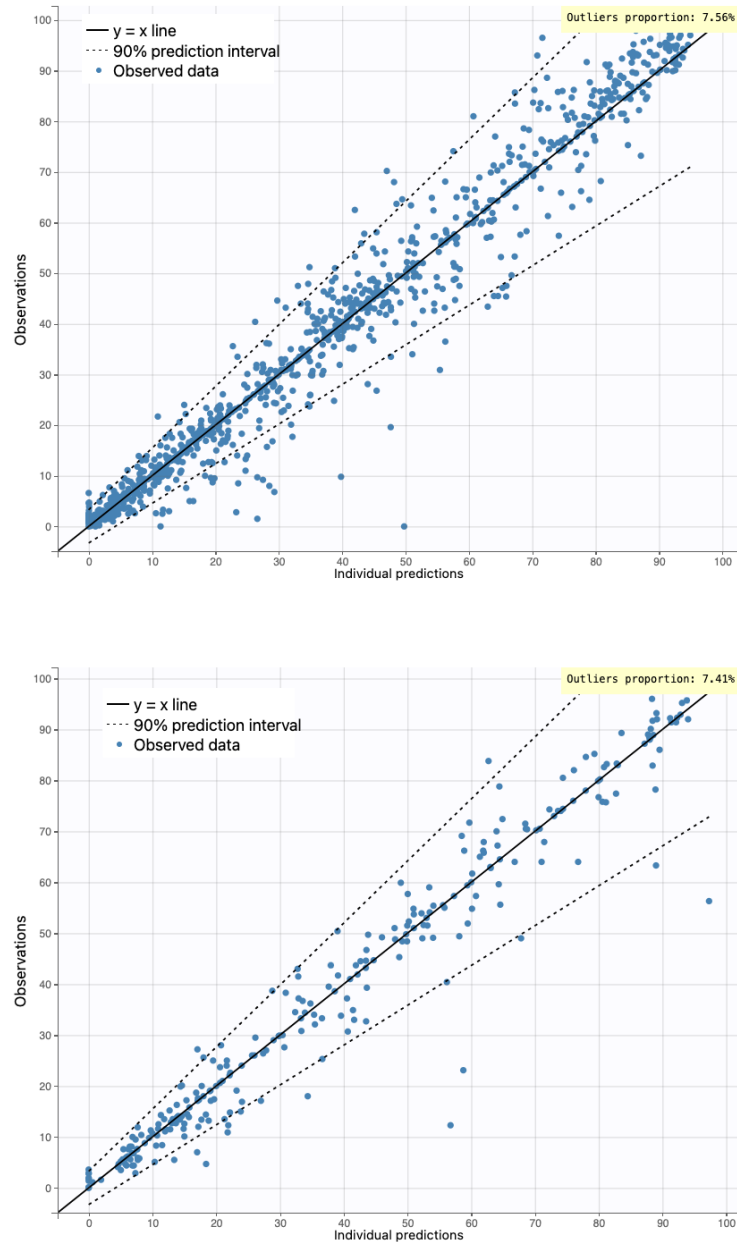


Figure 4.1: Observations vs. predictions in the base model (4.2.1) for an estimation done with 80% of patients (top) and applied to remaining 20% of patients (bottom).

4.3.2 Introducing mutations: IFN vs HU

Having verified the base model, we now proceed to include the three most prevalent additional mutations in our population: *TET2* with 48.8% patients in our data base carrying it, *DNMT3A* with 39.3% and *ASXL1* with 29.9%. For these three additional mutations, we study the impact on growth and treatment response for IFN and HU. This gives us important insight on the best treatment plan in the presence of these mutations. The effect of additional mutations on treatment Ruxo was not included as it was not possible to find good estimates for the related parameters. Indeed, when included, the estimation gave infinite confidence interval, giving no real information on the true values of these parameters.

Hence, we work with the set of additional mutations

$$\mathcal{M}_1 = \{TET2, DNMT3A, ASXL1\}$$

Table 4.4 shows the number of patients having each combination of mutations in this set.

Mutations	Number of individuals
TET2	57
ASXL1	46
DNMT3A	27
TET2 + ASXL1	28
TET2 + DNMT3A	24
ASXL1 + DNMT3A	12
TET2 + ASXL1 + DNMT3A	10
None	40

Table 4.4: Number of individuals carrying each combination of mutations (exclusively).

We include the effect of these three mutations on growth and treatment response in IFN and HU using (4.2.4). Then, the model we work with is given by

$$\begin{aligned} \frac{dP}{dt}(t) = C g(P(t)) & \left[\left(P(t) + \sum_{i \in \mathcal{M}_1} P^i(t) C_i \right) [1 - HU(t) K_{HU} - IFN(t - 182) K_{IFN}] \right. \\ & - \sum_{i \in \mathcal{M}_1} P^i(t) (HU(t) K_{HU}^i + IFN(t - 182) K_{IFN}^i) \\ & \left. - \sum_{i \in \mathcal{M}_1} \sum_{j \in \mathcal{M}_1} C_i (HU(t) K_{HU}^j + IFN(t - 182) K_{IFN}^j) P^{i,j}(t) \right] \\ & - Ruxo(t) P(t) K_{Ruxo}. \end{aligned} \quad (4.3.1)$$

This model has the base parameters $(C, \beta_C, \gamma, \beta_\gamma, K_{IFN}, K_{HU}, K_{RUXO})$, and the additional parameters $(C_{DNMT3A}, K_{IFN}^{DNMT3A}, K_{HU}^{DNMT3A}, C_{ASXL1}, K_{IFN}^{ASXL1}, K_{HU}^{ASXL1})$.

We estimate the parameters of the model using again the data set containing only 80% of the patients, to later validate the results found. We compare the prediction obtained with this estimation with the one obtained from the base model. There is a significant improvement in the prediction for certain individuals carrying at least one of the mutations in $\mathcal{M}_1$ as seen in Figure 4.2. We can observe in the figure that while the base model managed to approximate the general trend of the observations, it did not succeed to fully capture the dynamics. The addition of these three mutations allows to approximate much more closely the observations.

We perform cross-validation of these results on the data set containing the remaining 20% of the data, which was not used for the estimation. By comparing the prediction with the observations, we obtain a good fit of the data, with only small number of outliers outside the 90% prediction interval (10% of observations). Furthermore, we obtain as well an improvement in the prediction with respect to the base model for certain patients carrying at least one of these mutations as shown in Figure 4.3.

Results on additional mutations

Through the estimation, we obtain the following values on the parameters of the model, with their 95% confidence interval.

Parameter	Value	Confidence interval (95%)
C_{TET2}	12.76	[8.8 , 16.73]
K_{IFN}^{TET2}	-2.08	[-2.28, -1.87]
K_{HU}^{TET2}	-4.45	[-4.73, -4.17]
C_{DNMT3A}	-0.0041	[-0.016, 0.0077]
K_{IFN}^{DNMT3A}	-33.72	[-42.29, -25.16]
K_{HU}^{DNMT3A}	-6.41	[-8.89, -3.92]
C_{ASXL1}	5.93	[5.18, 6.68]
K_{IFN}^{ASXL1}	-37.62	[-43.8, -31.44]
K_{HU}^{ASXL1}	-2.32	[-2.9, -1.73]

Table 4.5: Estimation and 95% confidence interval of the parameters.

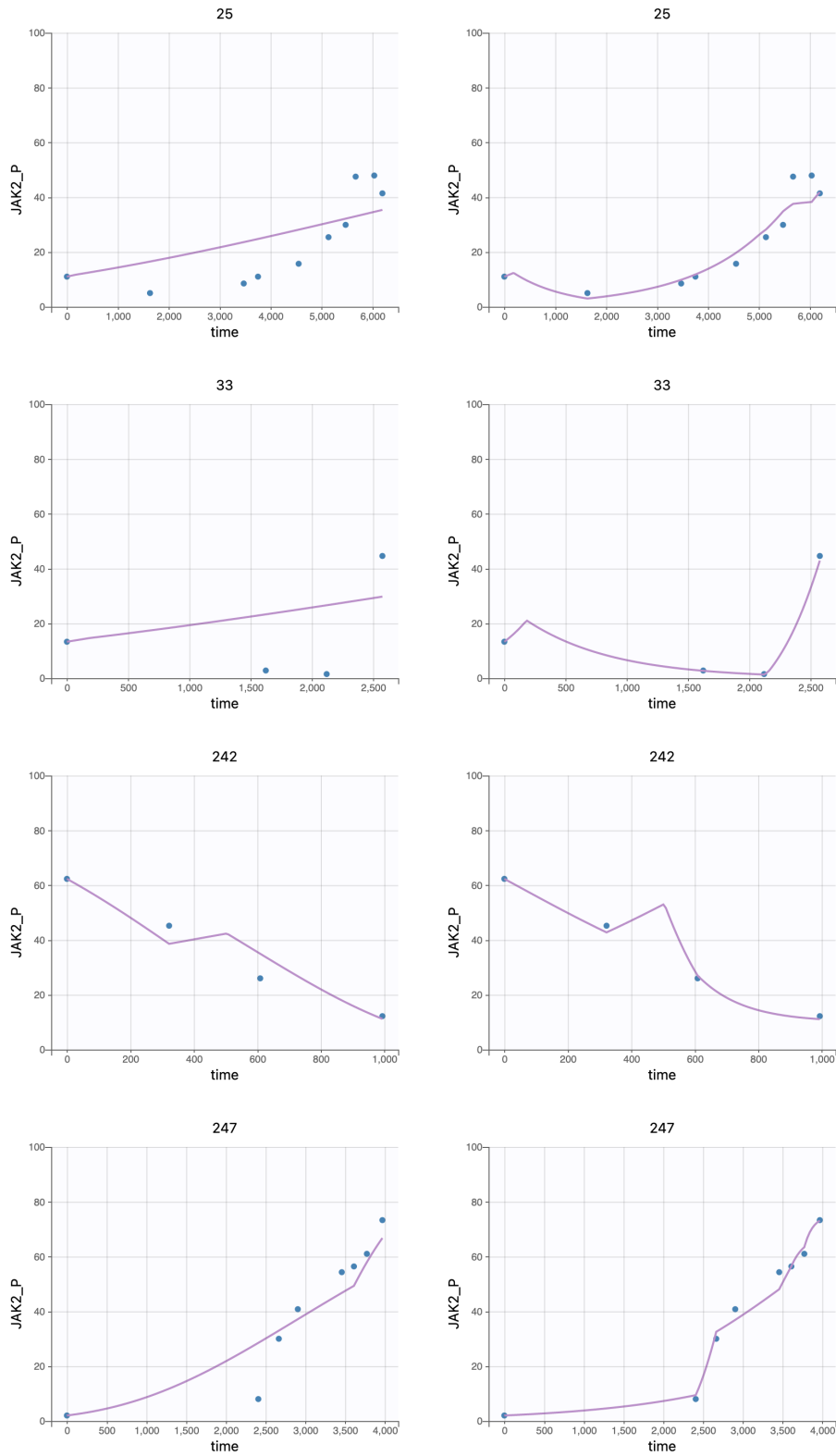


Figure 4.2: Observation points and estimation curve of four patients for the base model (4.2.1) (left) and for the full model with three additional mutations (4.3.1) (right).

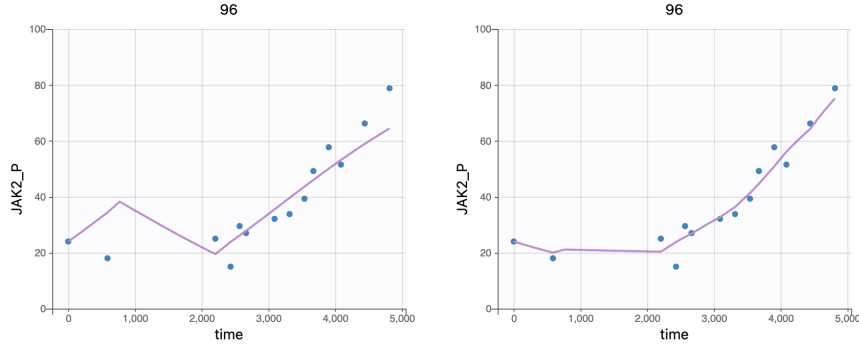


Figure 4.3: Observation points and estimation curve of a patient for the base model (4.2.1) (left) and for the full model with three additional mutations (4.3.1) (right).

This resulting estimation of the parameters can be interpreted as:

- $C_{TET2}, C_{ASXL1} > 0$: mutations $TET2$ and $ASXL1$ promote faster growth of the JAK2V617F population, with the effect of $TET2$ being much more important than the one of $ASXL1$.
- $C_{DNMT3A} \approx 0$: mutation $DNMT3A$ does not have a significant effect on the growth of the JAK2V617F population.
- $K_l^i < 0$ for $l \in \{IFN, HU\}$ and $i \in \{TET2, ASXL1, DNMT3A\}$: all three mutations induce a worse treatment response for IFN and HU.
- $K_{HU}^{TET2} < K_{IFN}^{TET2} < 0$: mutation $TET2$ has a more significant impact on the response to HU than the one to IFN.
- $K_{IFN}^i < K_{HU}^i < 0$ for $i \in \{ASXL1, DNMT3A\}$ mutations $DNMT3A$ and $ASXL1$ have a much more important impact on the response to IFN than the one to HU.

These results suggest that under the presence of $TET2$ or $ASXL1$ one should expect a faster disease evolution, particularly for $TET2$. When the $TET2$ mutation is present alone, treatment IFN should be more effective than HU. On the contrary, with the existence of $DNMT3A$ or $ASXL1$, one can expect a better response to HU than IFN. In the case where both $TET2$ and $DNMT3A$ or $TET2$ and $ASXL1$ are present, the best treatment will depend on the percentage of alleles with each mutation. Though the effect of $TET2$ on HU is much smaller than the ones of $DNMT3A$ and $ASXL1$ on IFN, indicating that HU would probably give the best results, regardless of the proportion of carrying each mutation.

4.4 Discussion and perspectives

In this work, we construct a mixed-effect model for the dynamics of the percentage of JAK2V617F alleles in individuals diagnosed with MPN. This model integrates the natural growth of the population through a generalized logistic function, the effect of the three main treatments of MPN, and the impact of additional mutations on growth speed and treatment response. We take the growth and treatment related parameters as random effects to account for inter-individual variability that can arise from individual characteristics, as well as from the different approaches in treatment for each patient. We consider the parameters associated to each additional mutations as fixed effects.

This model sets the framework to identify the impact of additional mutations, giving valuable information for treatment selection. We include the three most prevalent additional mutations in our cohort and study the effects on the response to treatments IFN and HU. The effect on treatment Ruxo is not considered, as it was not possible to find good estimates. Different possible solutions remain to be explored. A first step is to discuss if a modified model should be considered to include the mechanisms of this treatment differently. A simplification of the model could also be envisioned to facilitate the parameters estimation. As patients commonly have an accumulation of mutations, we aim to include more additional mutations and characterize not only the individual mutation effects, but the complete effect of different combinations of mutations, depending on the proportion of each.

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CHAPTER A

Appendix

A.1 Code from Chapter 2

This section presents the code for the simulations in Chapter 2

```
import numpy as np
from matplotlib.pyplot import *
import matplotlib.pyplot as plt
import math

#Define the function r and m and rate d

def r(x, z):
    return 0.02

def m(x, z):
    return 0.015

d = 0.005

##### STOCHASTIC SYSTEM #####

## N = number of compartments
## T = final time
```

```
## Nt = number of time divisions
## X0 = initial values

def stochastic(N, T, Nt, X0):
    dt = T/Nt
    X = np.zeros([Nt+1, N])
    X[0] = np.array(X0, dtype=float)

    for t in range(0, Nt):
        Xt = X[t].copy()

        for i in range(N):
            if i == 0:
                # LT-HSC
                lam_gain = N * r(1/N, Xt[-1]) * Xt[0] * dt
                lam_loss = N * m(1/N, Xt[-1]) * Xt[0] * dt
                gain = np.random.poisson(lam_gain.astype(np.float64))
                loss = np.random.poisson(lam_loss.astype(np.float64))
                X[t+1, i] = Xt[i] + gain / N - loss / N

            elif i < N - 1:
                # Middle compartments
                lam_gain1 = r(i/N, Xt[-1]) * Xt[i] * dt
                lam_gain2 = N * m((i-1)/N, Xt[-1]) * Xt[i-1] * dt
                lam_loss = N * m(i/N, Xt[-1]) * Xt[i] * dt
                gain1 = np.random.poisson(lam_gain1.astype(np.float64))
                gain2 = np.random.poisson(lam_gain2.astype(np.float64))
                loss = np.random.poisson(lam_loss.astype(np.float64))
                X[t+1, i] = max(Xt[i] + gain1 + gain2 - loss, 0)

            else:
                # Mature cells
                lam_gain = N * m((N-1)/N, Xt[-1]) * Xt[N-2] * dt
                lam_loss = N * d * Xt[-1] * dt
                gain = np.random.poisson(lam_gain.astype(np.float64))
                loss = np.random.poisson(lam_loss.astype(np.float64))
                X[t+1, i] = Xt[i] + gain / N - loss / N

    return X

##### PDE SYSTEM #####
```

```

## Nt = number of time divisions
## T = final time
## Nx = number of space divisions
## X0 = initial condition

def pde(Nt, T, Nx, X0):
    ## j = time
    ## i = maturation level

    Dt = T/Nt
    Dx = 1/Nx

    X = np.zeros([Nt+1, Nx])
    X[0] = np.array(X0, dtype=float)

    z = X0[Nx - 1]
    for j in range(1, Nt + 1):
        t = j * T/Nt

        for i in range(Nx):
            x = i/Nx

            if i == 0:
                X[j, i] = X[j - 1, i]
                    + Dt * (r(0, z) - m(0, z)) * X[j - 1, i]

            elif i < Nx - 1:
                X[j, i] = X[j - 1, i]
                    + (r(x, z) * X[j - 1, i]
                      - (m(x, z) * X[j - 1, i]
                        - m(x - Dx, z) * X[j - 1, i - 1]) * Nx) * Dt

            else:
                X[j, i] = X[j - 1, i] - Dt * d * X[j - 1, i]
                    + m(1, z) * X[j, Nx - 2]

        z = X[j, Nx - 1]
    return X

## Simulating stochastic system

```

```
T = 80 #Time interval
Nt = 20001 #Time divisions
Nx = 50 #number of compartments

Dt = T/Nt
Dx = 1/Nx

REPS = 50

X0 = np.zeros(Nx)
X0[0] = 50

X_502 = np.zeros(Nx)

for i in range(REPS):
    X = stochastic(Nx, T, Nt, X0)
    X_502 = X_502 + X * (1/REPS)

fig0, ax0 = plt.subplots(1, 1, sharex=True, figsize=(9, 6))
imag0 = ax0.imshow(X_502.T, interpolation='nearest', cmap=cm.YlGnBu,
                    origin='lower', extent=[0, T, 0, 1],
                    aspect='auto', vmin=0, vmax=223)

fig0.colorbar(imag0, pad=0.01, ax=ax0)

ax0.set_xlabel("Time")
ax0.set_ylabel("Maturity level")
plt.show()

## Simulating deterministic system

fig0, ax0 = plt.subplots(1, 1, sharex=True, figsize=(9, 6))
imag0 = ax0.imshow(XX.T, interpolation='nearest', cmap=cm.YlGnBu,
                    origin='lower', extent=[0, T, 0, 1],
                    aspect='auto', vmin=0.1, vmax = 223)

fig0.colorbar(imag0, pad=0.01, ax=ax0)

ax0.set_xlabel("Time")
ax0.set_ylabel("Maturity level")
```

```
plt.show()

plt.plot(np.linspace(0, 80, 2001), XX[:, -1])
plt.xlabel("Time")
plt.ylabel("Mature cells")
plt.show()
```

A.2 Code from Chapter 3

This section presents the code for the EM algorithm used in Chapter 3.

```
### EM ALGORITHM
library(pracma)

data <- read.csv("SMP.csv", header = TRUE, sep = ",")

D <- c()
for (i in length(data$X)){
  D <- c(D, rep(data$X[i], data$Cas_Total[i]))
}

##### DENSITIES #####
## Density function for T1 and T2
f1 <- function(x, params) {
  a <- params[1]
  b <- params[2]

  return((a + b * x) * exp(- a * x - b * x^2 / 2))
}

f2 <- function(x, params) {
  c <- params[3]
  l <- params[4]

  return(l^2 / c * M * exp(- l * x) * exp(- l / c * M * exp(- l * x)))
}

## Density function for TM as the integral
fun_int <- function(t, params){
```

```
fun <- function(s) {f2(s, params) * f1(t - s, params)}
aux <- try(integrate(fun, 0, t, subdivisions = 100L, rel.tol = 1e-13,
                    abs.tol = 1e-13), silent = TRUE)
if(inherits(aux, 'try-error')){
  r <- NA_real_
}
else {
  r <- max(c(aux$value, 0))
}
return(r)
}

## Probability of Z/X
prob_ZdX <- function(z, x, params){
  a <- params[1]
  b <- params[2]
  c <- params[3]
  l <- params[4]

  num <- l^2 / c * M * exp(- l * (x - z))
    * exp (- l / c * M * exp(-l * (x - z)))
    * (a + b * z) * exp(- a * z - b * z^2 / 2)

  den <- fun_int(x, params)

  return(num/den)
}

##### EXPECTATIONS #####
f_E1 <- function(x, params){

  fun <- function(s) {s * prob_ZdX(s, x, params)}

  aux <- try(integrate(fun, 0, x, subdivisions = 100L, rel.tol = 1e-13,
                    abs.tol = 1e-13), silent = TRUE)
  if(inherits(aux, 'try-error')){
    r <- NA_real_
  }
  else {
    r <- max(c(aux$value, 0))
  }
}
```

```
    return(r)
  }

f_E2 <- function(x, params){

  fun <- function(s) {s^2 / 2 * prob_ZdX(s, x, params)}

  aux <- try(integrate(fun, 0, x, subdivisions = 100L, rel.tol = 1e-13,
                      abs.tol = 1e-13), silent = TRUE)
  if(inherits(aux, 'try-error')){
    r <- NA_real_
  }
  else {
    r <- max(c(aux$value, 0))
  }
  return(r)
}

f_E3 <- function(x, params, l){

  fun <- function(s) {exp(l * s) * prob_ZdX(s, x, params)}

  aux <- try(integrate(fun, 0, x, subdivisions = 100L, rel.tol = 1e-13,
                      abs.tol = 1e-13), silent = TRUE)
  if(inherits(aux, 'try-error')){
    r <- NA_real_
  }
  else {
    r <- max(c(aux$value, 0))
  }
  return(r)
}

f_E4 <- function(x, params, a, b){

  fun <- function(s) {log(a + b * s) * prob_ZdX(s, x, params)}

  aux <- try(integrate(fun, 0, x, subdivisions = 100L, rel.tol = 1e-13,
                      abs.tol = 1e-13), silent = TRUE)
  if(inherits(aux, 'try-error')){
    r <- NA_real_
  }
}
```

```
  }  
  else {  
    r <- max(c(aux$value, 0))  
  }  
  return(r)  
}
```

```
##### Q HESSIAN AND GRADIENT #####
```

```
Q <- function(x, params, params_k){  
  r <- 0  
  
  N <- length(x)  
  
  a <- params[1]  
  b <- params[2]  
  c <- params[3]  
  l <- params[4]  
  
  ak <- params_k[1]  
  bk <- params_k[2]  
  ck <- params_k[3]  
  lk <- params_k[4]  
  
  for (i in 1:N){  
    Xi <- x[i]  
  
    E1 <- f_E1(Xi, params_k)  
    E2 <- f_E2(Xi, params_k)  
    E3 <- f_E3(Xi, params_k, l)  
    E4 <- f_E4(Xi, params_k, a, b)  
  
    r <- r + log(l^2 * M / c) - l * Xi + (1 - a) * E1 - b * E2 -  
      l * M / c * exp(- l * Xi) * E3 + E4  
  }  
  return(-r)  
}
```

```
hessian_Q <- function(x, params, params_k){  
  N <- length(x)
```

```
Q_h <- function(p){
  r <- 0
  a <- p[1]
  b <- p[2]
  c <- p[3]
  l <- p[4]

  ak <- params_k[1]
  bk <- params_k[2]
  ck <- params_k[3]
  lk <- params_k[4]

  for (i in 1:N){
    Xi <- x[i]

    E1 <- f_E1(Xi, params_k)
    E2 <- f_E2(Xi, params_k)
    E3 <- f_E3(Xi, params_k, l)
    E4 <- f_E4(Xi, params_k, a, b)

    r <- r + log(l^2 * M / c) - l * Xi + (1 - a) * E1 - b * E2 -
      l * M / c * exp(- l * Xi) * E3 + E4
  }
  return(-r)
}
hess <- hessian(Q_h, params)
return(hess)
}

grad_Q <- function(x, params, params_k){
  n <- 4
  N <- length(x)
  Q_g <- function(p){
    r <- 0
    a <- p[1]
    b <- p[2]
    c <- p[3]
    l <- p[4]

    ak <- params_k[1]
    bk <- params_k[2]
```

```
ck <- params_k[3]
lk <- params_k[4]
for (i in 1:N){
  Xi <- x[i]

  E1 <- f_E1(Xi, params_k)
  E2 <- f_E2(Xi, params_k)
  E3 <- f_E3(Xi, params_k, l)
  E4 <- f_E4(Xi, params_k, a, b)

  r <- r + log(l^2 * M / c) - l * Xi + (1 - a) * E1 - b * E2 -
    l * M / c * exp(- l * Xi) * E3 + E4
}
return(-r)
}
g <- grad(Q_g, params)
return(g)
}

##### LIKELIHOOD #####
LL <- function(x, params){
  N <- length(x)
  r <- 0
  for (i in 1:N){
    xi <- x[i]
    aux <- fun_int(xi, params)
    if(aux > 0){
      r <- r + log(aux)
    }
  }
  return(r)
}

##### EM #####

a0 <-
b0 <-
c0 <-
l0 <-

maxjter <- 50
```

```
params_k <- c(a0, b0, c0, l0)

dist <- 10
tol <- 0.1

while(dist > tol){
  H <- hessian_Q(D, params_k, params_k)
  g <- grad_Q(D, params_k, params_k)

  delta <- - solve(H) %*% g

  alfa <- 1/max(delta)
  params <- params_k + alfa * delta
  while(params[1] < 0 | params[2] < 0 | params[3] < 0 | params[4] < 0){
    alfa <- alfa/2
    params <- params_k + alfa * delta
  }

  f <- Q(D, params_k, params_k)
  fk <- Q(D, c(params), params_k)

  jter <- 0

  while(fk < f & jter < maxjter){
    alfa <- alfa/2
    params <- params_k + alfa * delta
    fk <- Q(D, params, params_k)
    jter <- jter + 1
  }
  dist <- abs(LL(D, params) - LL(D, params_k))
  print(dist)

  params_k <- params_k + alfa * delta
}
```

Titre : Modélisation stochastique de l'hématopoïèse et des syndrome myéloprolifératifs: limite hydrodynamique et analyse statistique

Mots clés : Processus de Markov, Hématopoïèse, Modèles à effets mixtes, Syndromes myéloprolifératifs

Résumé : Cette thèse combine la modélisation théorique et des approches guidées par les données afin de mieux comprendre l'hématopoïèse (formation des cellules sanguines) et les syndromes myéloprolifératifs (SMP, un groupe de cancers du sang). Tout d'abord, nous modélisons l'hématopoïèse avec une différenciation continue en étudiant la limite hydrodynamique d'un modèle compartimental stochastique lent-rapide. Ce modèle multi-échelle considère trois types de cellules (cellules souches, immatures et matures) et intègre la différenciation et la prolifération cellulaires, toutes deux régulées à travers une boucle de rétroaction. Lorsque le nombre de compartiments immatures tend vers l'infini, nous prouvons la convergence en loi des processus de taille des populations de cellules souches et matures, ainsi que de la mesure empirique de la dynamique des cellules immatures, convenablement renormalisés. Le système limite d'équations aux dérivées partielles consiste en un unique triplet impliquant des fonctions et une mesure couplées, solutions d'une équation déterministe à valeurs mesures avec des dynamiques de bord. Ensuite, à partir de données d'âge au diagnostic, nous caractérisons l'émergence structurée par l'âge et le développement des SMP issus de

la mutation *driver* la plus fréquente. Nous définissons un modèle pour la distribution de l'âge de diagnostic en le considérant comme la somme de deux composantes indépendantes : le temps nécessaire à l'apparition et entrée en prolifération de la mutation, et le temps correspondant à la durée d'expansion de la population clonale jusqu'au diagnostic. À travers un algorithme EM d'ajustement des données, ce travail met en évidence un mécanisme de vieillissement dans l'émergence de la mutation, tandis que le temps d'expansion de la maladie est constant. Le modèle proposé permet en outre d'obtenir des résultats quantitatifs sur ces durées, et de comparer deux types de SMP. Enfin, nous utilisons des données de séquençage longitudinales pour étudier les effets de différentes mutations additionnelles dans les SMP afin d'aider au choix de traitement. Un modèle à effets mixtes combinant la croissance naturelle de la population de cellules cancéreuses, la réduction de cette croissance par différents traitements, ainsi que les effets de chaque mutation additionnelle est construit. Ce modèle est ajusté aux données à l'aide du logiciel Monolix, ce qui permet d'identifier des résistances aux traitements liées à des mutations additionnelles.

Title : Stochastic modeling of hematopoiesis and Myeloproliferative Neoplasms: hydrodynamic limit and statistical analysis

Keywords : Markov processes, Hematopoiesis, Mixed effect models, Myeloproliferative Neoplasms

Abstract : This thesis combines theoretical modeling and data-driven approaches to deepen the understanding of hematopoiesis (blood cells formation) and Myeloproliferative Neoplasms (MPN, a group of blood cancers). We first model hematopoiesis with continuum differentiation by studying the hydrodynamic limit of a stochastic slow-fast compartmental model. The multiscale model considers three types of cells (stem, immature and mature cells) and integrates cell differentiation and cell proliferation, both regulated through a feedback loop. As the number of immature compartments goes to infinity, the convergence in law of the stem and mature cells population size processes and of the empirical measures of the immature cells dynamics, conveniently rescaled, is proven. The partial differential equations limiting system consists of a unique triplet involving coupled functions and a measure, which are solutions of a deterministic measure valued equation with boundary dynamics. Then, through age of detection data,

we characterize the age-structured emergence and the development of MPNs arising from the most common driver mutation. A model is defined for the distribution of the age of diagnostic by considering it as the sum of two independent components: the time for the mutation to occur and enter proliferation, and the time corresponding to the expansion of the clonal population until diagnosis. Through an EM algorithm to fit the data, this work proves an aging mechanism in the emergence of the mutation. The proposed model further allows to obtain quantitative results on these times and compare two types of MPN. Lastly, we use longitudinal sequencing data to study the effects of different additional mutations in MPN to support treatment selection. A mixed-effect model combining natural growth of the cancerous cells population, the reduction of this growth by different treatments, and the effects of each additional mutation is constructed. This model is fitted to data using the software Monolix to identify the effects of additional mutations.